SCIENTIFIC JOURNAL PUBLICATION TREND TRACKING SYSTEM

Group 6

I. INTRODUCTION

1 Overview

1.1 Project Information

- Project Name: Scientific Journal Publication Trend Tracking System
- Group Name: Group 6
- Software Type: Web Application
- GitHub Repository:

  `https://github.com/NguyenNgocBaoTram10/Scientific-Journal-Publication-Trend-Tracking-System`

1.2 Project team

Full Name	Role	Student ID	Email
Lê Thanh Tú	Leader	049206004778	tult4778@ut.edu.vn
Nguyễn Ngọc Bảo Trâm	Member	082306008015	tramnnb8015@ut.edu.vn
Lê Thị Huyền Trân	Member	077306003635	tranlth3635@ut.edu.vn
Trần Trung Hiếu	Member	089206008473	hieutt8473@ut.edu.vn
Trần Văn Học	Member	052206005618	hoctv5618@ut.edu.vn
Nguyễn Đoàn Nhật Quang	Member	079206010585	quangndn0585@ut.edu.vn

Bảng 1: Project Team Information

2 Project Scope and Limitations

2.1 Major Features

2.1.1 Major Features for User

- Register and log in to the system.
- Search scientific publications using keywords, authors, journals, or research fields.
- View detailed publication information.
- Save publications to bookmarks.

- Follow research topics of interest.
- Receive notifications about new publications and research trends.
- View publication trend analysis and statistical dashboards.
- Generate and export analytical reports (PDF/CSV).

2.1.2 Major Features for Administrator

- Log in and log out.
- Manage user accounts.
- Configure academic API sources (Semantic Scholar, OpenAlex, and Crossref).
- Monitor the data synchronization process.
- Manage system notifications.
- Generate system reports.

2.2 Project Limitations

2.2.1 Limited Use of Full-Text Data

- The system only uses publication metadata such as title, abstract, keywords, authors, and publication year.
- It does not process full-text content, which limits the depth of analysis and understanding of research papers.

2.2.2 Dependence on External APIs

- The system relies on external APIs such as OpenAlex, and Crossref.
- Any API downtime, structural changes, or request limitations may affect system stability and data collection performance.

2.2.3 Simplified Trend Analysis and Update Frequency

- Trend analysis is mainly based on publication frequency and growth rate.
- It does not apply advanced Machine Learning or AI models for prediction.
- Data is synchronized periodically (daily or weekly), so real-time trends may not be immediately reflected.

II. PROJECT MANAGEMENT PLAN

1 Overview

1.1 Scope and estimation

#	WBS Item	Complexity	Duration (Days)
1. Project Initiation			
1.1	Define project scope	Medium	1
1.2	Collect and analyze requirements	Complex	2
2. Project Management and Environment Setup			
2.1	Set up development environment (GitHub, Database, Backend, Frontend)	Simple	1
2.2	Design database (ERD, Database Schema)	Complex	2
2.3	Configure security (JWT, RBAC, Password Hashing)	Complex	2
3. System Design			
3.1	Design Use Case Diagram and Activity Diagram	Medium	1
3.2	Design Sequence Diagram	Medium	1
3.3	Design DFD and ERD	Medium	1
4. System Development			
4.1	User Registration and Login	Medium	2
4.2	Search Scientific Publications	Complex	2
4.3	View Publication Details	Medium	1
4.4	Bookmark Management	Medium	1
4.5	Research Topic Following	Medium	1
4.6	Notification Management	Medium	1
4.7	Publication Trend Analysis	Complex	3
4.8	Develop Statistical Dashboard	Complex	2
4.9	Display Trending Topics	Medium	1
4.10	Generate Statistical Reports (PDF/CSV)	Complex	2

#	WBS Item	Complexity	Duration (Days)
5. Administration Functions			
5.1	User Management	Complex	2
5.2	Configure Academic API Sources	Complex	2
5.3	Synchronize Data from Semantic Scholar, OpenAlex, and Crossref	Complex	3
6. Testing			
6.1	Prepare Test Plan	Medium	1
6.2	Design Test Cases	Medium	2
6.3	Execute Testing	Complex	2
6.4	Fix Defects After Testing	Complex	2
7. Project Finalization			
7.1	Complete Project Documentation	Medium	2
7.2	Deploy the System	Medium	1
		Total Duration	**45 Working Days**

1.2 Project Objective

- **Timeline:** The project must be completed before **July 3, 2026**.
- **Allocated Effort:** 45 man-days.

1.3 Project Risks

#	Risk Description	Impact	Possibilit	Response Plan
1	External academic APIs (Semantic Scholar, OpenAlex, Crossref) become unavailable or change their endpoints.	The system cannot retrieve publication data, causing incomplete search results and inaccurate trend analysis.	High	Implement API fallback mechanisms, monitor API status regularly, and cache previously retrieved data whenever possible.

#	Risk Description	Impact	Possibility	Response Plan
2	Large volumes of publication data lead to slow system performance.	Users experience long response times when searching publications or viewing dashboards.	Medium	Optimize database indexing, implement pagination, caching, and asynchronous background synchronization.
3	Security vulnerabilities expose user accounts or stored data.	Unauthorized access, data leakage, and loss of user trust.	Medium	Use JWT authentication, RBAC authorization, password hashing, HTTPS, and regular security testing.
4	Trend analysis results become inaccurate because of outdated or incomplete datasets.	Generated reports and statistics may mislead users' research decisions.	Medium	Schedule automatic data synchronization and validate collected data before analysis.
5	System integration between frontend, backend, and database encounters compatibility issues.	Development delays and unexpected system failures during deployment.	Medium	Perform integration testing continuously and maintain version compatibility among all components.
6	Insufficient testing before deployment leaves critical defects unresolved.	System crashes, incorrect functionality, and poor user experience.	High	Prepare comprehensive Test Plans, Test Cases, conduct functional, integration, and user acceptance testing before release.
7	Project schedule delays due to underestimated implementation effort.	Failure to complete the project within the planned timeline.	Medium	Monitor project progress weekly, prioritize critical features, and adjust the implementation schedule when necessary.

2 Project Deliverables

Sprint	Week	Duration	Deliverables	Note
Sprint 1	Week 1	14/05 – 20/05/2026	• Define project scope • Analysis requirement	
Sprint 2	Week 2	21/05 – 27/05/2026	• Finish Product Overview • Finish User Requirement • Finish Functional Requirement	
Sprint 3	Week 3	28/05 – 03/06/2026	• Finish Database Design • Finish Software design description	
Sprint 4	Week 4	04/06 – 10/06/2026	• Finish Software requirement specification	
Sprint 5	Week 5	11/06 – 17/06/2026	• Finish Software design description (continued)	
Sprint 6	Week 6	18/06 – 24/06/2026	• Finish Software testing document • Finish Release package and user guide	
Sprint 7	Week 7	25/06 – 01/07/2026	• Testing • Finish document	

3. **Responsibility Assignment**

Do ~D; Review ~R; Support ~S

Responsibility	Thanh Tú	Bảo Trâm	Huyền Trân	Trung Hiếu	Văn Học	Nhật Quang
Project Planning & Tracking	S	D	S	S	S	S
Prepare Project Introduction Document	R	D	R	R	R	R
Prepare Software Requirement Document	S	D	S	S	S	S
Prepare Software Design Document	D	D	D	D	D	D
Prepare Test Document	D	D	D	D	D	D
Prepare Software User Guides	D	D	D	D	D	D
Prepare Final Report	D	D	D	D	D	D

4. **Project Communications**

Communication Item	Who	Purpose	When	Type
Team Weekly Meeting	Group	• Report work project • Assign work for next week • Control project deadline	Every Sunday	Meeting on Google Meet
Weekly Report	Member	• Report work process	1 day / week	Report via Zalo
Group Meeting	Group	• Report work process	Every Thursday	Offline in class

5. Configuration Management

5.1 Document Management

- Document tool: Confluence
- Task management tool: Jira

5.2 Source Code Management

- Source code management tool: GitHub

5.3 Tools & Infrastructure

Category	Tools / Infrastructure
Programming Languages	PHP, HTML5, CSS3, JavaScript
Database	MySQL
Database Administration	phpMyAdmin
Academic APIs	OpenAlex API, Crossref REST API
API Testing	Postman
Version Control	GitHub
Project Management	Jira, Confluence
Diagramming	Draw.io, PlantUML
IDE / Code Editor	Visual Studio Code
Local Server	WAMP Server

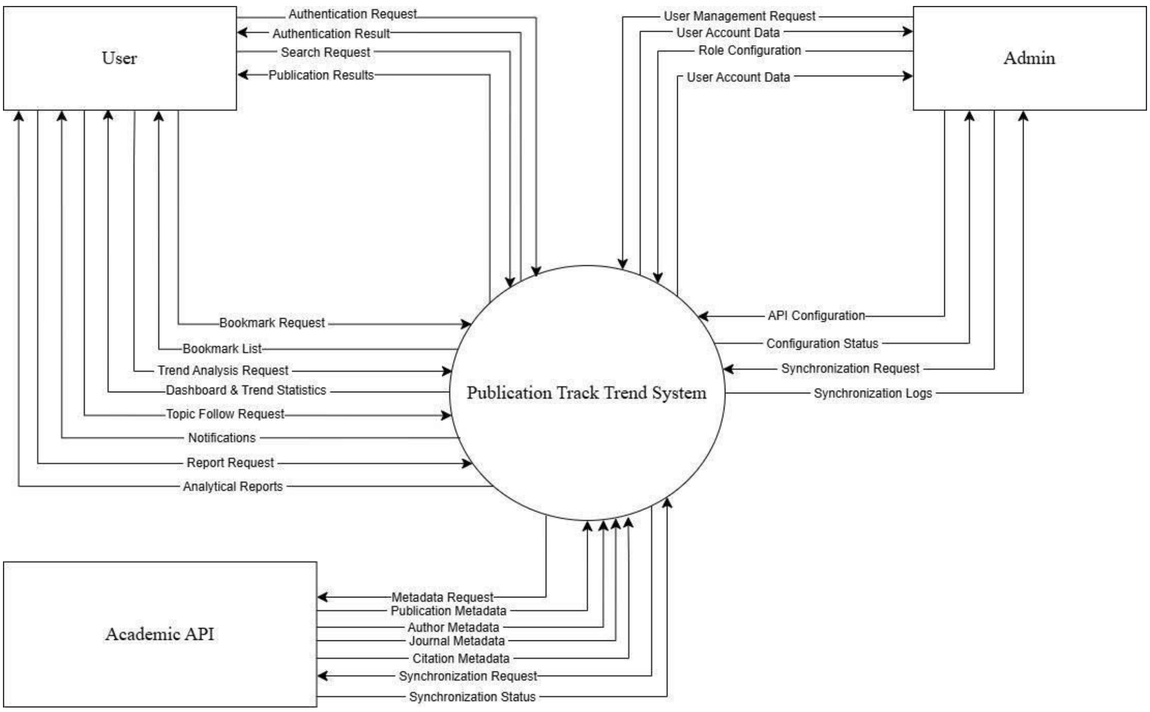
III. SOFTWARE REQUIREMENT SPECIFICATION

1 Product Overview

The Scientific Journal Publication Trend Tracking System (SJPTTS) is a web-based application developed to help researchers, students, lecturers, and academic institutions search, monitor, and analyze scientific publication trends. The system integrates academic data from OpenAlex, and Crossref to provide comprehensive and up-to-date publication information.

The system allows users to search scientific publications using keywords, authors, journals, research fields. Users can view detailed publication information, bookmark publications, follow research topics, receive notifications about newly published papers, analyze publication trends through interactive dashboards, and export reports in PDF or CSV format.

Administrators can manage user accounts, configure academic API sources, synchronize publication data, monitor system operations, and maintain the overall performance of the platform. The system aims to provide an efficient solution for discovering scientific literature and supporting research through publication trend analysis.



Hình 1: Publication Trend Tracking System

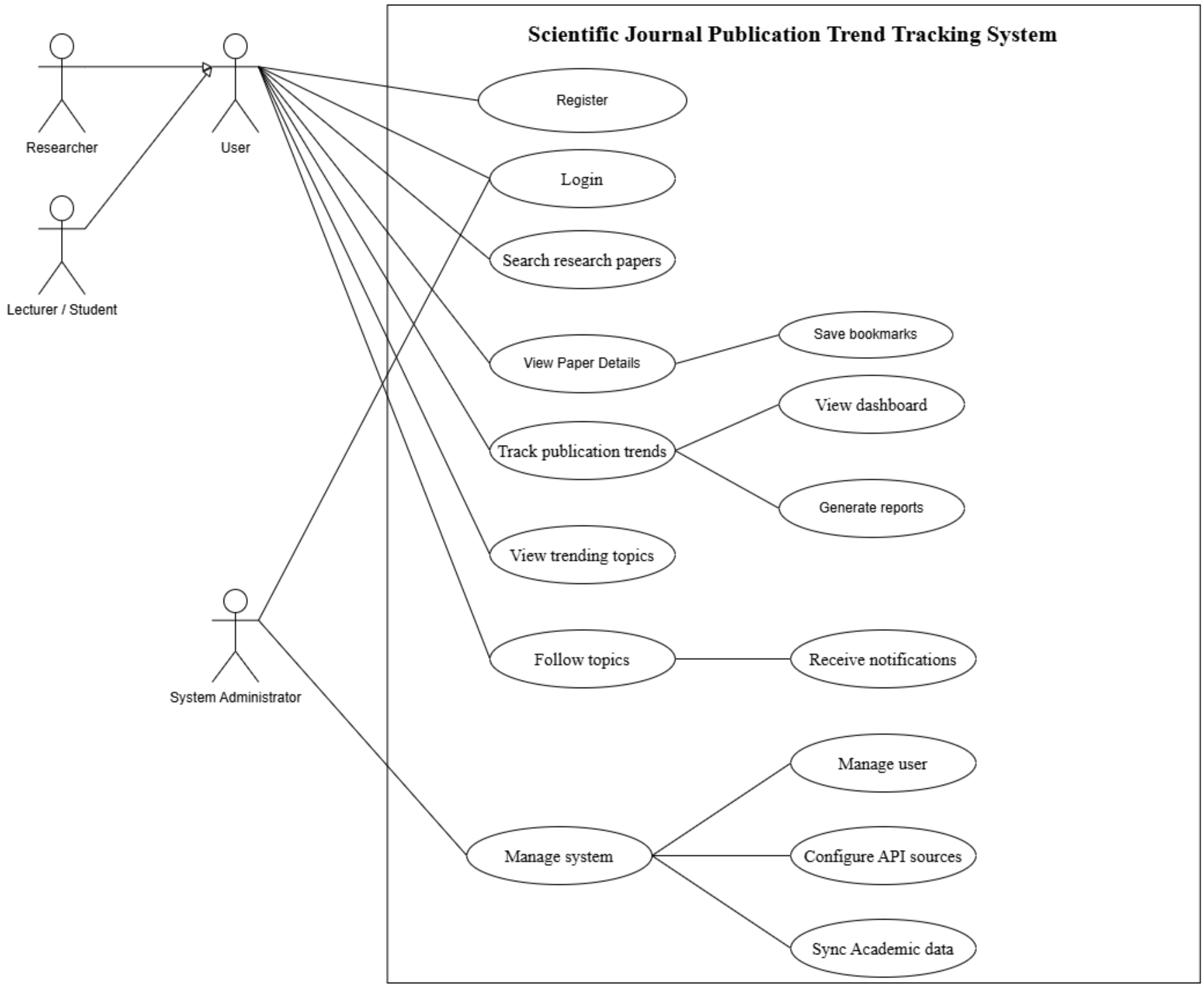
2 User Requirements

2.1 Actors

#	**Actor**	**Description**
1	User	System user who can search scientific publications, view publication details, save bookmarks, follow research topics, receive notifications, and analyze publication trends.
2	Administrator	System administrator responsible for managing users, configuring the system, setting up and validating academic data sources (OpenAlex, Crossref), synchronizing data, and monitoring system operations.
3	Publication Trend Tracking System	The core system that handles all functionalities such as paper search, trend analysis, dashboard generation, trending topic detection, bookmark management, notification delivery, and report generation.
4	Academic API	External services such as OpenAlex and Crossref that provide scientific publication metadata (authors, journals, citations) for system synchronization and analysis.

2.2 Use Case Diagrams

2.2.1 Diagram



Hình 2: Use Case Diagram

2.2.2 Use Case Descriptions

Main UC Group	**Main Agent (Actor)**	**Sub-UCs**	**Brief Description of the Process**	**Output**
Manage Account	User	Register, Login	Users create an account and authenticate to access the Scientific Journal Publication Trend Tracking System.	A registered account is created and a secure user session is established.

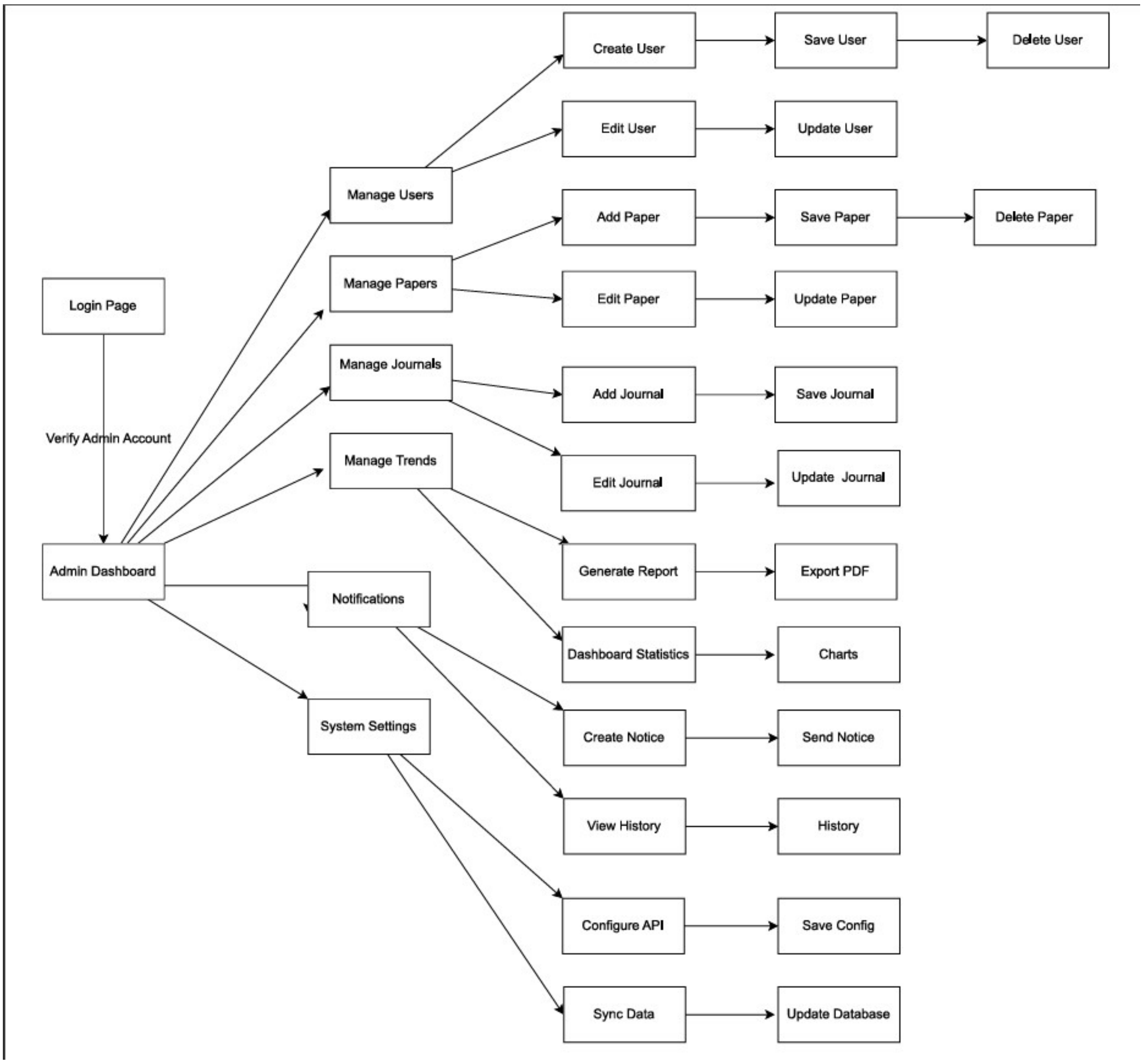
Main UC Group	Main Agent (Actor)	Sub-UCs	Brief Description of the Process	Output
Search Publications	User	Search Research Papers, View Paper Details	Users search scientific publications using keywords, author names, or journal names and view detailed publication metadata.	Matching publications and detailed publication information are displayed.
Manage Bookmarks	User	Save Bookmarks	Users save selected publications into their personal bookmark collection for future reference.	The selected publication is stored in the user's bookmark list.
Track Publication Trends	User	Track Publication Trends, View Dashboard, View Trending Topics	Users analyze publication trends by keyword, topic, author, journal, and time range. The system generates statistical charts and identifies trending research topics.	Trend statistics, dashboards and trending topics are displayed.
Follow Topics	User	Follow Topics, Receive Notifications	Users follow research topics to receive notifications about tracked activities	The selected topic is added to the user's followed-topic list and notifications become available.
Generate Reports	User	Generate Reports	Users generate analytical reports based on publication trend analysis and selected report parameters.	An analytical report summarizing publication trends is generated.
Manage System	Administrator	Manage System, Configure API Sources, Synchronize Academic Data	Administrators manage system settings, configure academic APIs, synchronize publication metadata and maintain system operations.	System settings and publication metadata are successfully updated.
Manage Users	Administrator	Manage Users	Administrators manage user accounts, roles, permissions and account status.	User information and access permissions are updated.
Configure API Sources	Administrator	Configure API Sources	Administrators configure, update and test external academic APIs such as OpenAlex, Crossref and Semantic Scholar.	API configuration settings are successfully validated and stored.

Main UC Group	Main Agent (Actor)	Sub-UCs	Brief Description of the Process	Output
Synchronize Academic Data	Administrator	Synchronize Academic Data	Administrators synchronize publication metadata from external academic APIs into the local database for searching and trend analysis.	The publication metadata database is updated with the latest synchronized records.

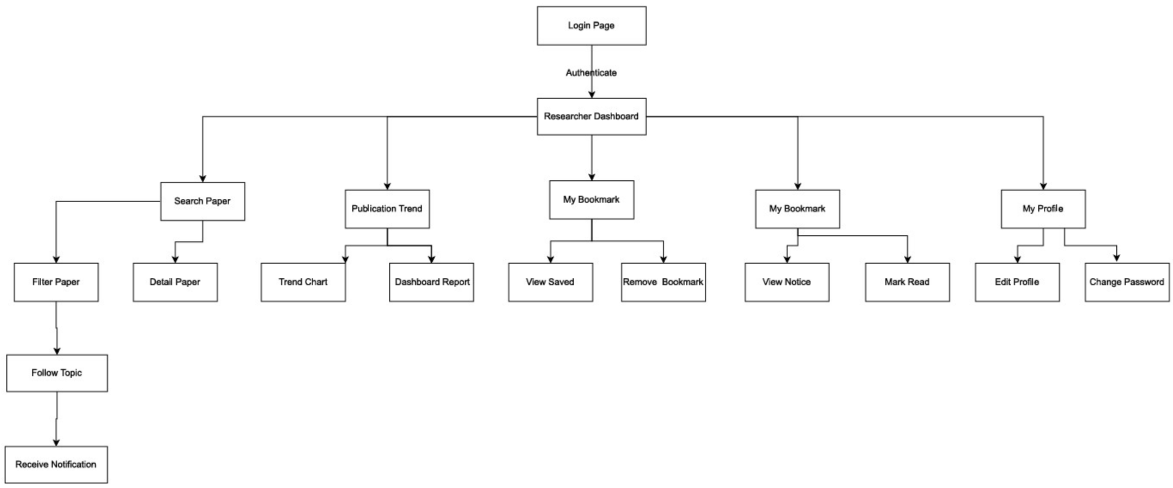
3 Functional Requirement

3.1 System Functional Overview

3.1.1 Screens Flow



Hình 3: Screens Flow



Hình 4: Screen Description

Functional Group	Screen Name	Detailed Description
User Authentication	Login	Purpose: Allows users to log into the system. Components: Username, Password, Login button, and Register link. Flow: The user enters credentials → The system authenticates the user → The main system interface is displayed. Rule: Username and password are required; invalid credentials will generate an error message.
User Authentication	Register	Purpose: Allows new users to create an account. Components: Full Name, Email, Username, Password, and Register button. Flow: The user enters the required information → The system validates the email and username → A new account is created → The user is redirected to the Login page. Rules: Email and username must be unique; passwords are securely hashed before storage.
Paper Management	Search Research Papers	Purpose: Allows users to search scientific publications by keyword, author, or journal. Components: Search box, filters, Search button, and result list. Flow: The user enters search criteria → The system searches publication metadata → Matching publications are displayed.

Functional Group	Screen Name	Detailed Description
Paper Management	View Paper Details	Purpose: Displays detailed metadata of a selected research paper. Components: Title, Authors, Abstract, Keywords, Publication Year, Journal Name, Citation Information, and Save Bookmark button. Flow: The user selects a paper → The system retrieves publication metadata → Paper details are displayed.
Paper Management	Bookmarks	Purpose: Manages the user's bookmarked papers. Components: Bookmark list, Delete Bookmark button. Flow: The user views or removes bookmarked papers. Rule: Duplicate bookmarks are not allowed.
Publication Trend Analysis	Track Publication Trends	Purpose: Analyzes publication trends based on keywords, research topics, authors, or journals. Components: Search filters, time range selector, and Analyze button. Flow: The user enters analysis criteria → The system analyzes publication metadata → Statistical results and charts are displayed.
Publication Trend Analysis	Dashboard	Purpose: Displays an overview of publication statistics. Components: Publications by year chart, Topic distribution chart, Journal distribution chart, and Growth trend chart. Flow: The system generates dashboard visualizations after trend analysis is completed.
Publication Trend Analysis	Trending Topics	Purpose: Displays the most popular research topics. Flow: The system analyzes publication data → Ranks research topics → Displays trending topics.
Topic Management	Follow Topics	Purpose: Allows users to follow research topics of interest. Components: Topic list and Follow button. Flow: The user selects a topic → The system stores it in the followed-topic list.
Notifications	Notifications	Purpose: Displays system notifications or notifications when following a new topic. Flow: The system generates notifications → The user views notification details and marks them as read.

Functional Group	Screen Name	Detailed Description
Reporting	Generate Reports	Purpose: Generates reports based on publication trend analysis. Flow: The user clicks "Generate Report"→ The system displays options: Preview & Print / Save as PDF, Direct PDF Download, and Download Report (HTML) → The system executes the request based on the selected option and completes the generation, display, or export of the analytical report.
System Administration	Manage System	Purpose: Provides administrative functions for system management. Components: User Management, API Configuration, and Academic Data Synchronization. Flow: The administrator selects a management function → The system performs the requested operation.
System Administration	Manage Users	Purpose: Manages user accounts. Components: User list, Change User Role, Activate Account, and Deactivate Account. Flow: The administrator selects a user → Updates account information → Saves the changes.
System Administration	Configure API Sources	Purpose: To define and manage external academic data sources. Flow: The administrator enters source details (Name, URL, Priority, Status, Description). The administrator clicks "Add API Source"to save it to the system. The system displays the list, allowing the administrator to track, edit, or delete entries via the "Action"column.
System Administration	Synchronize Academic Data	Purpose: Synchronize publication data from external APIs into the system. Components: API List, Synchronize button, Synchronization Log. Workflow: Click Synchronize → The system retrieves data from the APIs → Updates the database → Displays the synchronization result.

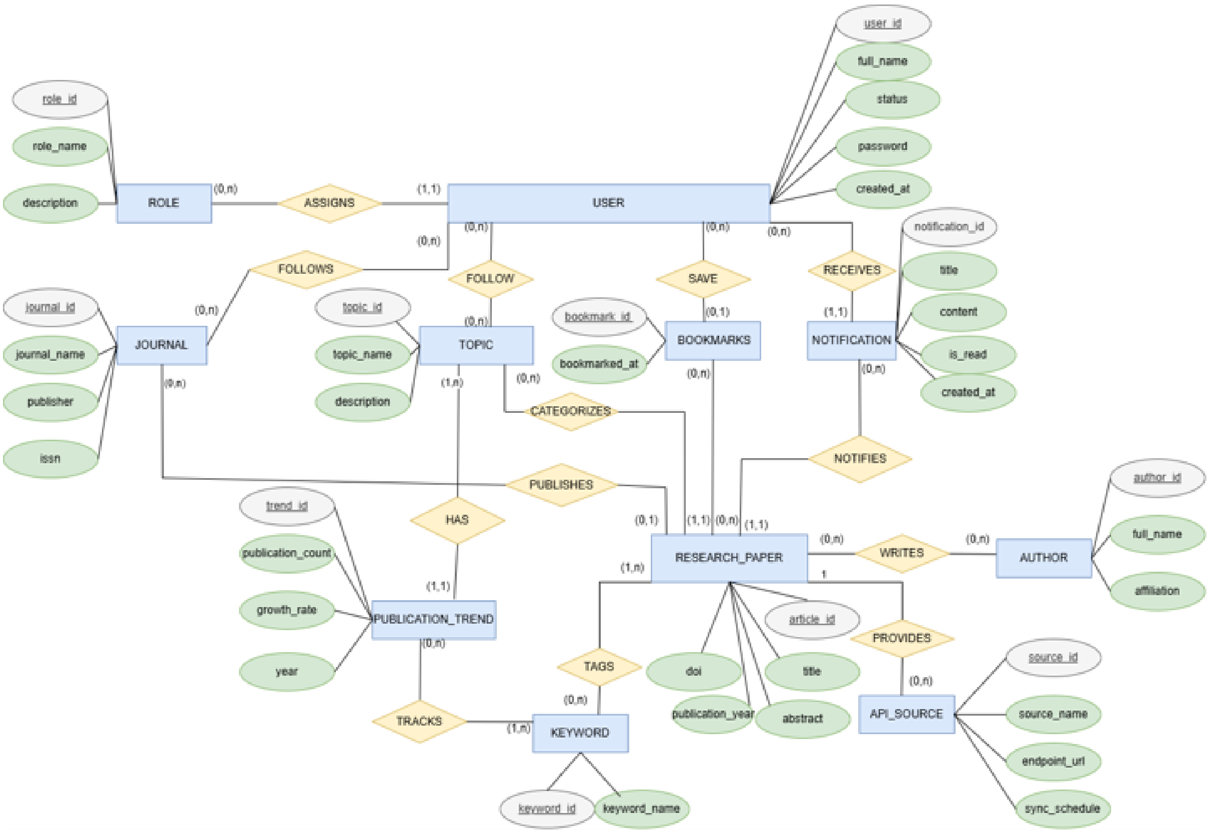
3.1.2 Screen Authorization

Screen	Researcher	Lecturer/Student	System Administrator
Login	x	x	x
Logout	x	x	x
Register	x	x	
Search Research Papers	x	x	
View Paper Details	x	x	
Save Bookmarks	x	x	
Track Publication Trends	x	x	
View Dashboard	x	x	
Generate Reports	x	x	
View Trending Topics	x	x	
Follow Topics	x	x	
Receive Notifications	x	x	
View Profile	x	x	
Edit Profile	x	x	
View Notifications	x	x	
Create Report	x	x	
Manage System			x
Manage Users			x
Configure API Sources			x
Sync Academic Data			x

3.1.3 Non-Screen Functions

#	Feature	System Function	Description
1	Automated Academic Data Synchronization	Scheduled Job	Scheduled job that automatically retrieves publication metadata directly from external academic APIs (OpenAlex and Crossref). The retrieved data is used for publication search, statistical analysis, and publication trend tracking.
2	Automated Trend Calculation	Scheduled Job	Scheduled job that automatically recalculates publication trend statistics after each data synchronization from external academic APIs. The system analyzes total publications, total citations, Open Access rate, and research topic trends, then updates the data displayed on the dashboard and Trending Topics.

3.2 Entity Relationship Diagram



Hình 5: Entity Relationship Diagram (ERD)

3.3 Authentication

3.3.1 Login

Role: User

Function trigger:

- The user enters a username and password to log in.

Function description:

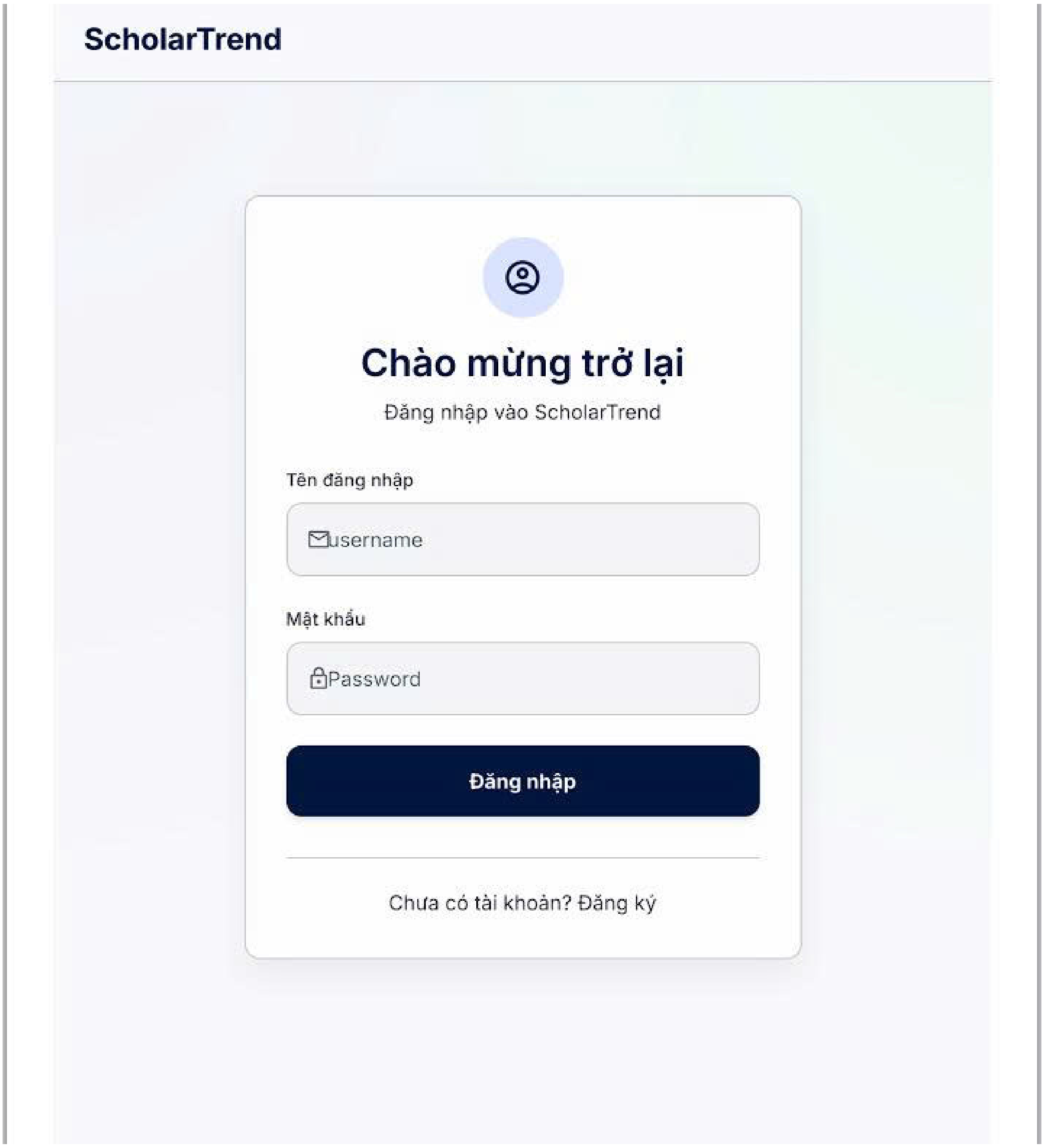
- **Purpose:** Allow users to access the system.

- **Data processing:** Authenticate the entered username and password against the system database.

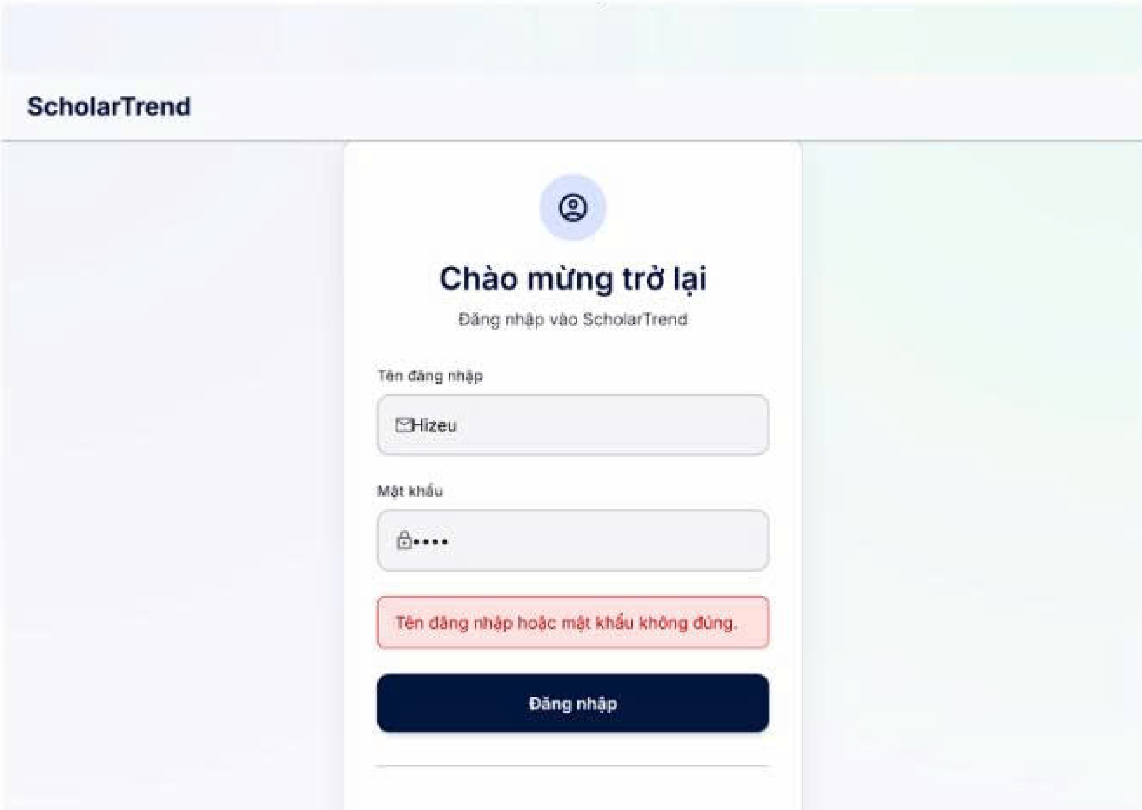
Function Details:

- If the login is successful, redirect the user to the Home page.

- If the login fails, display the error message: “Tên đăng nhập hoặc mật khẩu không đúng.”


ScholarTrend
Chào mừng trở lại
Đăng nhập vào ScholarTrend
Tên đăng nhập
username
Mật khẩu
Password
Đăng nhập
Chưa có tài khoản? Đăng ký

Hình 6:


ScholarTrend
Chào mừng trở lại
Đăng nhập vào ScholarTrend
Tên đăng nhập
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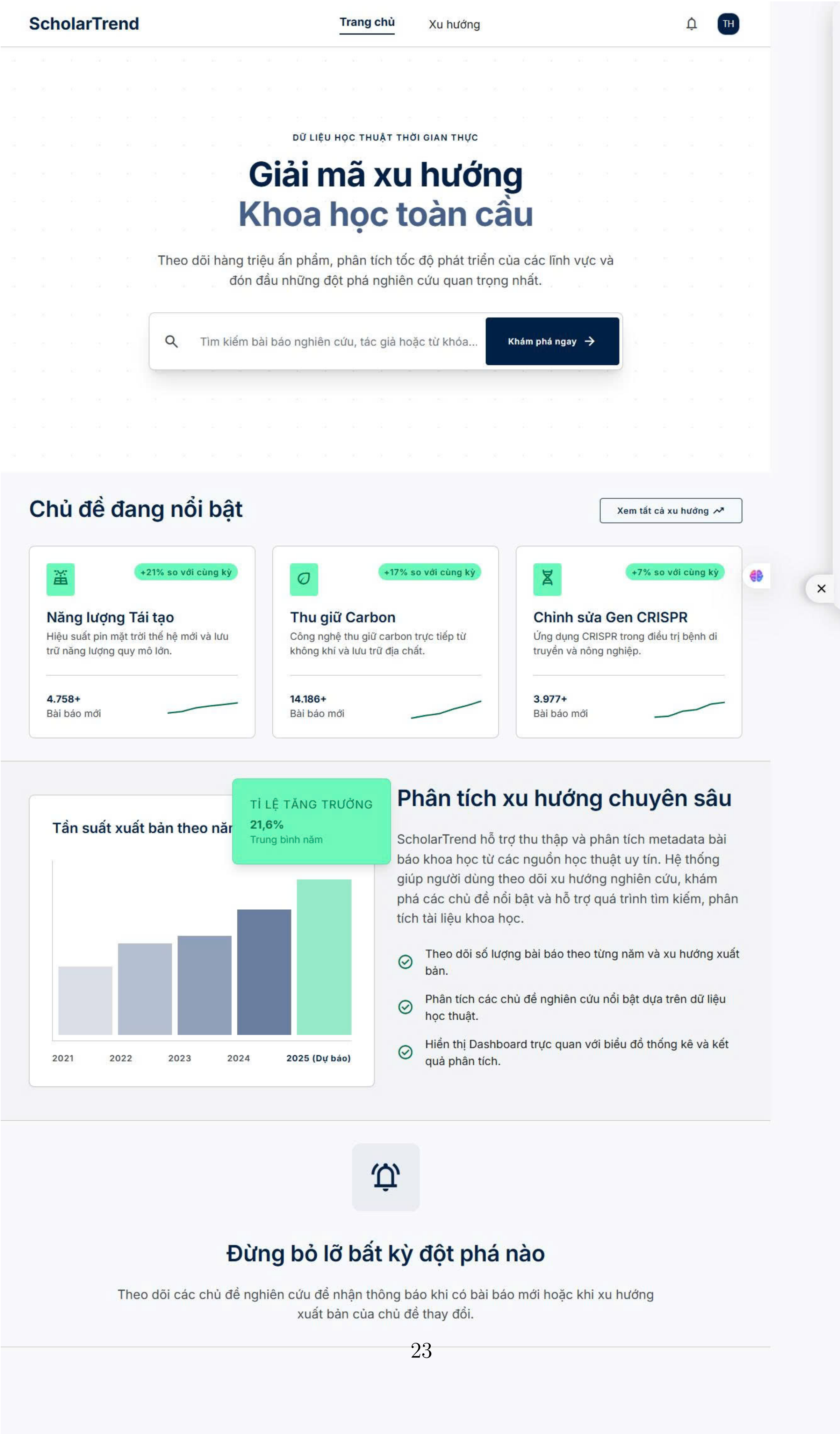
Mật khẩu
Tên đăng nhập hoặc mật khẩu không đúng.
Đăng nhập

Hình 7:


ScholarTrend
Trang chủ
Xu hướng
TH
DỮ LIỆU HỌC THUẬT THỜI GIAN THỰC
Giải mã xu hướng
Khoa học toàn cầu
Theo dõi hàng triệu ấn phẩm, phân tích tốc độ phát triển của các lĩnh vực và đón đầu những đột phá nghiên cứu quan trọng nhất.
Tìm kiếm bài báo nghiên cứu, tác giả hoặc từ khóa...
Khám phá ngay
Chủ đề đang nổi bật
Xem tất cả xu hướng
+21% so với cùng kỳ
Năng lượng Tái tạo
Hiệu suất pin mặt trời thế hệ mới và lưu trữ năng lượng quy mô lớn.
4.758+
Bài báo mới
+17% so với cùng kỳ
Thu giữ Carbon
Công nghệ thu giữ carbon trực tiếp từ không khí và lưu trữ địa chất.
14.186+
Bài báo mới
+7% so với cùng kỳ
Chỉnh sửa Gen CRISPR
Ứng dụng CRISPR trong điều trị bệnh di truyền và nông nghiệp.
3.977+
Bài báo mới
Tần suất xuất bản theo năr
TỈ LỆ TĂNG TRƯỞNG
21,6%
Trung bình năm
2021
2022
2023
2024
2025 (Dự báo)
Phân tích xu hướng chuyên sâu
ScholarTrend hỗ trợ thu thập và phân tích metadata bài báo khoa học từ các nguồn học thuật uy tín. Hệ thống giúp người dùng theo dõi xu hướng nghiên cứu, khám phá các chủ đề nổi bật và hỗ trợ quá trình tìm kiếm, phân tích tài liệu khoa học.
Theo dõi số lượng bài báo theo từng năm và xu hướng xuất bản.
Phân tích các chủ đề nghiên cứu nổi bật dựa trên dữ liệu học thuật.
Hiển thị Dashboard trực quan với biểu đồ thống kê và kết quả phân tích.
Đừng bỏ lỡ bất kỳ đột phá nào
Theo dõi các chủ đề nghiên cứu để nhận thông báo khi có bài báo mới hoặc khi xu hướng xuất bản của chủ đề thay đổi.

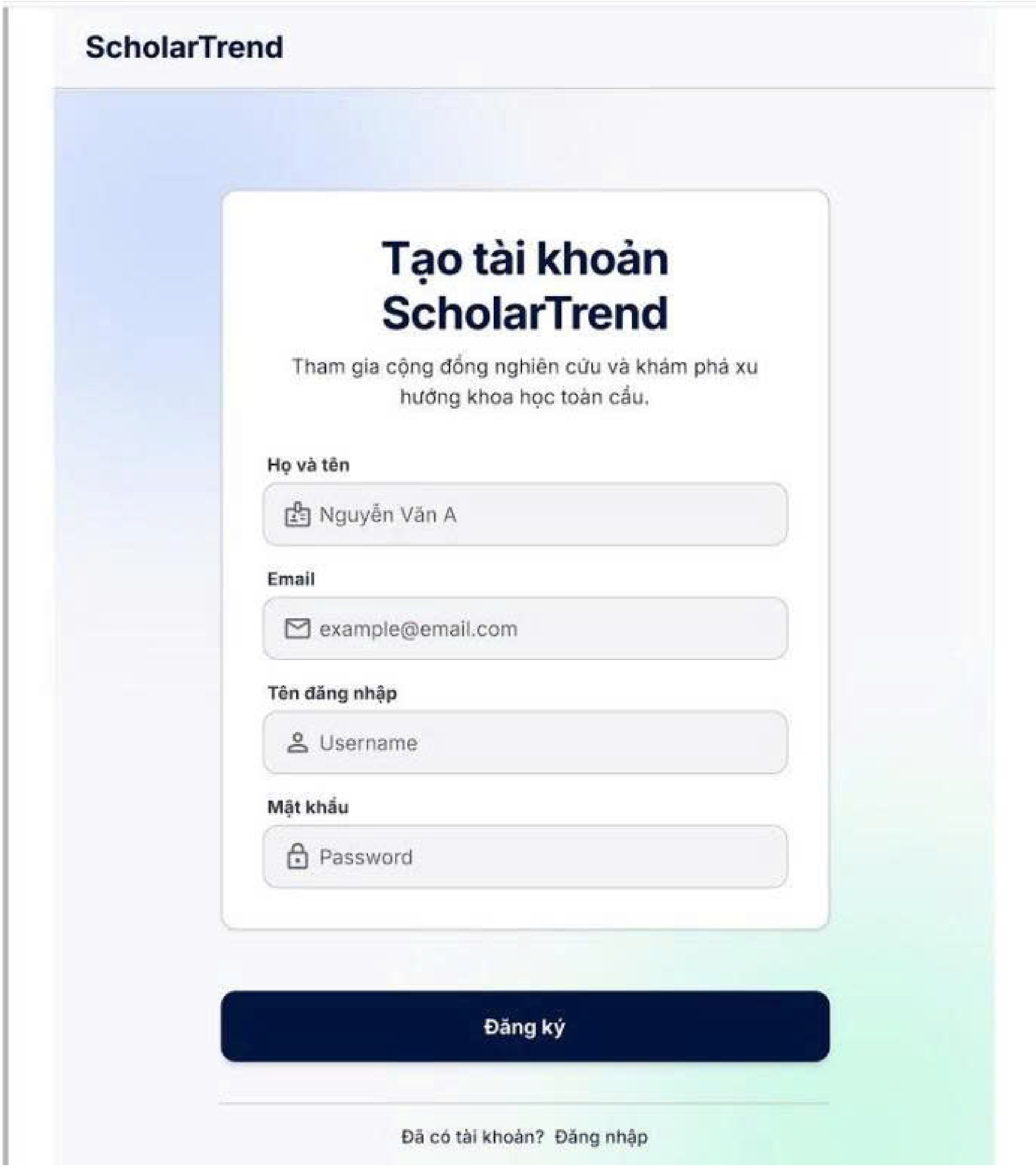
3.3.2 Register

Role: User
Function trigger: The user clicks Đăng kí.
Function description:

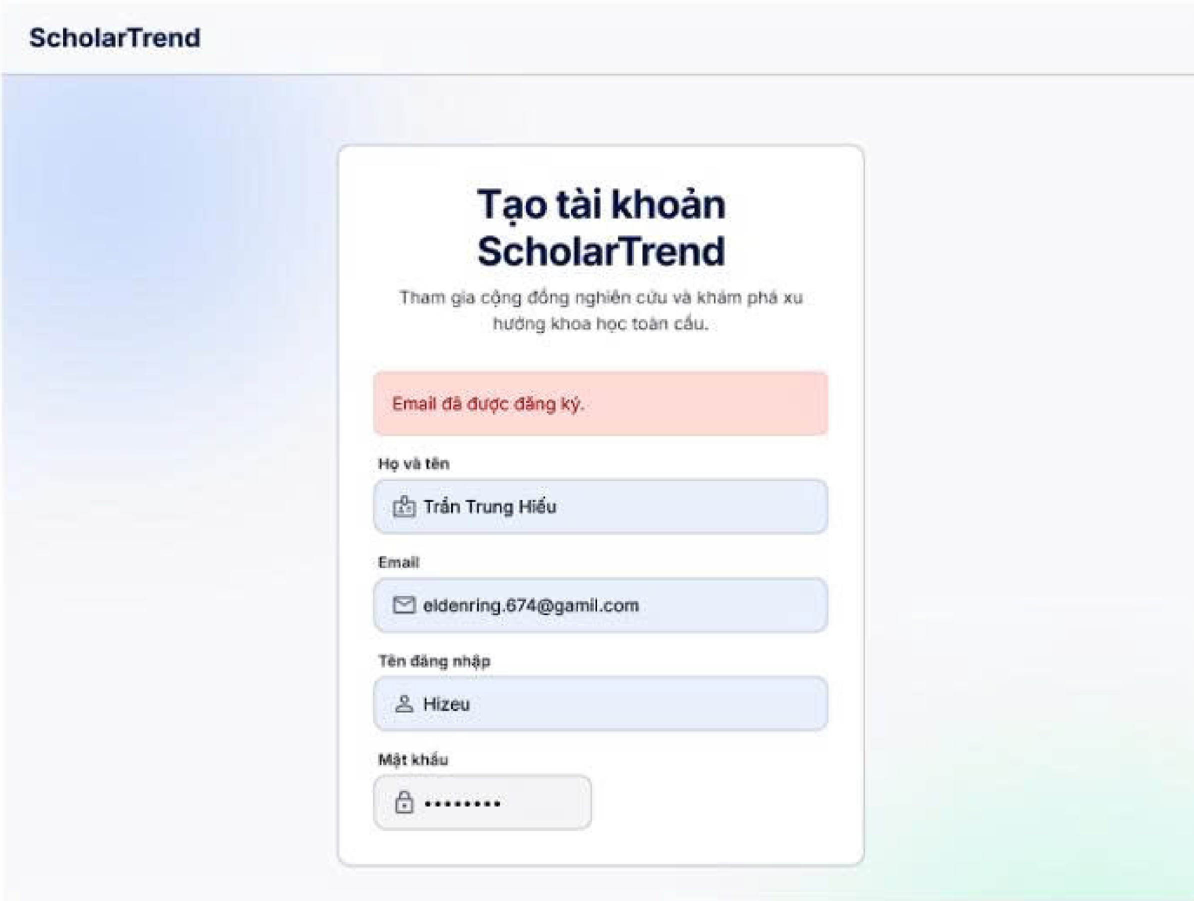
- Purpose: Create a new user account.
- Data processing: Validate user information and store it in the database.

Function Details:

- Whether the username already exists
- Validate password requirements.
- Create a new account.
- Display a success message.
- Sign up and check your email.


ScholarTrend
Tạo tài khoản ScholarTrend
Tham gia cộng đồng nghiên cứu và khám phá xu hướng khoa học toàn cầu.
Họ và tên
Nguyễn Văn A
Email
example@email.com
Tên đăng nhập
Username
Mật khẩu
Password
Đăng ký
Đã có tài khoản? Đăng nhập

Hình 9:



Hình 10:

3.3.3 Profile Management

Update Profile

Role: User

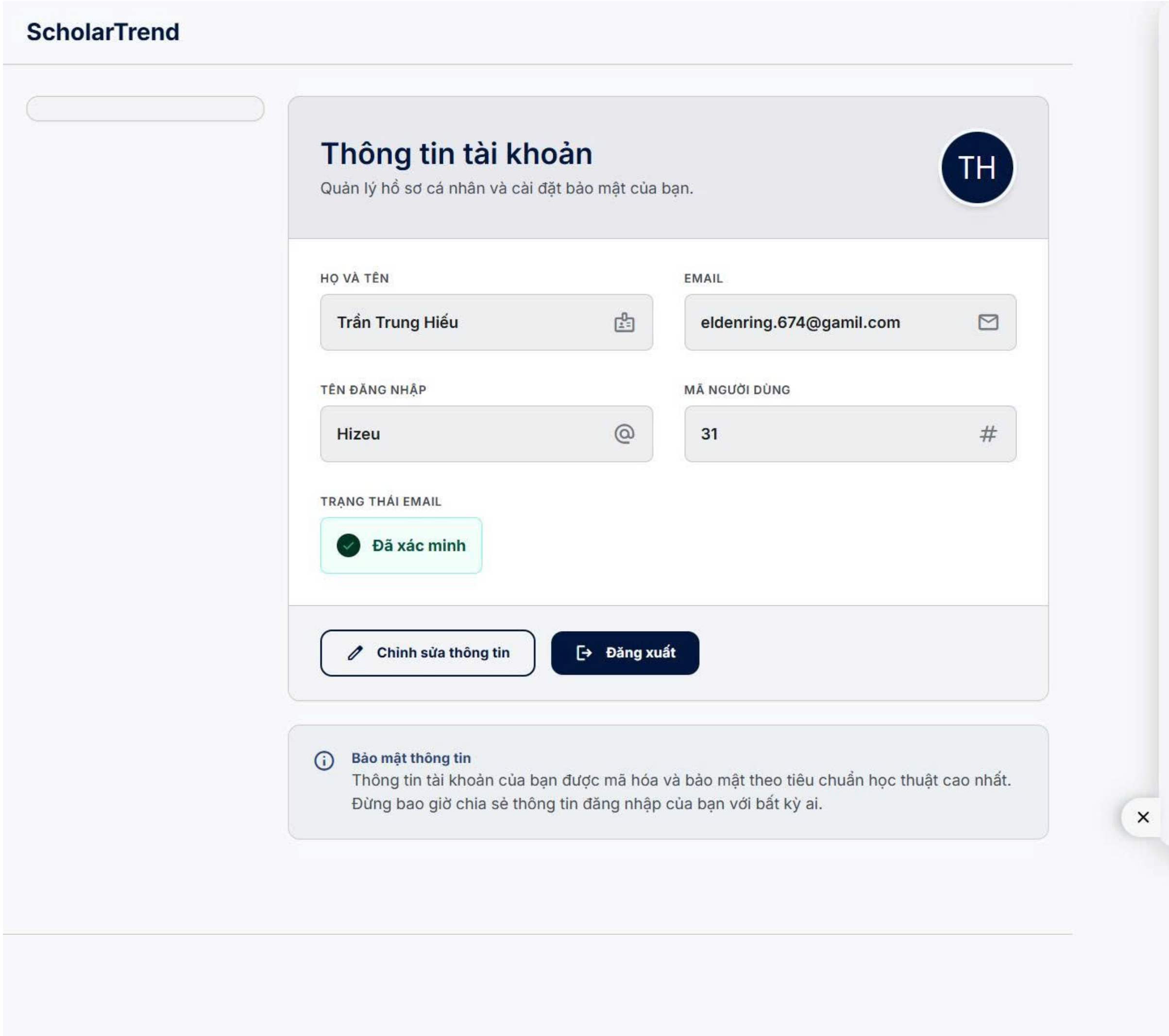
Function trigger: The user opens the Profile page and clicks Chỉnh sửa thông tin.

Function description:

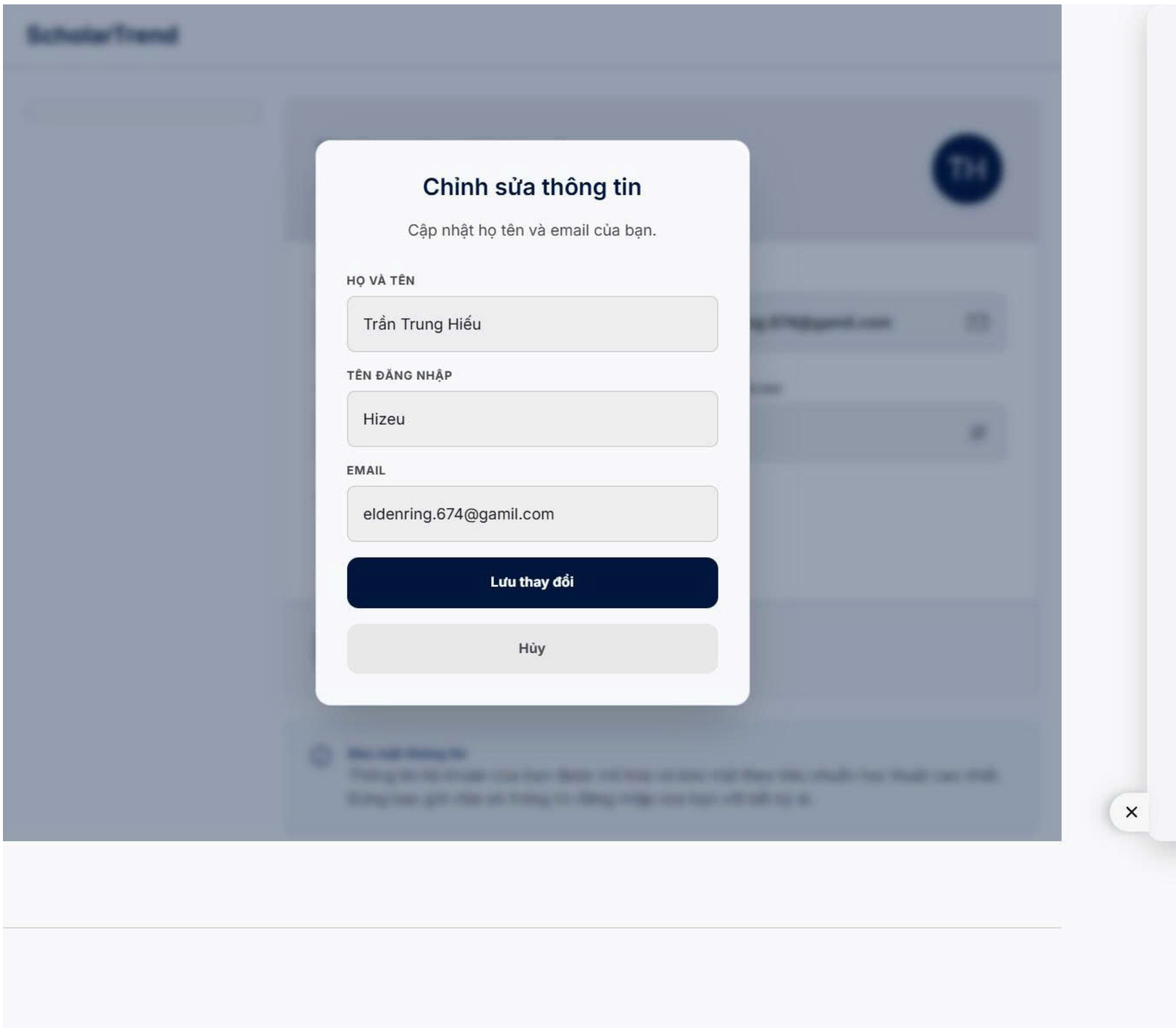
- Purpose: Allow users to update their personal information.
- Data processing: Validate and save the updated information.

Function Details:

- Update name, Email and other personal information.
- Display a success message after saving.


ScholarTrend
Thông tin tài khoản
Quản lý hồ sơ cá nhân và cài đặt bảo mật của bạn.
TH
HỌ VÀ TÊN
Trần Trung Hiếu
EMAIL
eldenring.674@gamil.com
TÊN ĐĂNG NHẬP
Hizeu
MÃ NGƯỜI DÙNG
31
TRẠNG THÁI EMAIL
Đã xác minh
Chỉnh sửa thông tin
Đăng xuất
Bảo mật thông tin
Thông tin tài khoản của bạn được mã hóa và bảo mật theo tiêu chuẩn học thuật cao nhất.
Đừng bao giờ chia sẻ thông tin đăng nhập của bạn với bất kỳ ai.

Hình 11:



Hình 12:

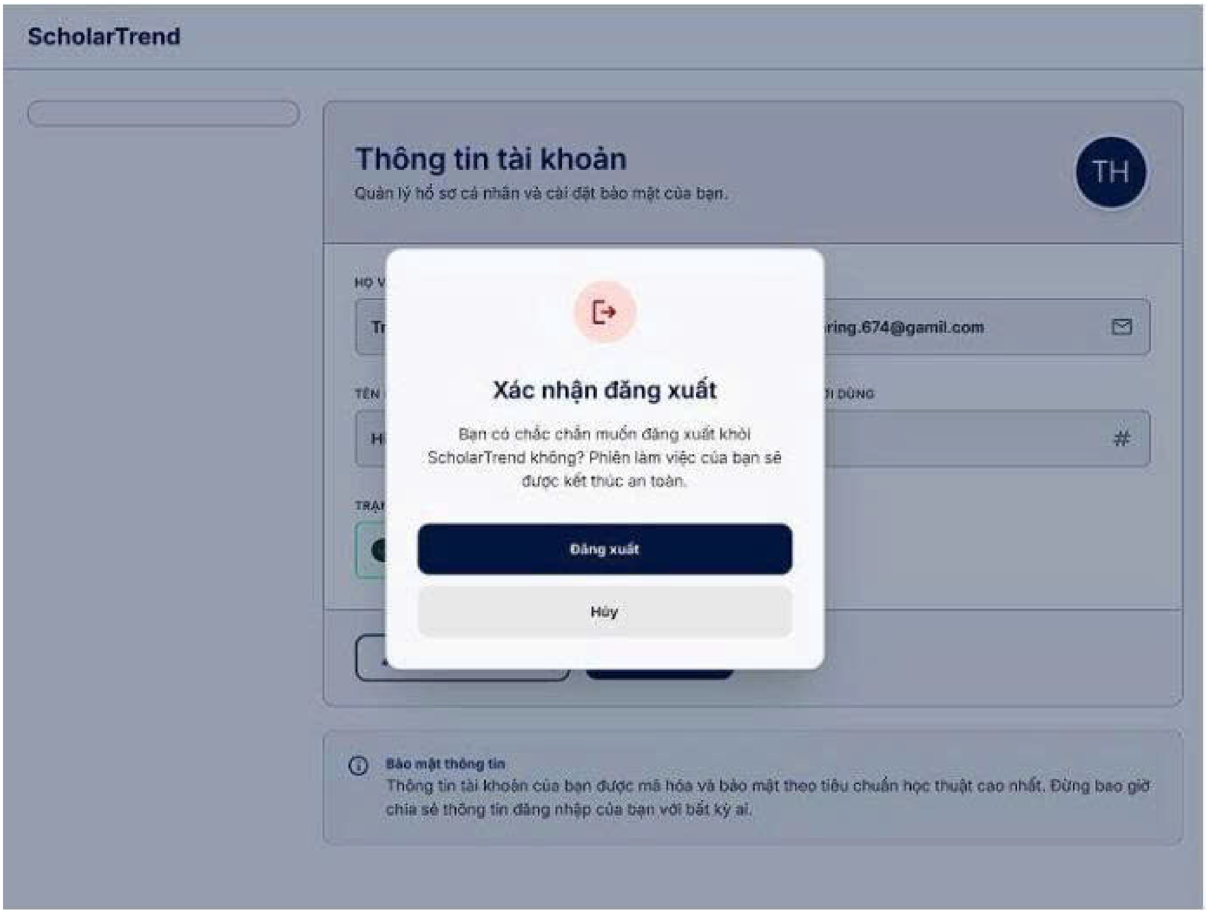
3.3.4 Logout

Role: User
Function trigger: The user opens the Profile page and clicks Đăng xuất.
Function description:

- Purpose: Allow the user to securely log out of the system.
- Data processing: End the current user session.

Function Details:

- Display a confirmation dialog asking whether the user wants to Đăng xuất.
- If the user selects Đăng xuất, the system logs the user out and redirects them to the Đăng nhập page.
- If the user selects Hủy, the dialog is closed and the user remains on the current page.



Hình 13:

3.3.5 Search Papers

Role: User

Function Trigger

User opens *"Tìm kiếm"* → Enters a keyword → Selects filters (Publication Year, Open Access, Paper Type) → Clicks **Search**.

Function Description

- **Purpose:**

  Allow users to search scientific papers from academic databases based on keywords and filtering conditions.

- **Data Processing:**

  Validate the search input, send the request to academic databases (e.g., Crossref API), retrieve matching publications, apply selected filters, and display the search results.

Function Details

- **Validation:**

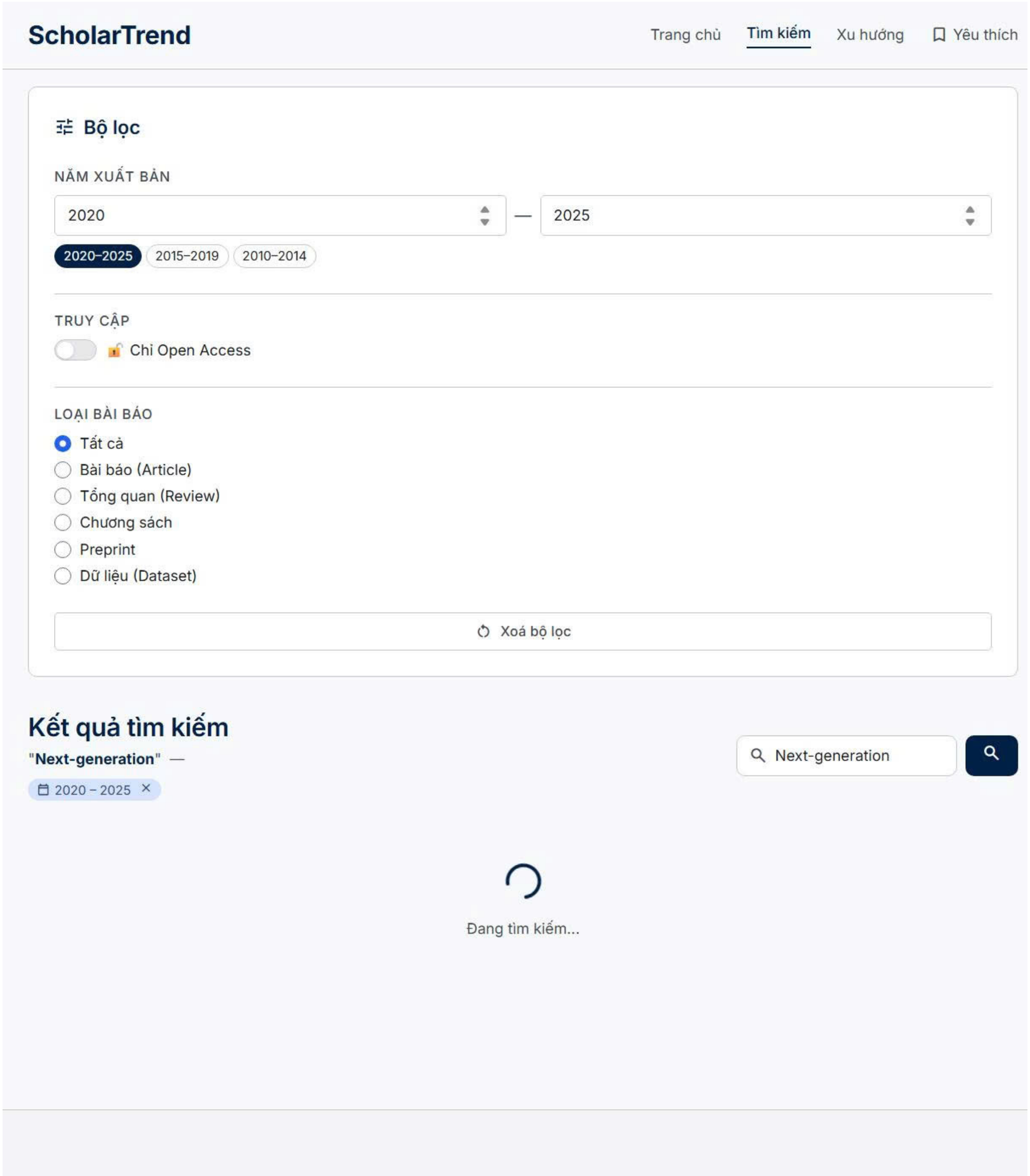
  – Keyword cannot be empty.

- Start year must not be greater than End year.

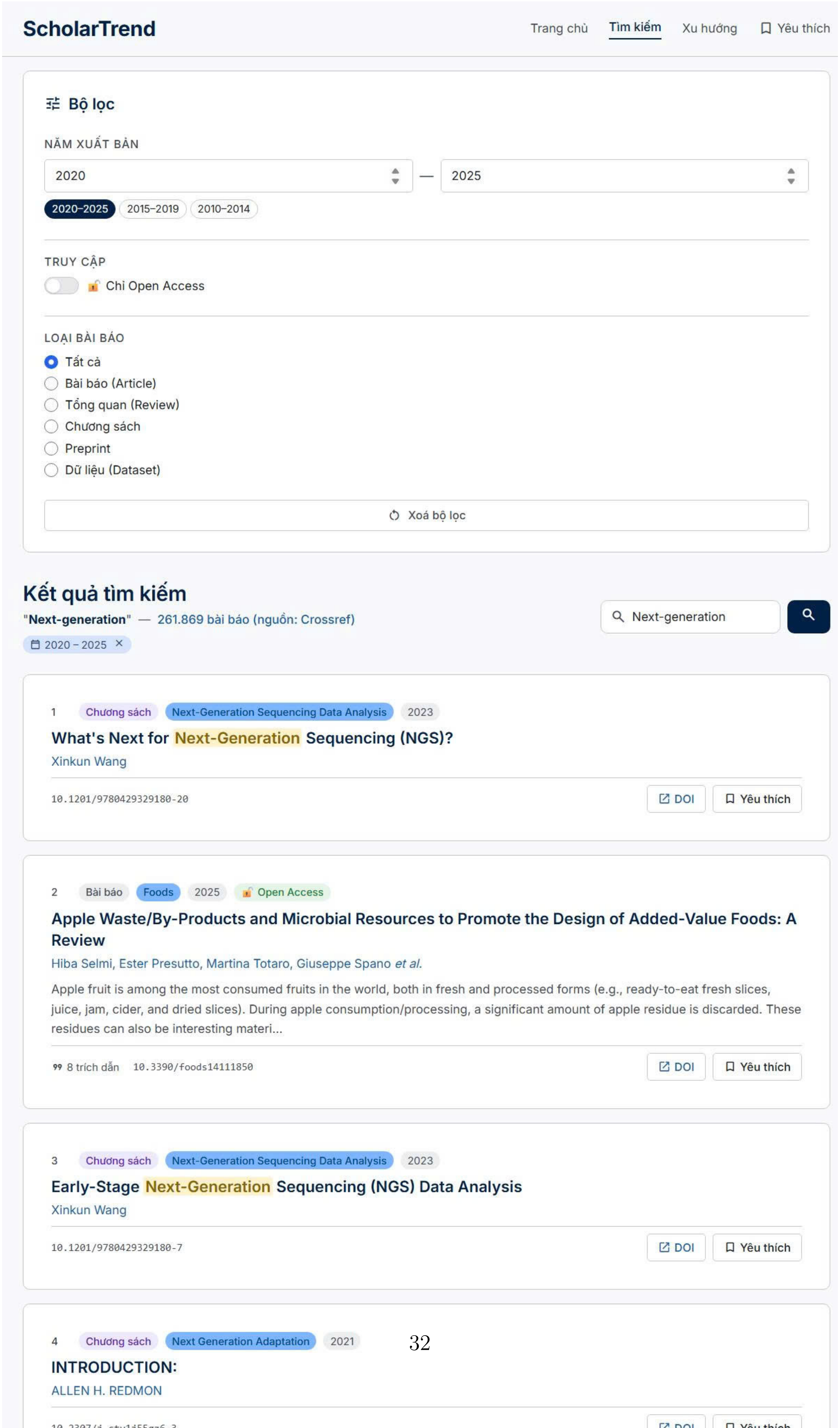
- **Business Rules:**
  - Search supports Title, Keyword and DOI.
  - Users may filter by publication year.
  - Users may filter by paper type.
  - Users may choose to display only Open Access publications.
  - Results are displayed with pagination (10 items per page).
  - Each result includes Title, Authors, Year, DOI and Citation Count (if available), with the option to save the paper to Favorites.

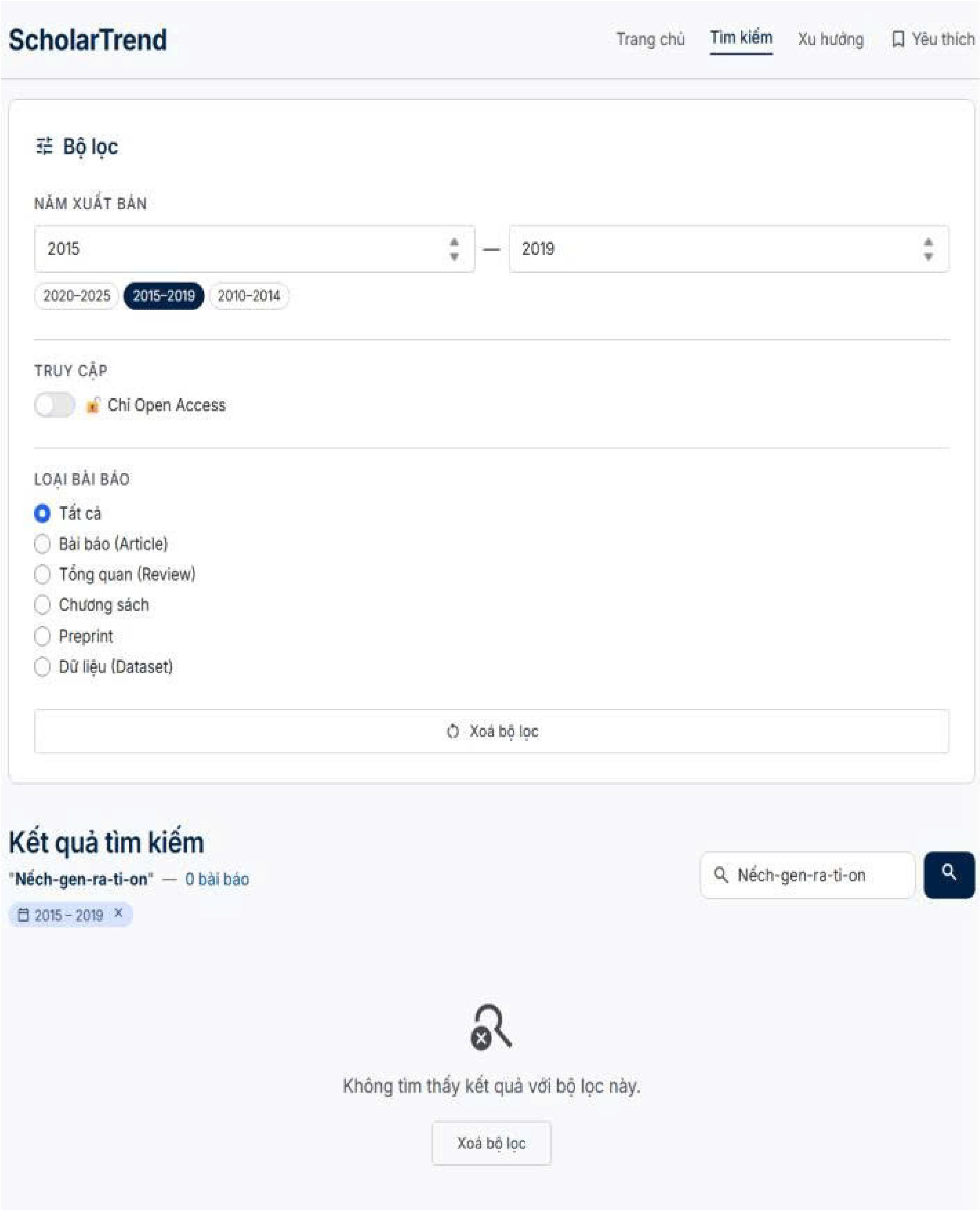
- **Normal Case:**
  - User enters a keyword and selects filters.
  - User clicks the **Search** button.
  - System queries the academic database.
  - Matching publications are returned and displayed.
  - User can view paper details, open the DOI link, or save the paper to Favorites.

- **Abnormal Case:**
  - No papers found → Display *"No matching papers found."*
  - Invalid year range → Request the user to correct the input.
  - API unavailable → Display *"Unable to retrieve publication data. Please try again later."*


ScholarTrend
Trang chủ
Tìm kiếm
Xu hướng
Yêu thích
Bộ lọc
NĂM XUẤT BẢN
2020
2025
2020–2025
2015–2019
2010–2014
TRUY CẬP
Chỉ Open Access
LOẠI BÀI BÁO
Tất cả
Bài báo (Article)
Tổng quan (Review)
Chương sách
Preprint
Dữ liệu (Dataset)
Xoá bộ lọc
Kết quả tìm kiếm
"Next-generation" —
2020 – 2025
Next-generation
Đang tìm kiếm...

Hình 14:


ScholarTrend
Trang chủ
Tìm kiếm
Xu hướng
Yêu thích
Bộ lọc
NĂM XUẤT BẢN
2020
—
2025
2020–2025
2015–2019
2010–2014
TRUY CẬP
Chỉ Open Access
LOẠI BÀI BÁO
Tất cả
Bài báo (Article)
Tổng quan (Review)
Chương sách
Preprint
Dữ liệu (Dataset)
Xoá bộ lọc
Kết quả tìm kiếm
"Next-generation" — 261.869 bài báo (nguồn: Crossref)
Next-generation
2020 – 2025 ×
1 Chương sách Next-Generation Sequencing Data Analysis 2023
What's Next for Next-Generation Sequencing (NGS)?
Xinkun Wang
10.1201/9780429329180-20
DOI
Yêu thích
2 Bài báo Foods 2025 Open Access
Apple Waste/By-Products and Microbial Resources to Promote the Design of Added-Value Foods: A Review
Hiba Selmi, Ester Presutto, Martina Totaro, Giuseppe Spano et al.
Apple fruit is among the most consumed fruits in the world, both in fresh and processed forms (e.g., ready-to-eat fresh slices, juice, jam, cider, and dried slices). During apple consumption/processing, a significant amount of apple residue is discarded. These residues can also be interesting materi...
8 trích dẫn 10.3390/foods14111850
DOI
Yêu thích
3 Chương sách Next-Generation Sequencing Data Analysis 2023
Early-Stage Next-Generation Sequencing (NGS) Data Analysis
Xinkun Wang
10.1201/9780429329180-7
DOI
Yêu thích
4 Chương sách Next Generation Adaptation 2021
INTRODUCTION:
ALLEN H. REDMON

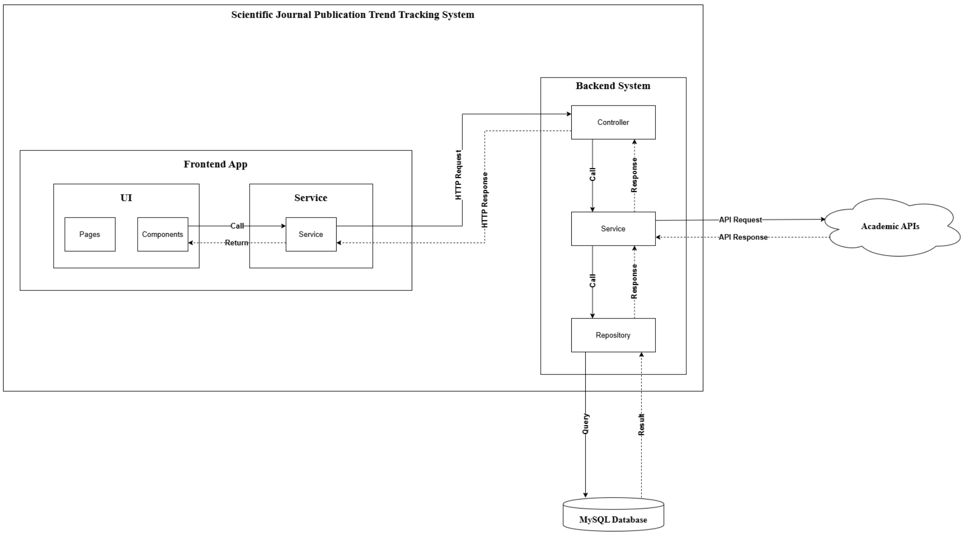

ScholarTrend
Trang chủ
Tìm kiếm
Xu hướng
Yêu thích
Bộ lọc
NĂM XUẤT BẢN
2015
2019
2020–2025
2015–2019
2010–2014
TRUY CẬP
Chỉ Open Access
LOẠI BÀI BÁO
Tất cả
Bài báo (Article)
Tổng quan (Review)
Chương sách
Preprint
Dữ liệu (Dataset)
Xoá bộ lọc
Kết quả tìm kiếm
"Nếch-gen-ra-ti-on" — 0 bài báo
2015 – 2019
Nếch-gen-ra-ti-on
Không tìm thấy kết quả với bộ lọc này.
Xoá bộ lọc

Hình 16:

IV. Software Design Description

1 System Design

1.1 System Architecture



1.1.1 Frontend App

The frontend application is developed using HTML, CSS, and JavaScript to provide the user interface for interacting with the system. It is responsible for presenting data, handling user interactions, and communicating with the backend system through HTTP requests.

- **Pages**: Responsible for rendering the main user interfaces, including Login Page, Dashboard Page, Publication Search Page, Trend Analysis Page, Bookmark Page, and Notification Page.
- **Components**: Reusable user interface elements used to build application pages, such as search bars, filter panels, publication cards, charts, tables, and navigation menus.
- **Service**: Responsible for managing communication between the frontend and backend system through HTTP requests and responses, including sending requests and processing returned data.

1.1.2 Backend System

The backend system follows a layered architecture consisting of Controller Layer, Service Layer, and Repository Layer. This architecture improves maintainability, scalability, and separation of concerns.

- **Controller Layer**: Handles incoming HTTP requests from the frontend, processes request validation, and forwards requests to the appropriate service layer.

- **Service Layer**: Contains business logic for processing publication data, trend analysis, user personalization, and system administration.
- **Repository Layer**: Responsible for interacting with the database, executing queries, and managing data persistence.

1.1.3 External Systems

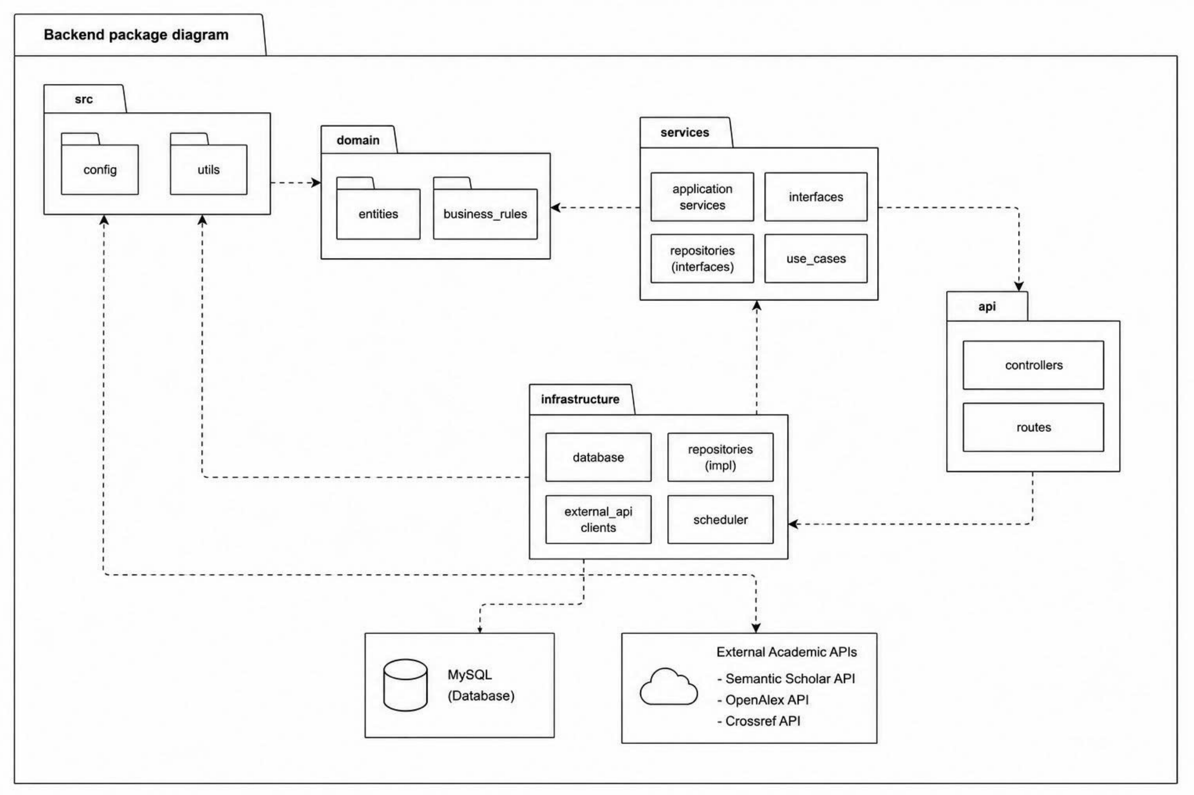
The system integrates with external services to retrieve academic publication data and store system information.

- **Academic APIs**: External academic data sources such as Semantic Scholar API, OpenAlex API, and Crossref API are used to retrieve publication metadata.
- **MySQL Database**: Stores user information, publication metadata, bookmarks, followed topics, notifications, and system configuration data.

1.2 Package Diagram

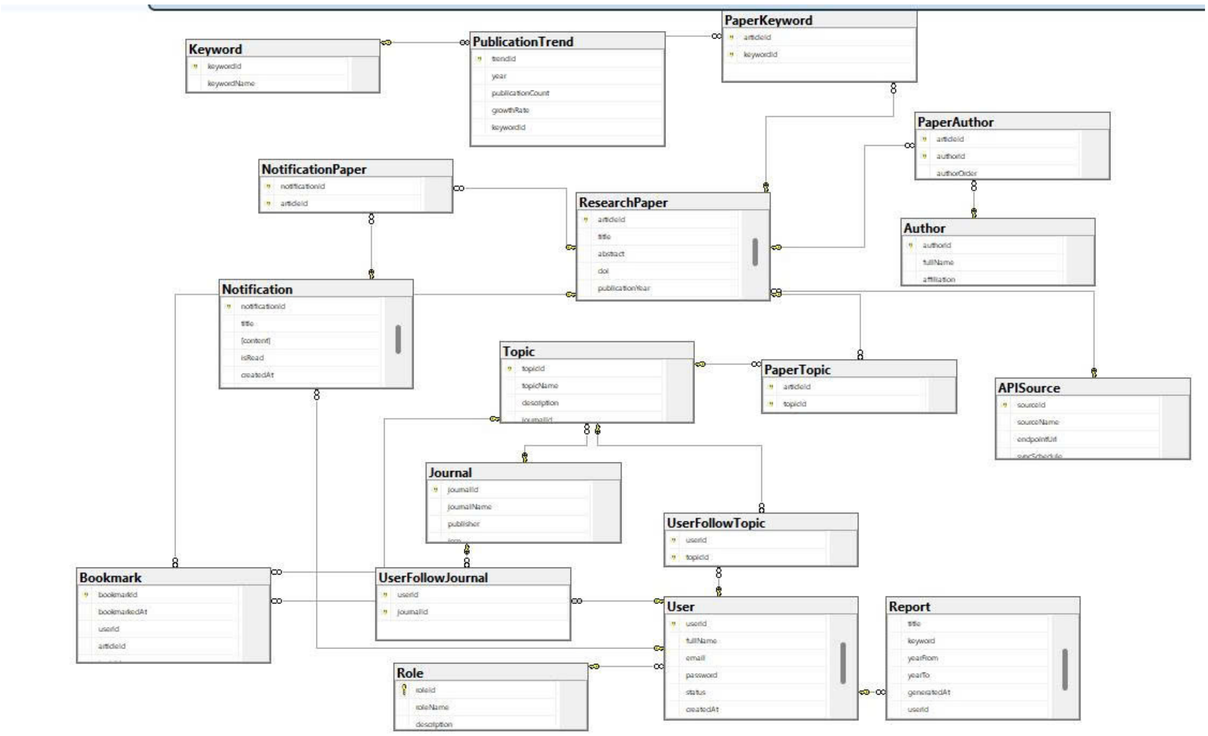
1.2.1 Front End

1.2.2 Back End



Hình 17: **Backend package diagram**

2 Database Design



Hình 18: Database Design

Entities Description

2.1 Role Table

Field	Type	Description	PK	FK
roleId	INT	Role ID	Yes	No

roleName	NVARCHAR(100)	Role name	No	No
description	NVARCHAR(255)	Description	No	No

2.2 User Table

Field	Type	Description	PK	FK
userId	INT	User ID	Yes	No
fullName	NVARCHAR(150)	Full name	No	No
email	NVARCHAR(150)	Email	No	No
password	NVARCHAR(255)	Password	No	No
status	NVARCHAR(50)	Status	No	No
createdAt	DATETIME2	Created time	No	No
roleId	INT	Role reference	No	Yes → Role

2.3 Journal Table

Field	Type	Description	PK	FK
journalId	INT	Journal ID	Yes	No
journalName	NVARCHAR(200)	Journal name	No	No
publisher	NVARCHAR(150)	Publisher	No	No
issn	NVARCHAR(20)	ISSN	No	No

2.4 UserFollowJournal (M:N)

Field	Type	Description	PK	FK
userId	INT	User	Yes (Composite)	Yes → User

journalId	INT	Journal	Yes (Composite)	Yes → Journal

2.5 Topic Table

Field	Type	Description	PK	FK
topicId	INT	Topic ID	Yes	No
topicName	NVARCHAR(150)	Topic name	No	No
description	NVARCHAR(MAX	Description	No	No
journalId	INT	Journal reference	No	Yes → Journal

2.6 UserFollowTopic (M:N)

Field	Type	Description	PK	FK
userId	INT	User	Yes (Composite)	Yes → User
topicId	INT	Topic	Yes (Composite)	Yes → Topic

2.7 APISource Table

Field	Type	Description	PK	FK
sourceId	INT	Source ID	Yes	No
sourceName	NVARCHAR(150)	Source name	No	No
endpointUrl	NVARCHAR(500)	API URL	No	No
syncSchedule	NVARCHAR(100)	Sync schedule	No	No

2.8 Author Table

Field	Type	Description	PK	FK
authorId	INT	Author ID	Yes	No
fullName	NVARCHAR(150)	Full name	No	No
affiliation	NVARCHAR(255)	Affiliation	No	No

2.9 Keyword Table

Field	Type	Description	PK	FK
keywordId	INT	Keyword ID	Yes	No
keywordName	NVARCHAR(100)	Keyword	No	No

2.10 Research Paper Table

Field	Type	Description	PK	FK
articleId	INT	Paper ID	Yes	No
title	NVARCHAR(500)	Title	No	No
abstract	NVARCHAR(MAX	Abstract	No	No
doi	NVARCHAR(200)	DOI	No	No
publicationYear	INT	Year	No	No
sourceId	INT	API Source	No	Yes → APISource

2.11 PaperTopic (M:N)

Field	Type	Description	PK	FK

articleId	INT	Paper	Yes (Composite)	Yes → Research-Paper
topicId	INT	Topic	Yes (Composite)	Yes → Topic

2.12 PaperKeyword (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Composite)	Yes → Research-Paper
keywordId	INT	Keyword	Yes (Composite)	Yes → Keyword

2.13 PaperAuthor (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Composite)	Yes → Research-Paper
authorId	INT	Author	Yes (Composite)	Yes → Author
authorOrder	INT	Author order	No	No

2.14 PublicationTrend Table

Field	Type	Description	PK	FK
trendId	INT	Trend ID	Yes	No
year	INT	Year	No	No
publicationCount	INT	Count	No	No
growthRate	FLOAT	Growth rate	No	No
keywordId	INT	Keyword	No	Yes → Keyword

2.15 Report Table

Field	Type	Description	PK	FK
reportId	INT	Report ID	Yes	No
title	NVARCHAR(300)	Title	No	No
keyword	NVARCHAR(200)	Keyword	No	No
yearFrom	INT	From year	No	No
yearTo	INT	To year	No	No
generatedAt	DATETIME2	Created time	No	No
userId	INT	User	No	Yes → User

2.16 Notification Table

Field	Type	Description	PK	FK
notificationId	INT	Notification ID	Yes	No
title	NVARCHAR(300)	Title	No	No
content	NVARCHAR(MAX	Content	No	No
isRead	BIT	Read status	No	No
createdAt	DATETIME2	Created time	No	No

userId	INT	User	No	Yes → User

2.17 NotificationPaper (M:N)

Field	Type	Description	PK	FK
notificationId	INT	Notification	Yes (Composite)	Yes → Notification
articleId	INT	Paper	Yes (Composite)	Yes → ResearchPaper

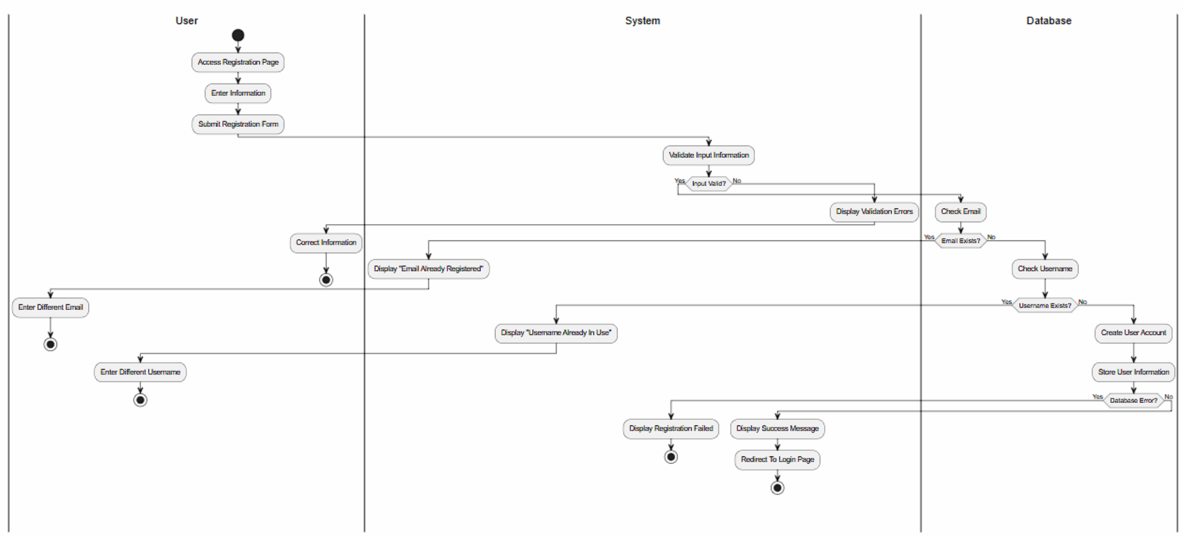
2.18 Bookmark Table

Field	Type	Description	PK	FK
bookmarkId	INT	Bookmark ID	Yes	No
bookmarkedAt	DATETIME2	Time saved	No	No
userId	INT	User	No	Yes → User
articleId	INT	Paper	No	Yes → ResearchPaper
topicId	INT	Topic	No	Yes → Topic

3 Detail Design

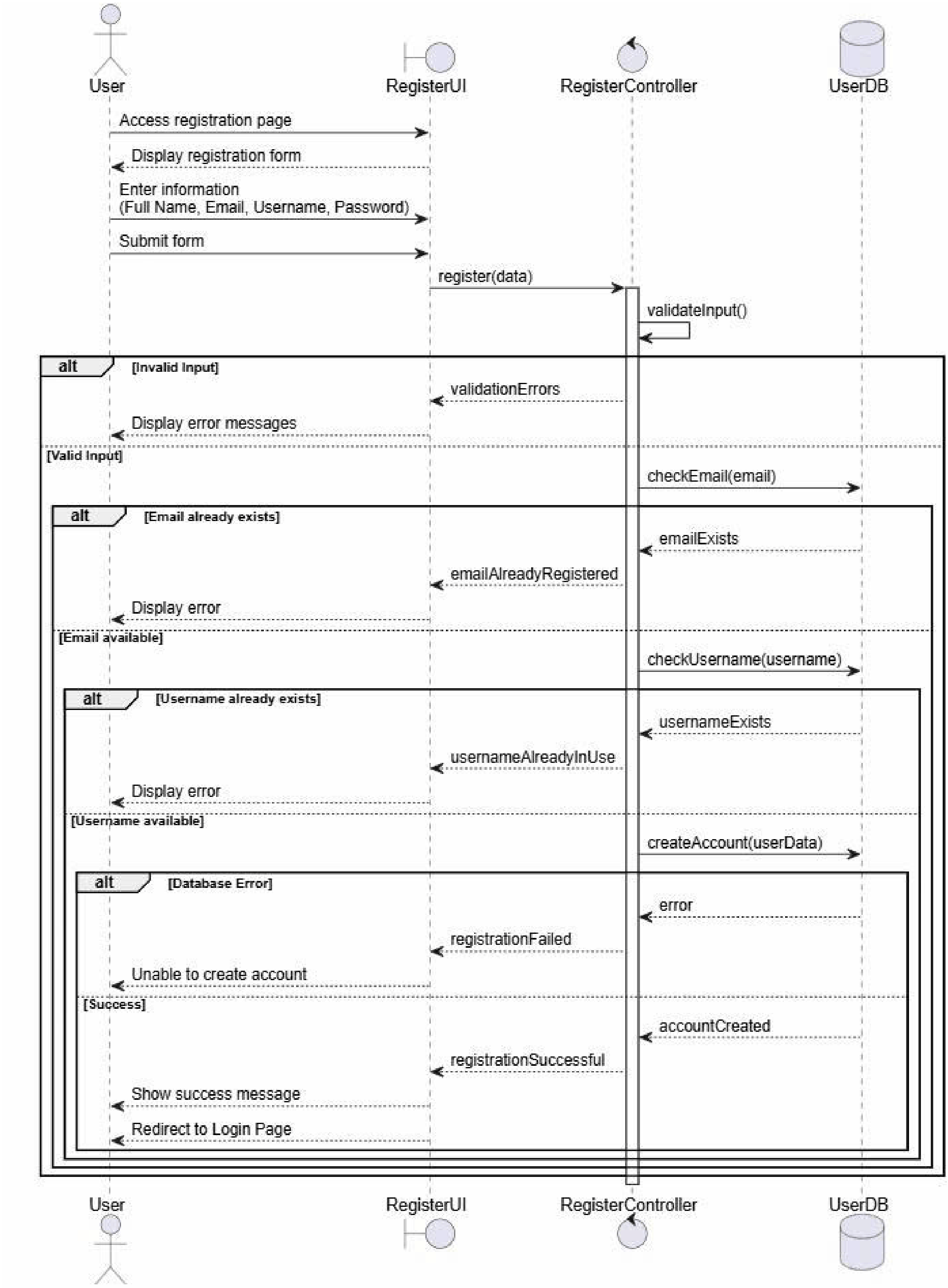
3.1 Register

3.1.1 Activity diagram



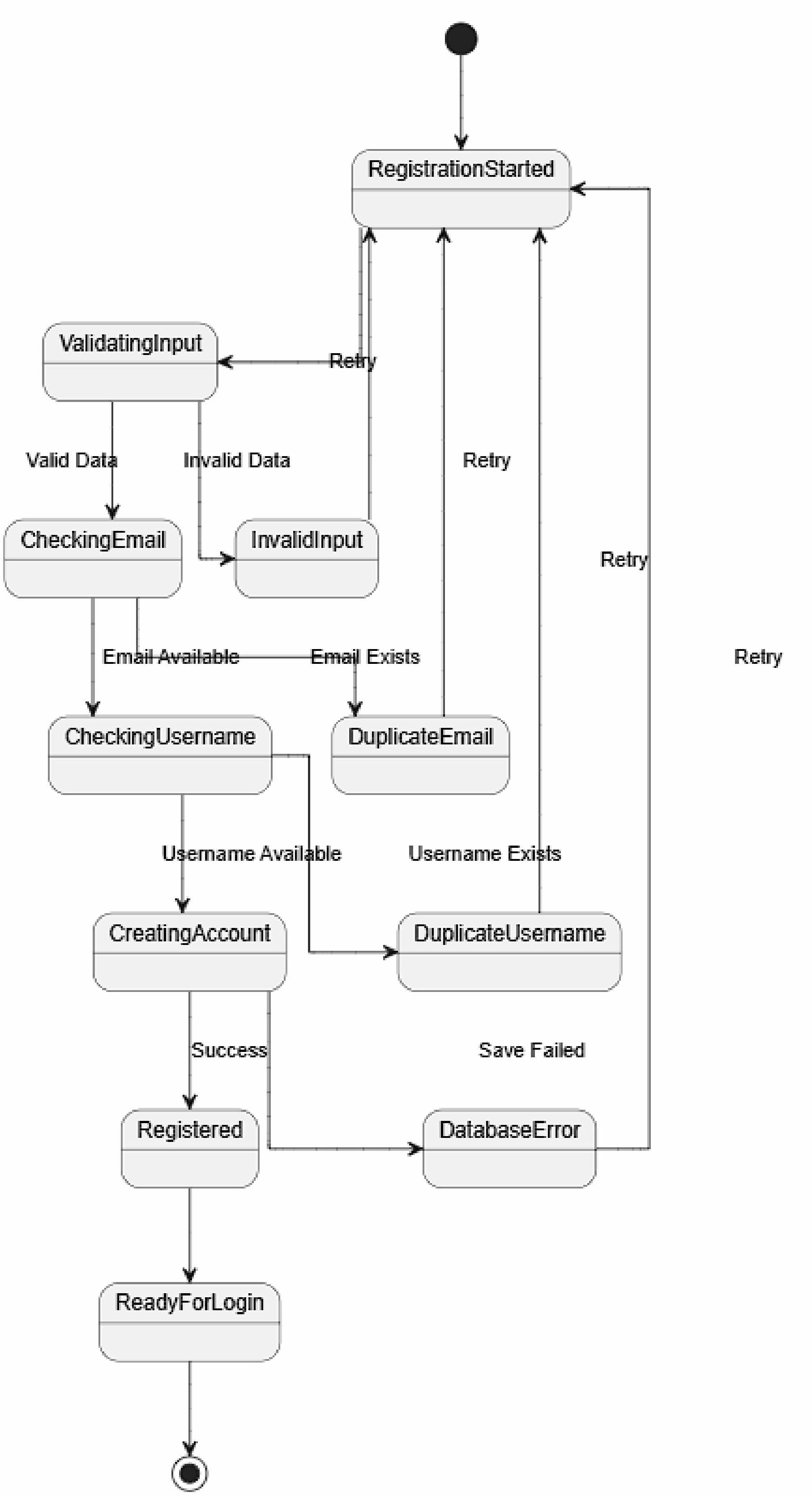
Hình 19: Activity diagram

3.1.2 Sequence Diagram



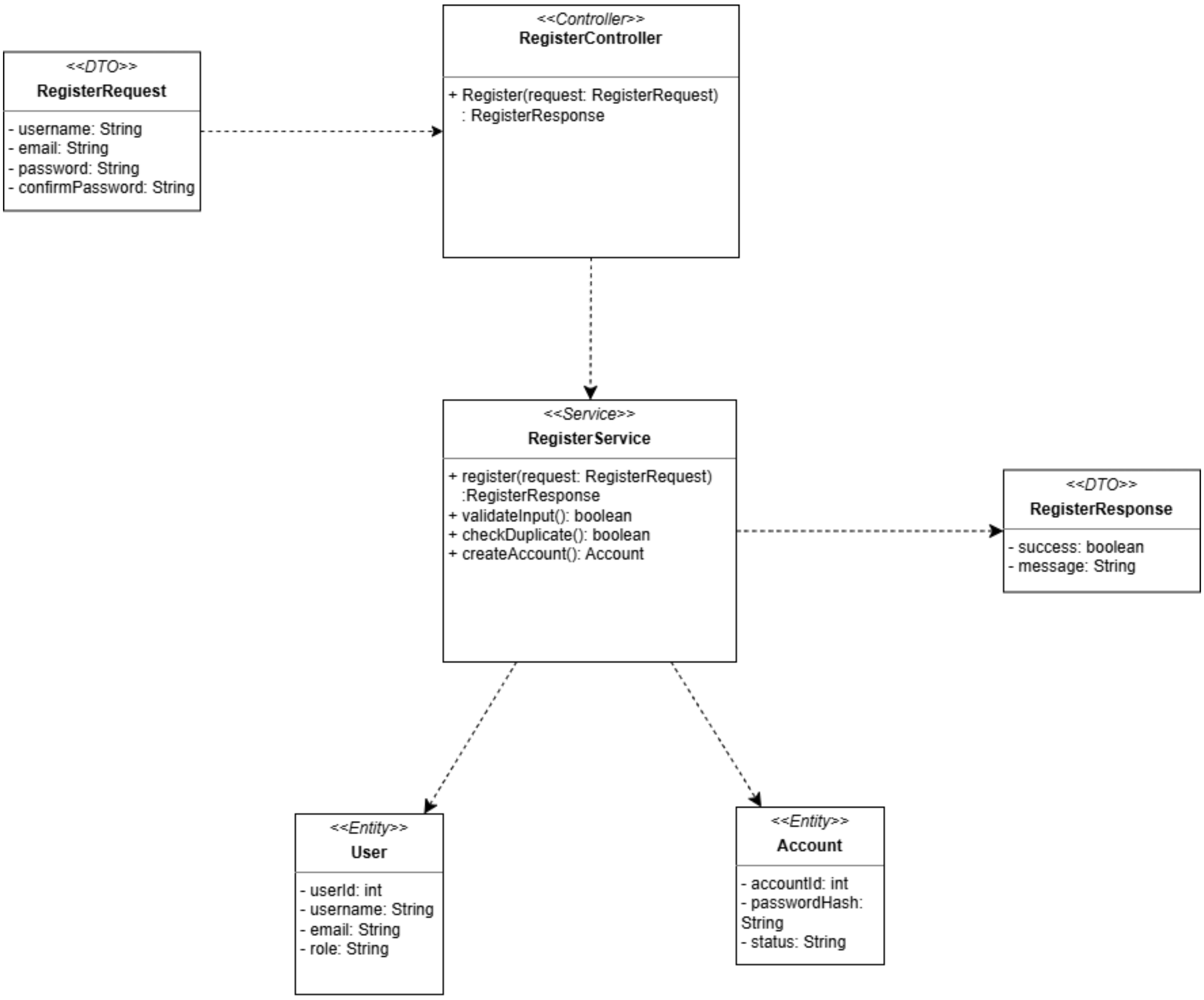
Hình 20: Sequence Diagram

3.1.3 State Diagram



Hình 21: State diagram

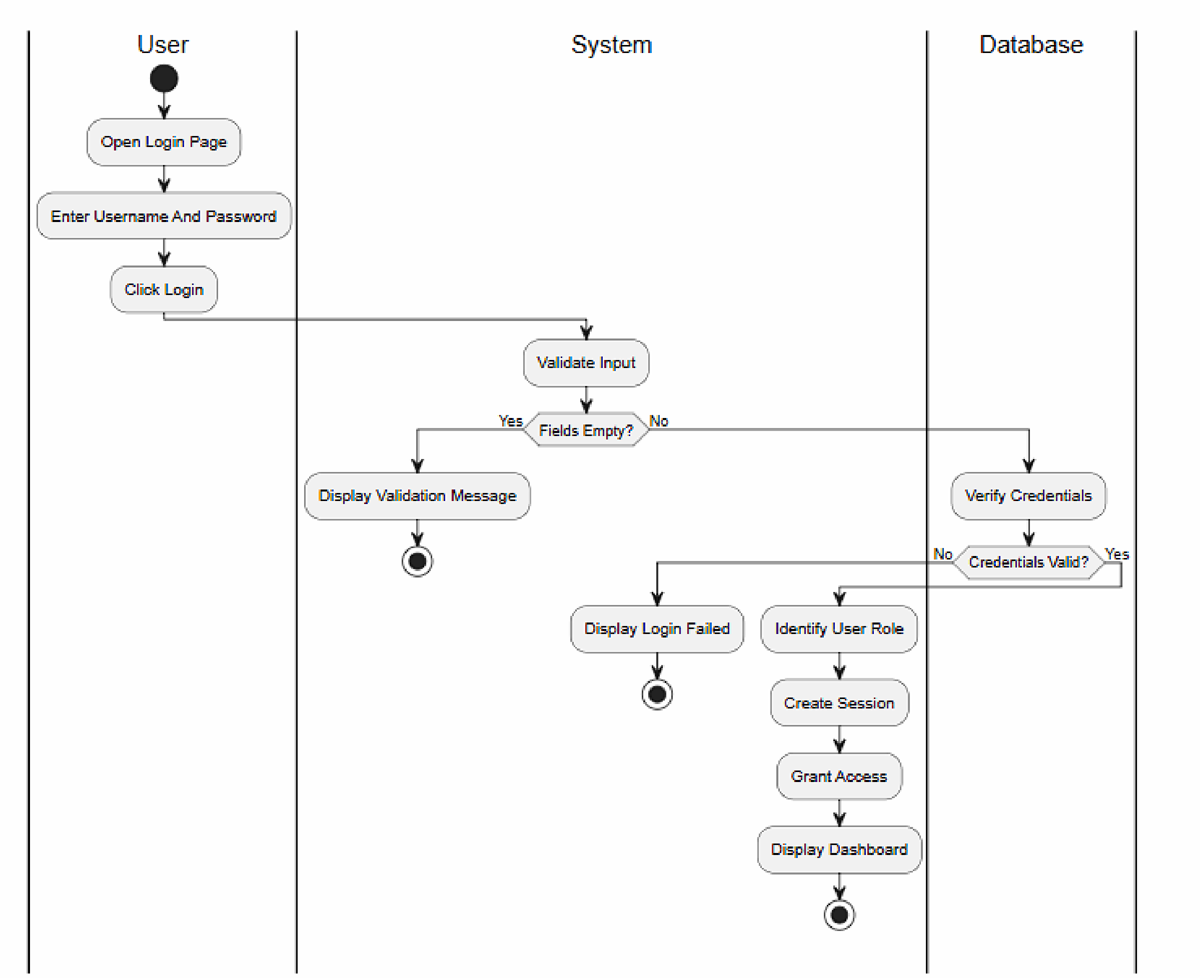
3.1.4 Class Diagram



Hình 22: Class Diagram: Register

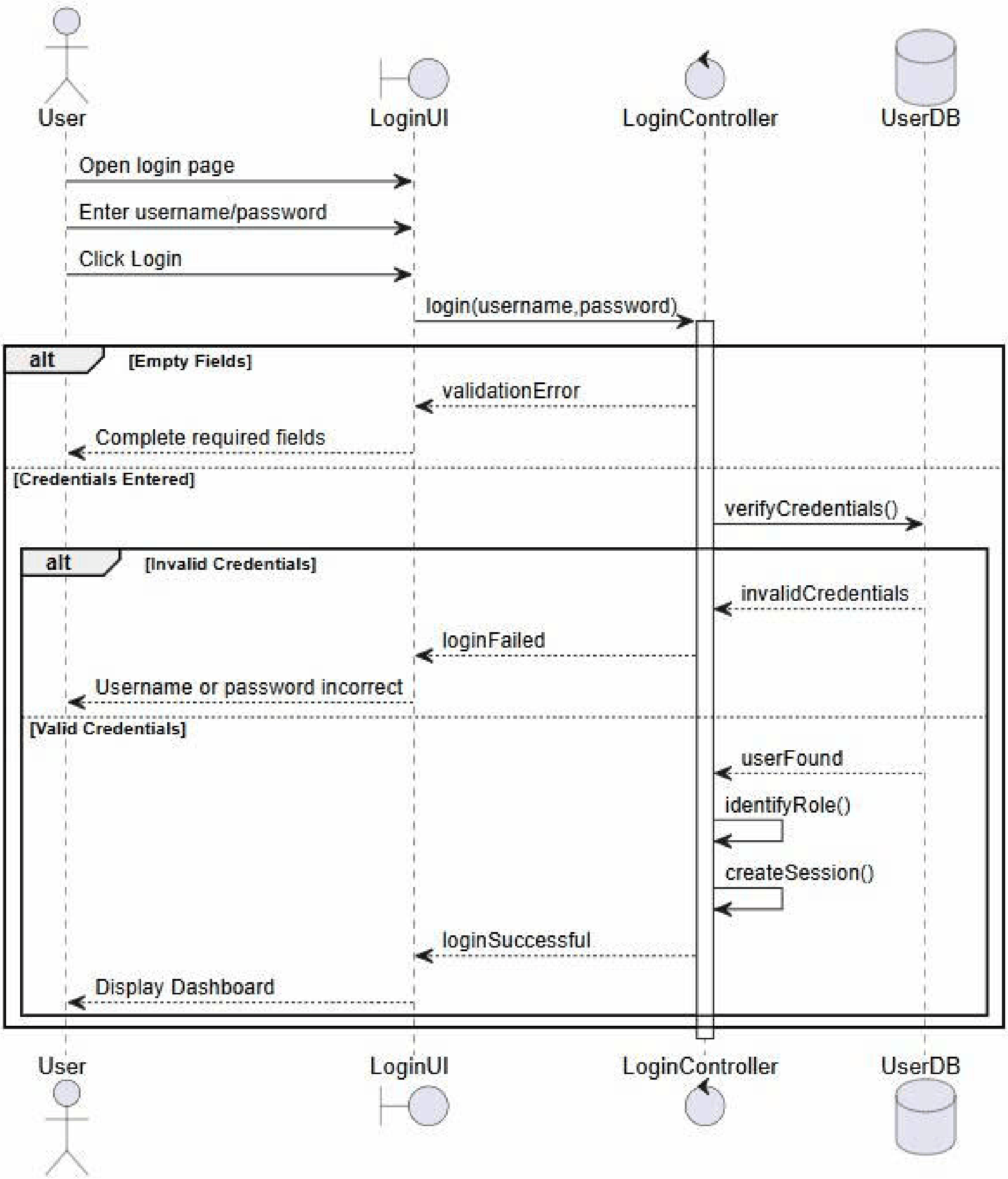
3.2 Login

3.2.1 Activity Diagram



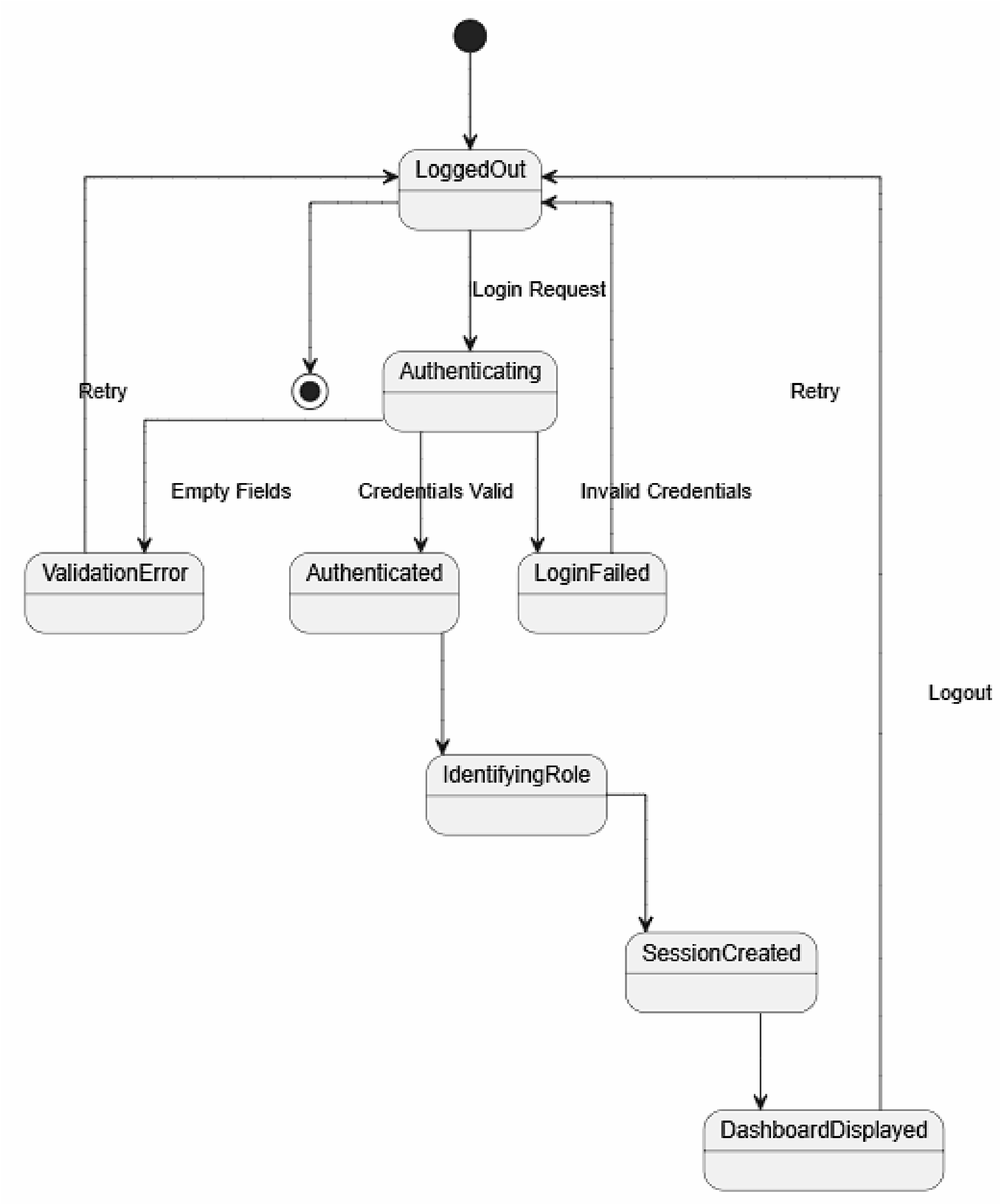
Hình 23: Activity diagram

3.2.2 Sequence Diagram



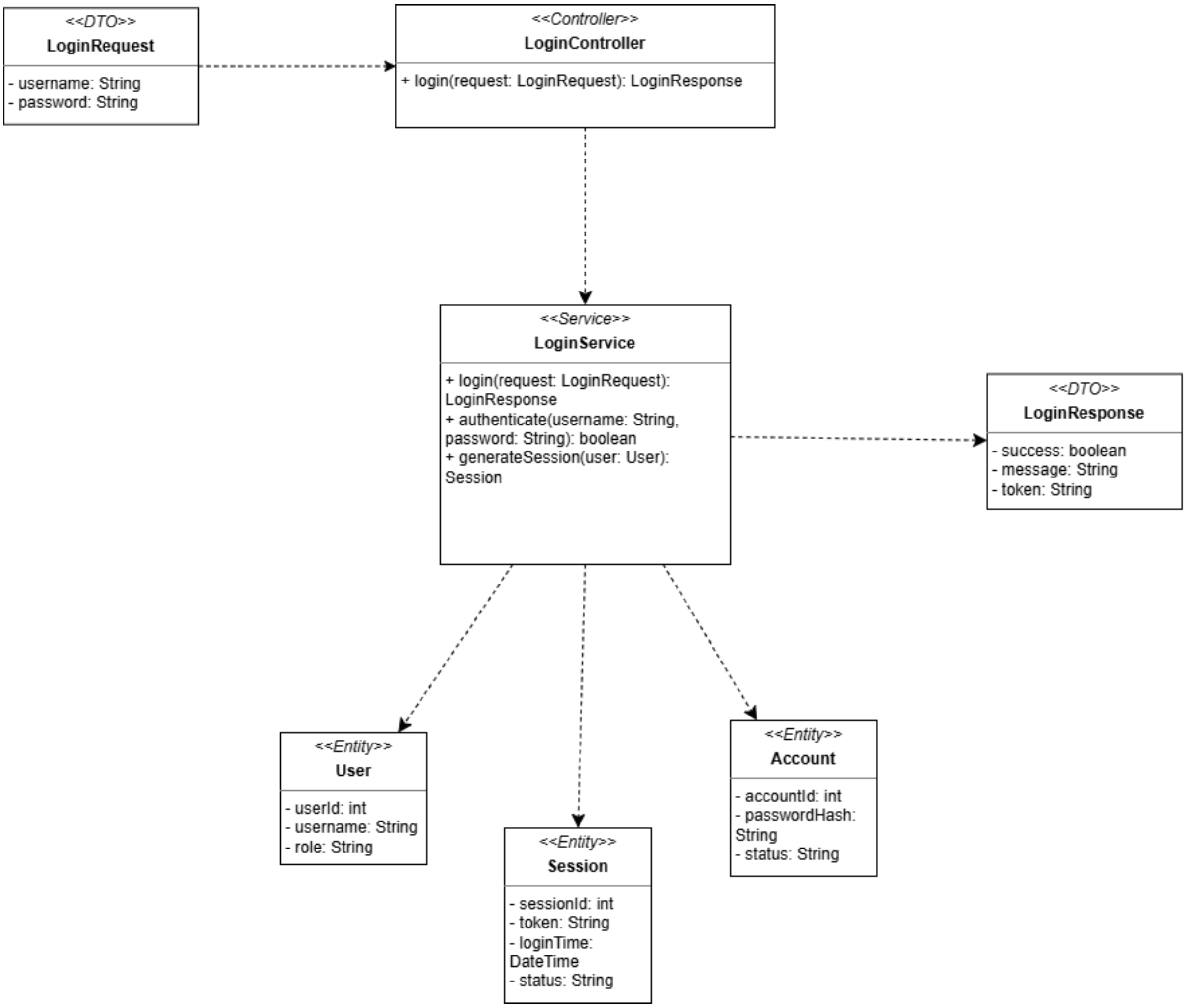
Hình 24: Sequence Diagram

3.2.3 State Diagram



Hình 25: State diagram

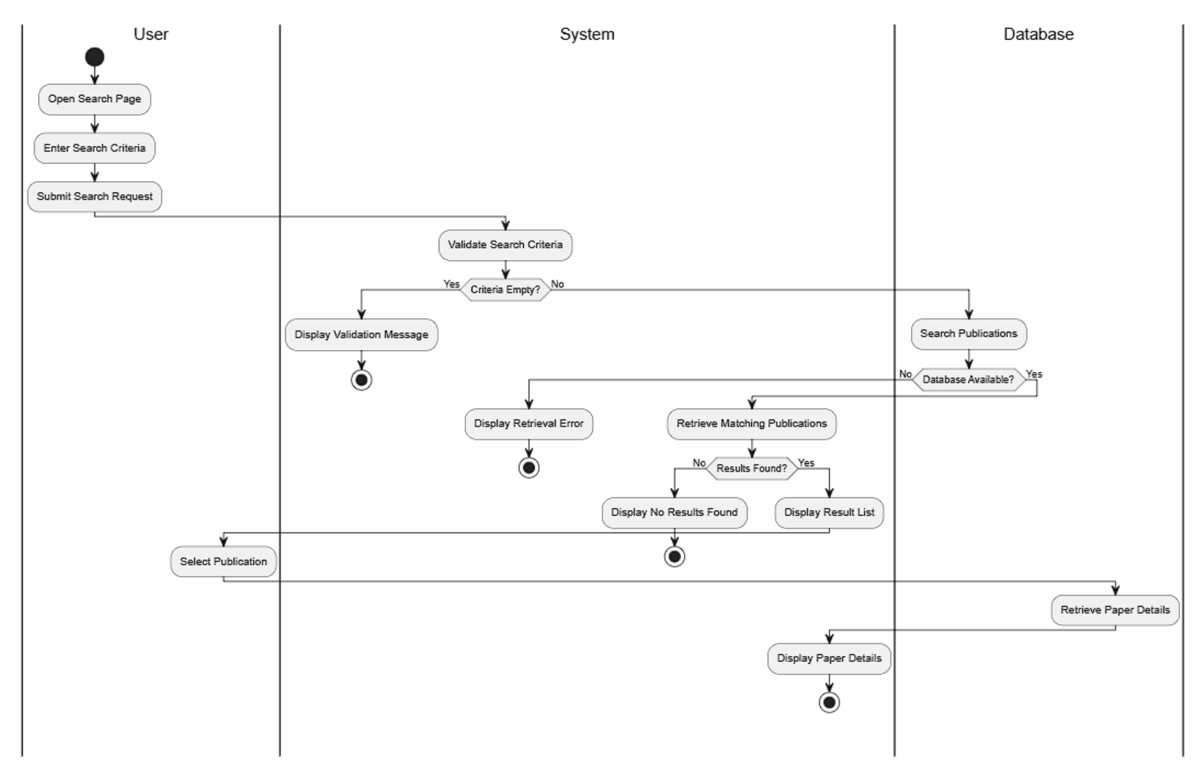
3.2.4 Class Diagram



Hình 26: Class Diagram: Login

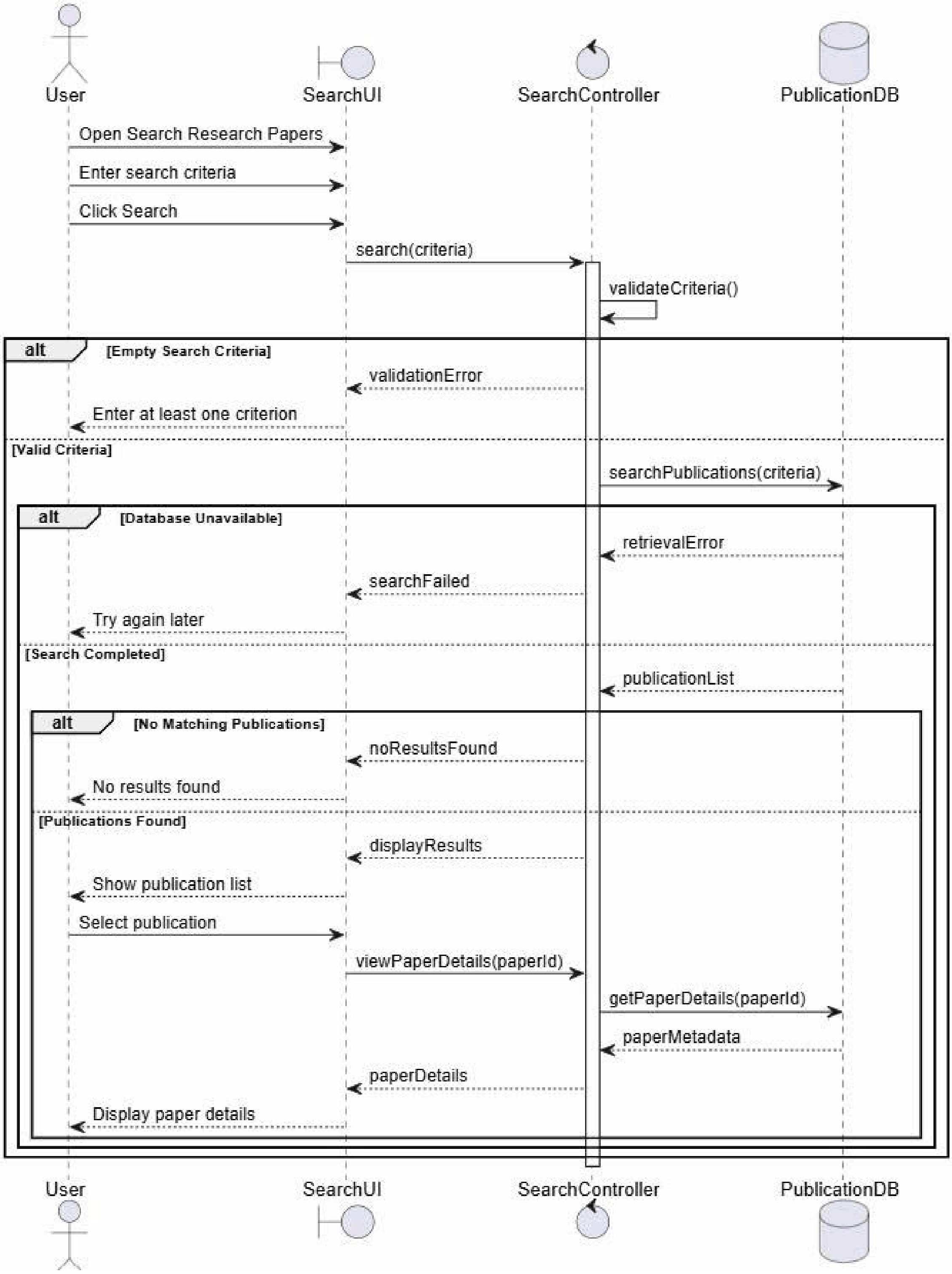
3.3 Search Research Papers

3.3.1 Activity Diagram



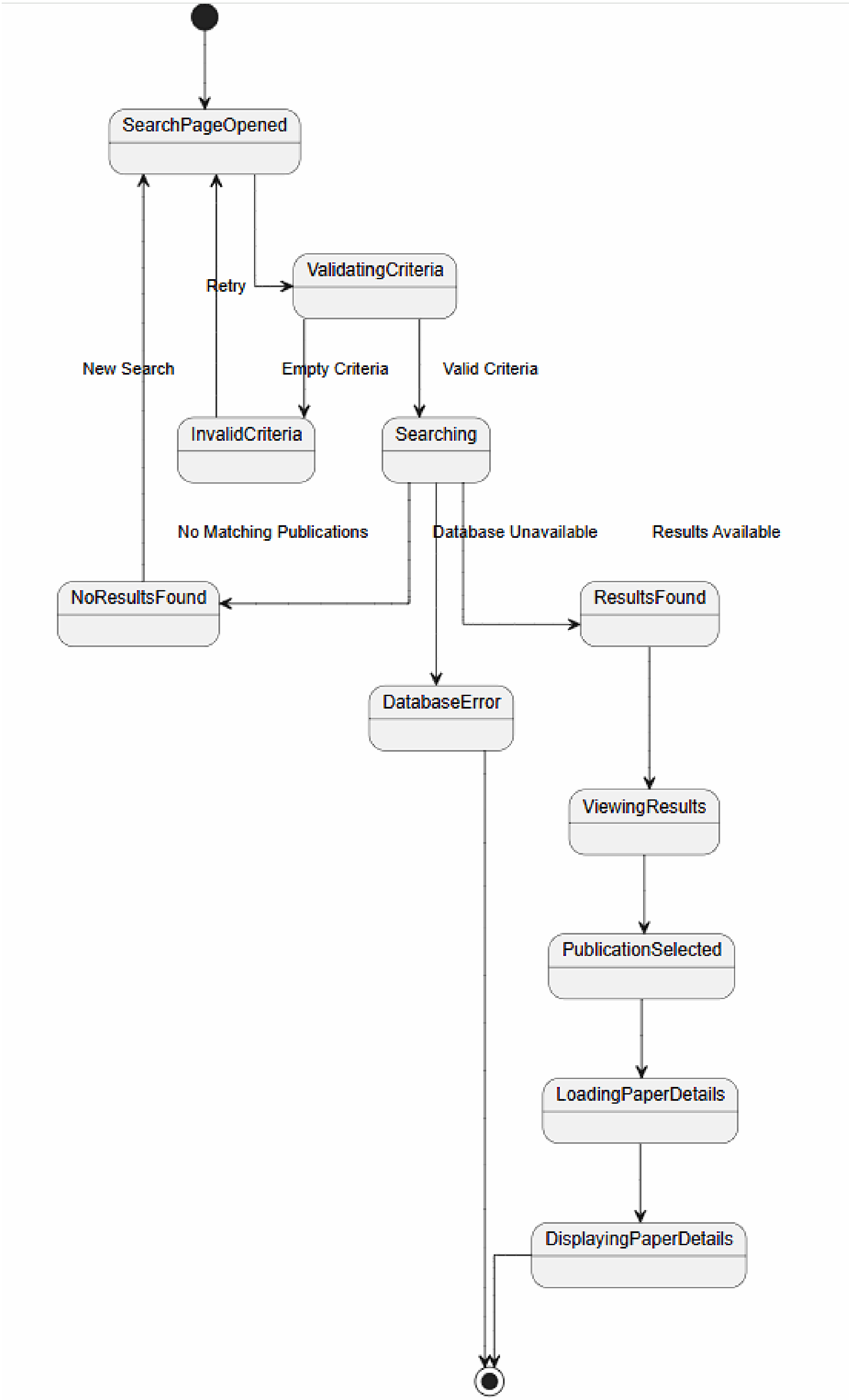
Hình 27: Activity diagram

3.3.2 Sequence Diagram



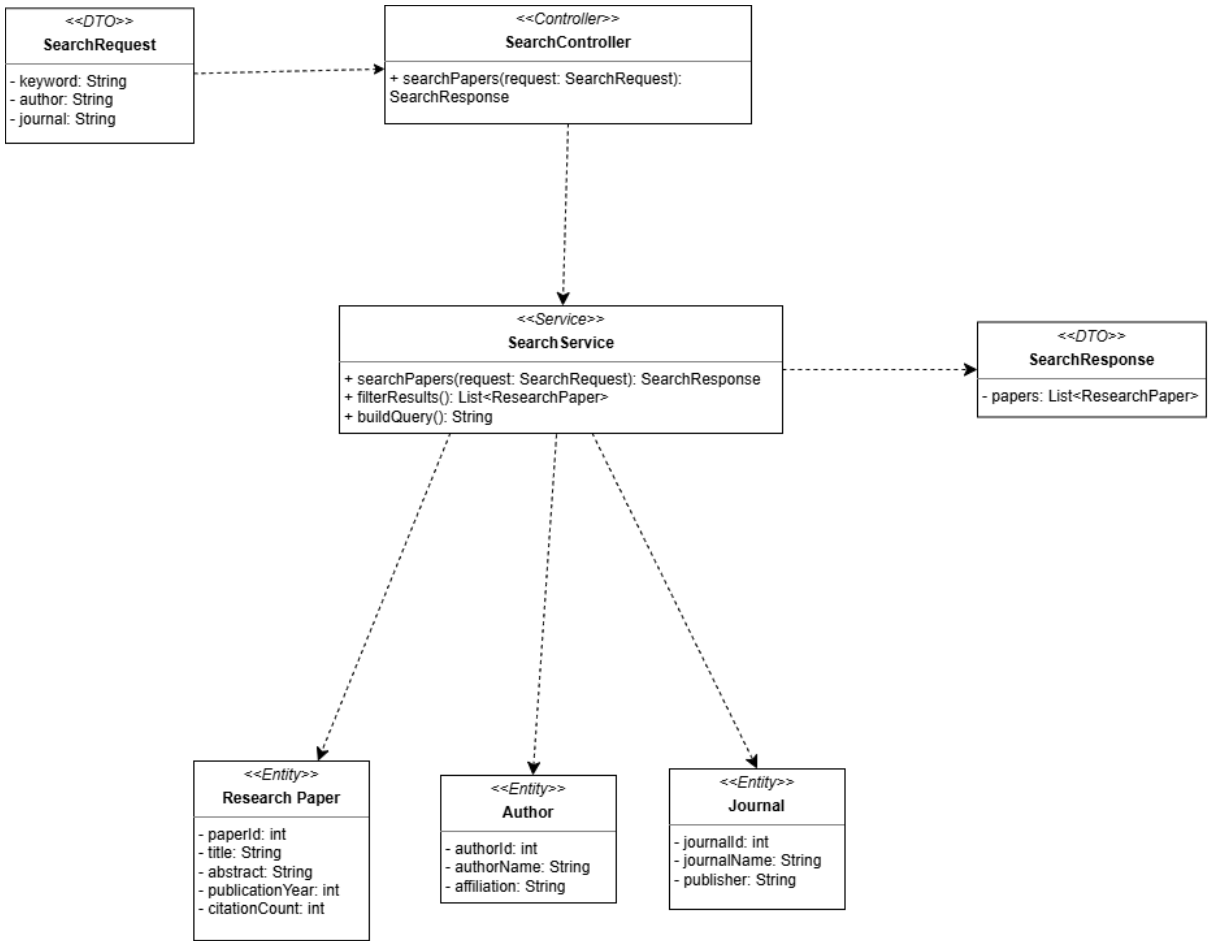
Hình 28: Sequence Diagram

3.3.3 State Diagram



Hình 29: State diagram

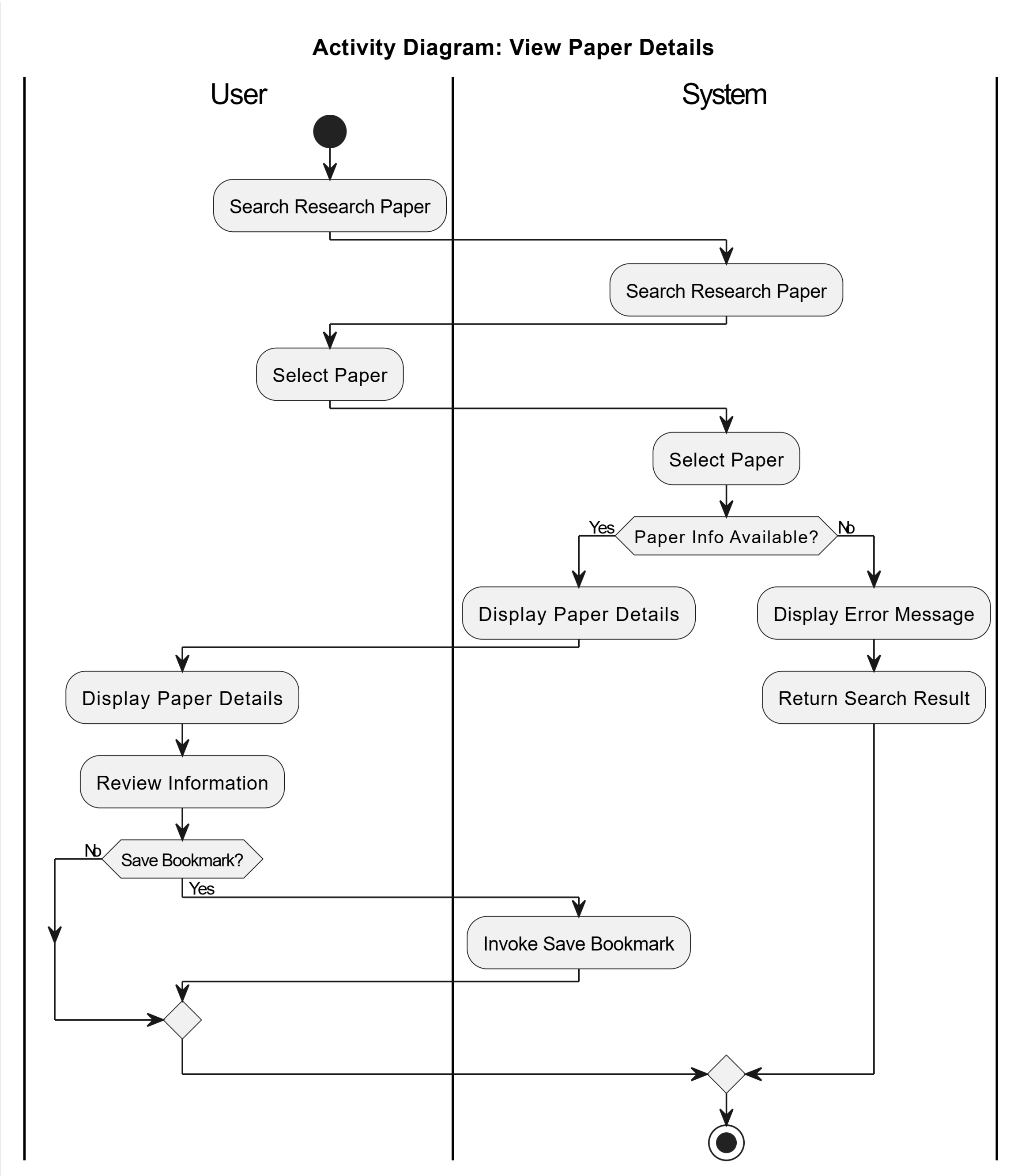
3.3.4 Class Diagram



Hình 30: Class Diagram: Search Research Papers

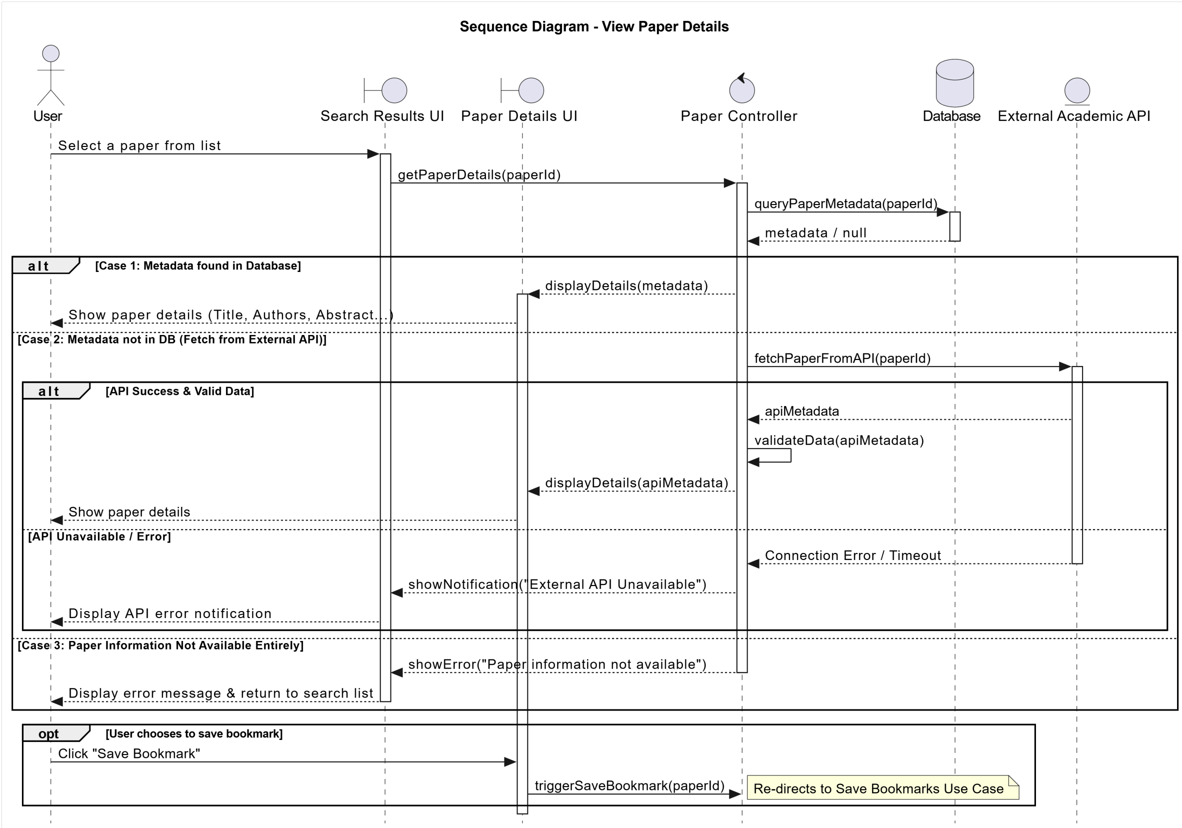
3.4 View Paper Details

3.4.1 Activity Diagram



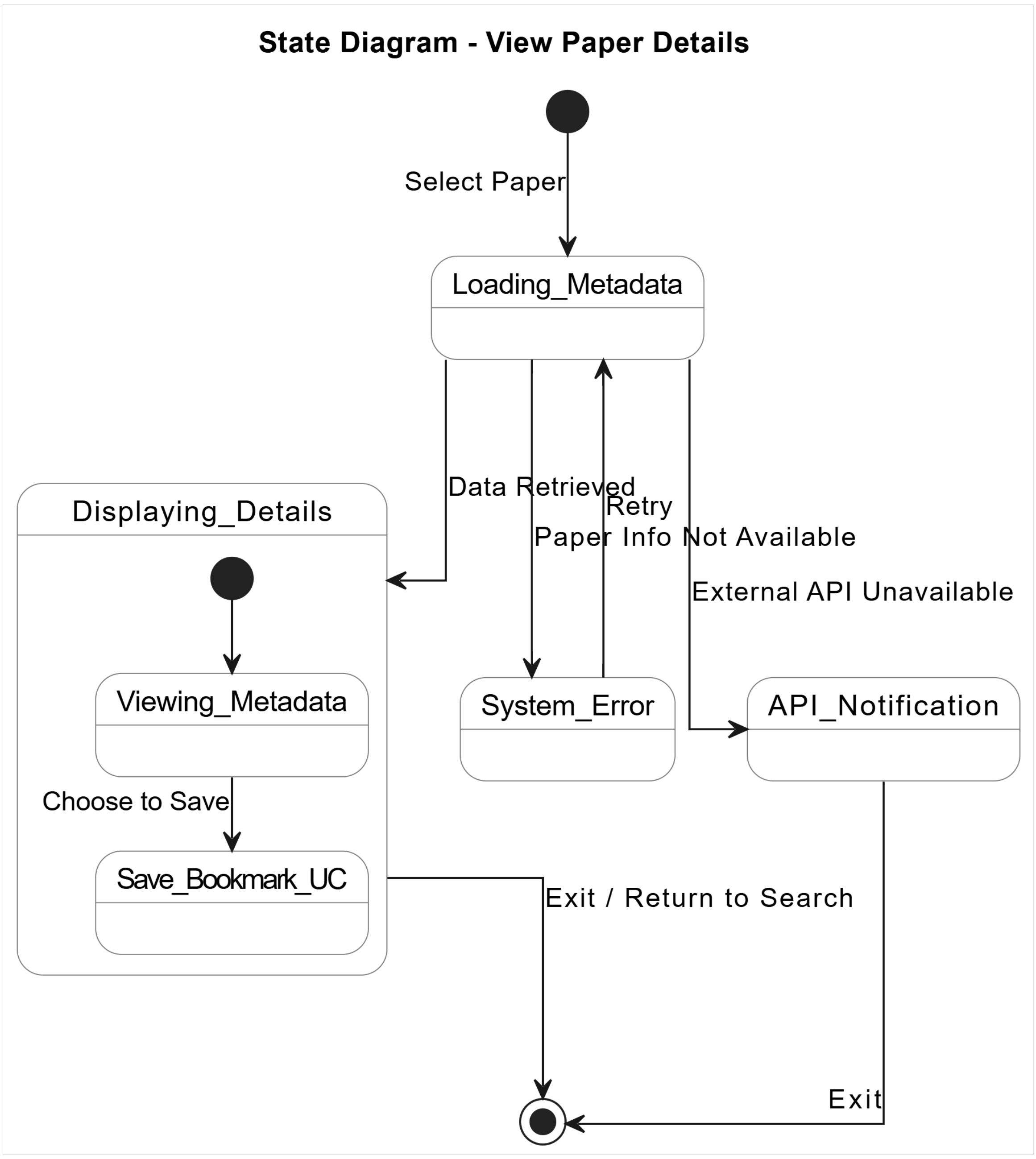
Hình 31: Activity Diagram: View Paper Details

3.4.2 Sequence Diagram



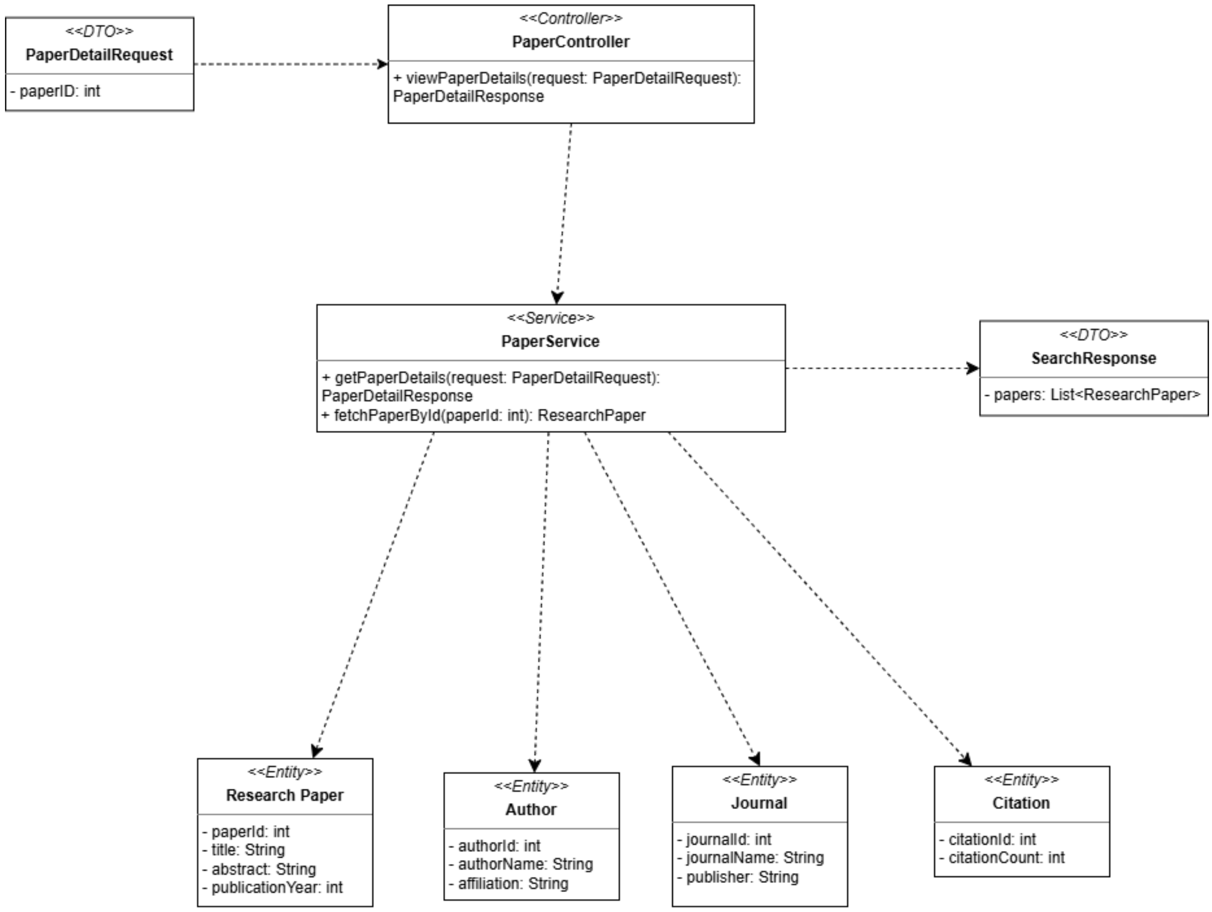
Hình 32: Sequence Diagram: View Paper Details

3.4.3 State Diagram



Hình 33: State Diagram: View Paper Details

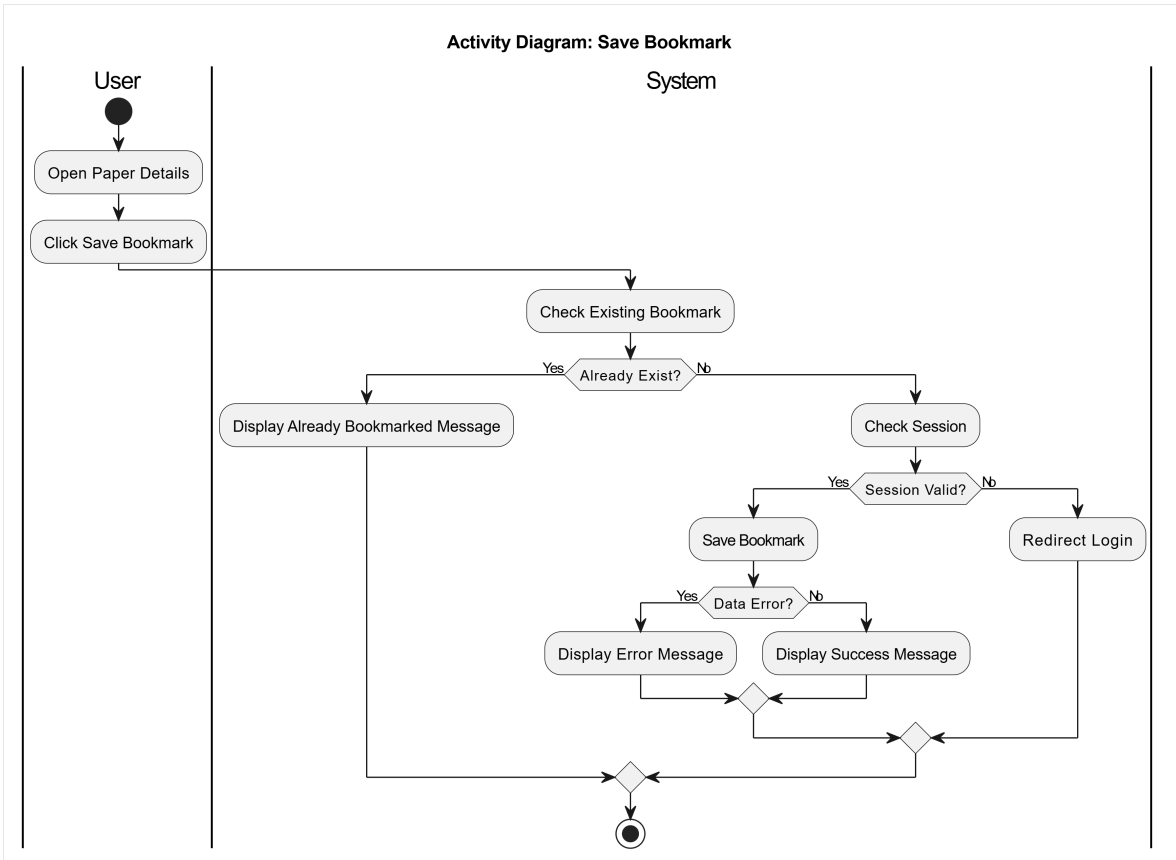
3.4.4 Class Diagram



Hình 34: Class Diagram: View Paper Detail

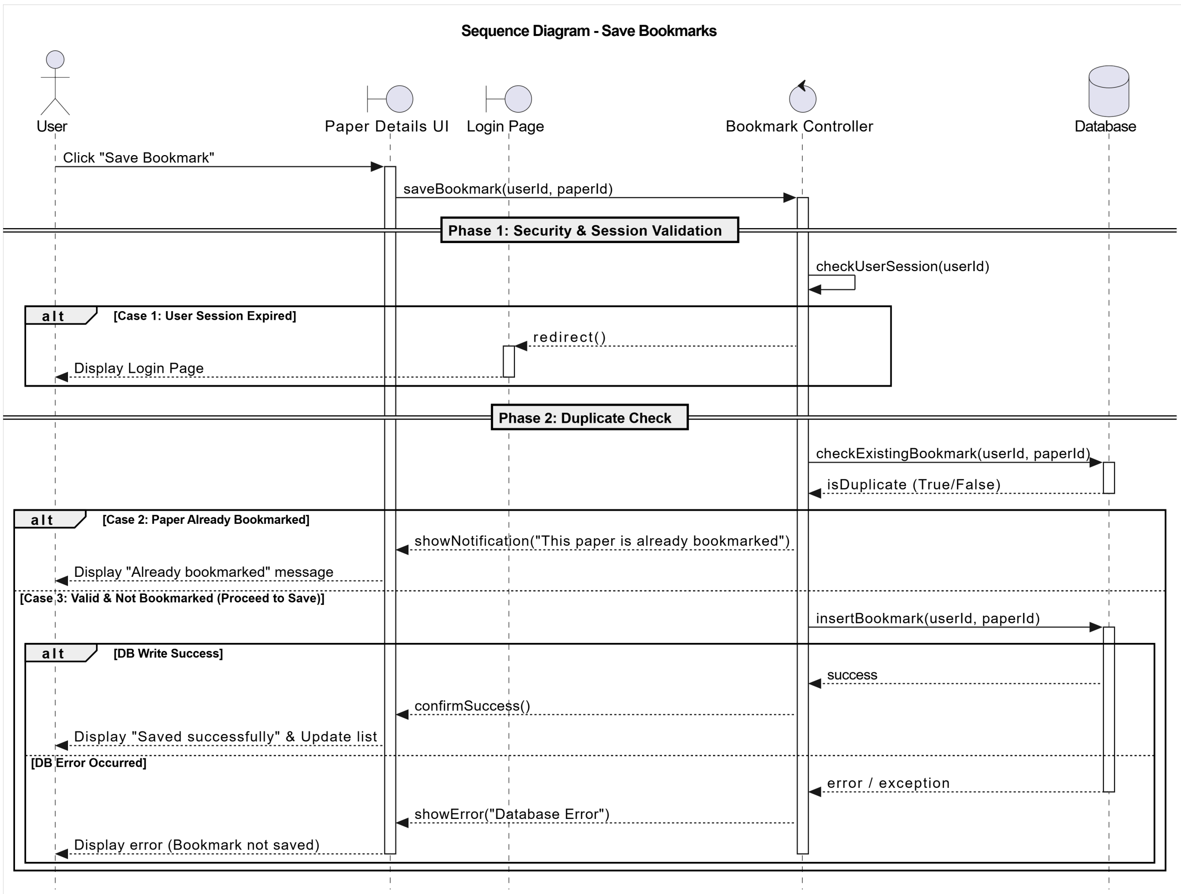
3.5 Save Bookmarks

3.5.1 Activity Diagram



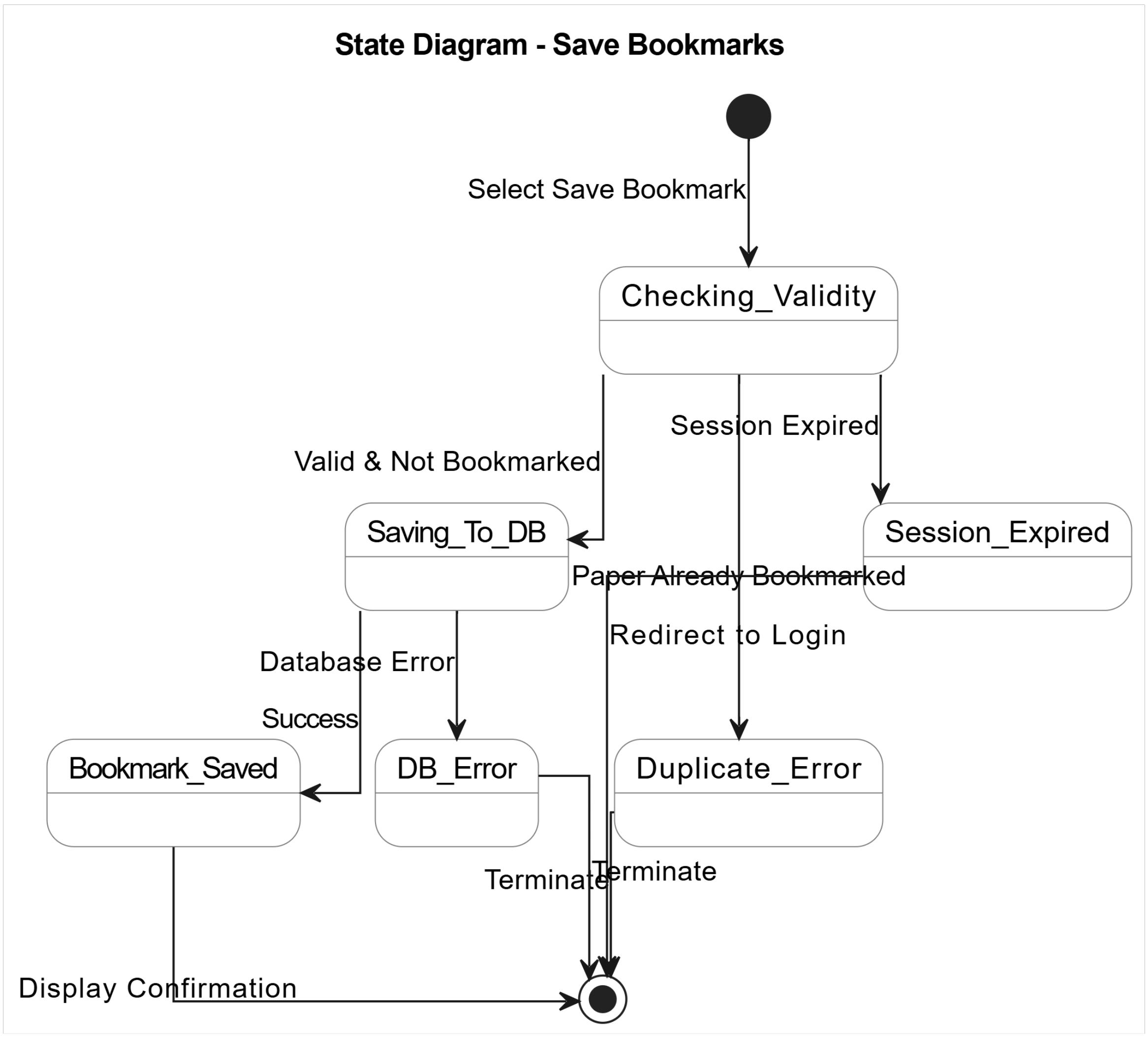
Hình 35: Activity Diagram: Save Bookmarks

3.5.2 Sequence Diagram



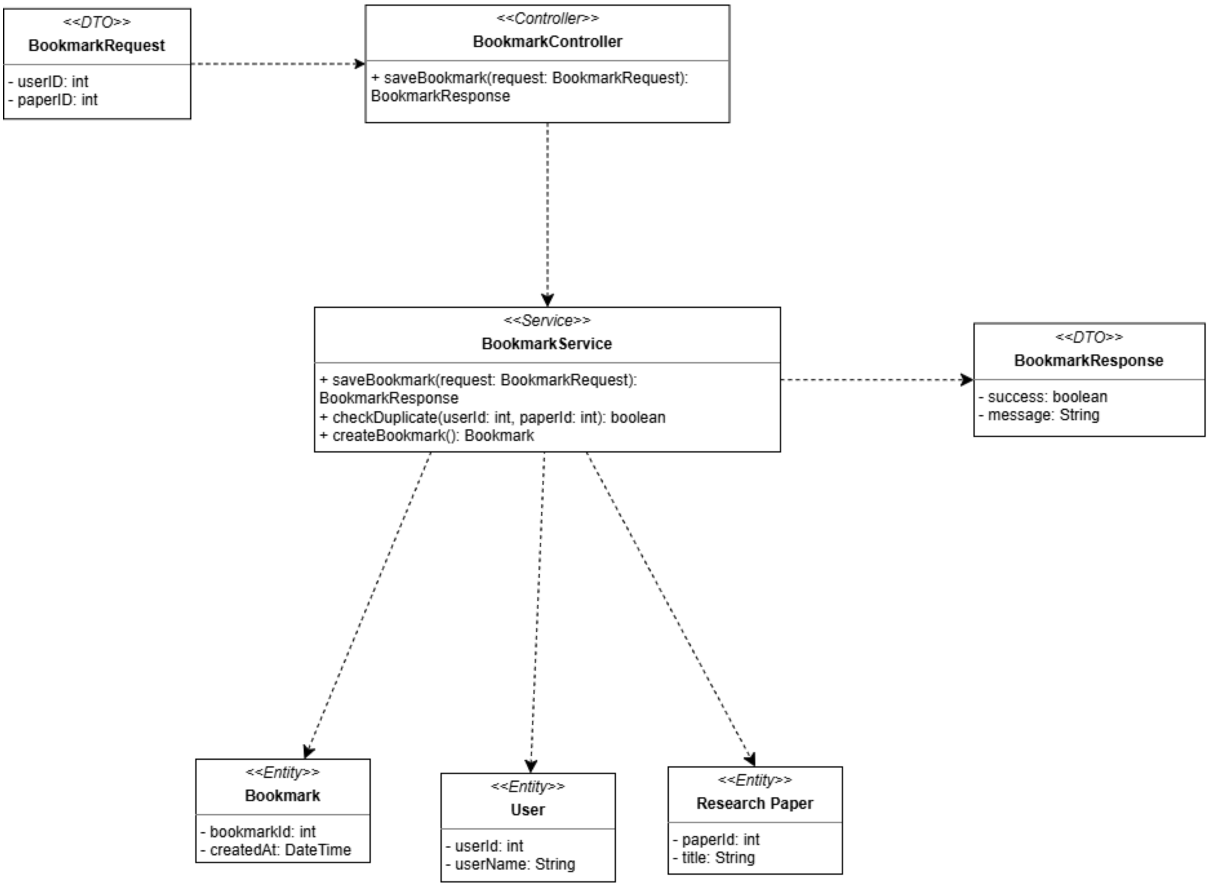
Hình 36: Sequence Diagram: Save Bookmarks

3.5.3 State Diagram



Hình 37: State Diagram: Save Bookmarks

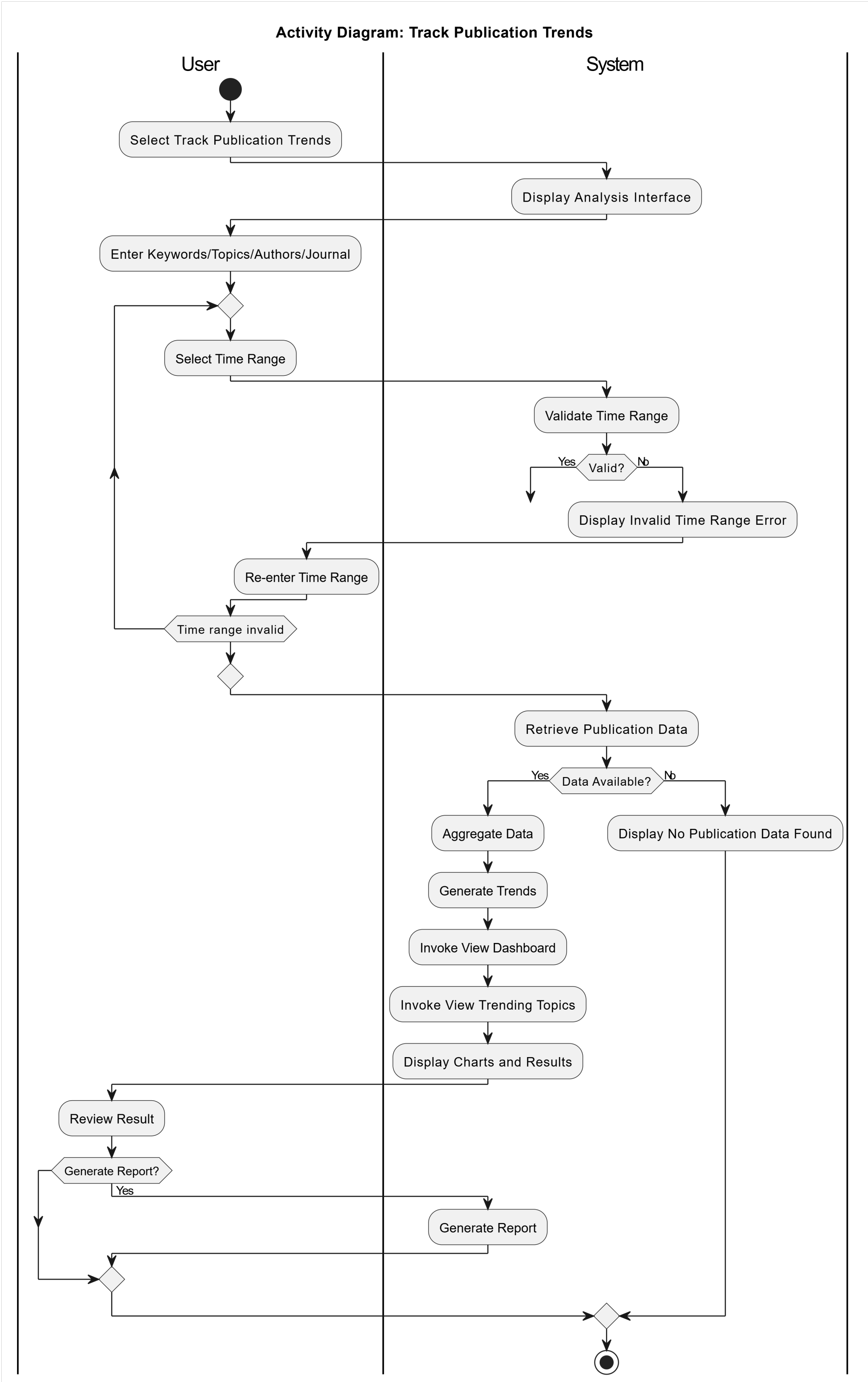
3.5.4 Class Diagram



Hình 38: Class Diagram: Save Bookmarks

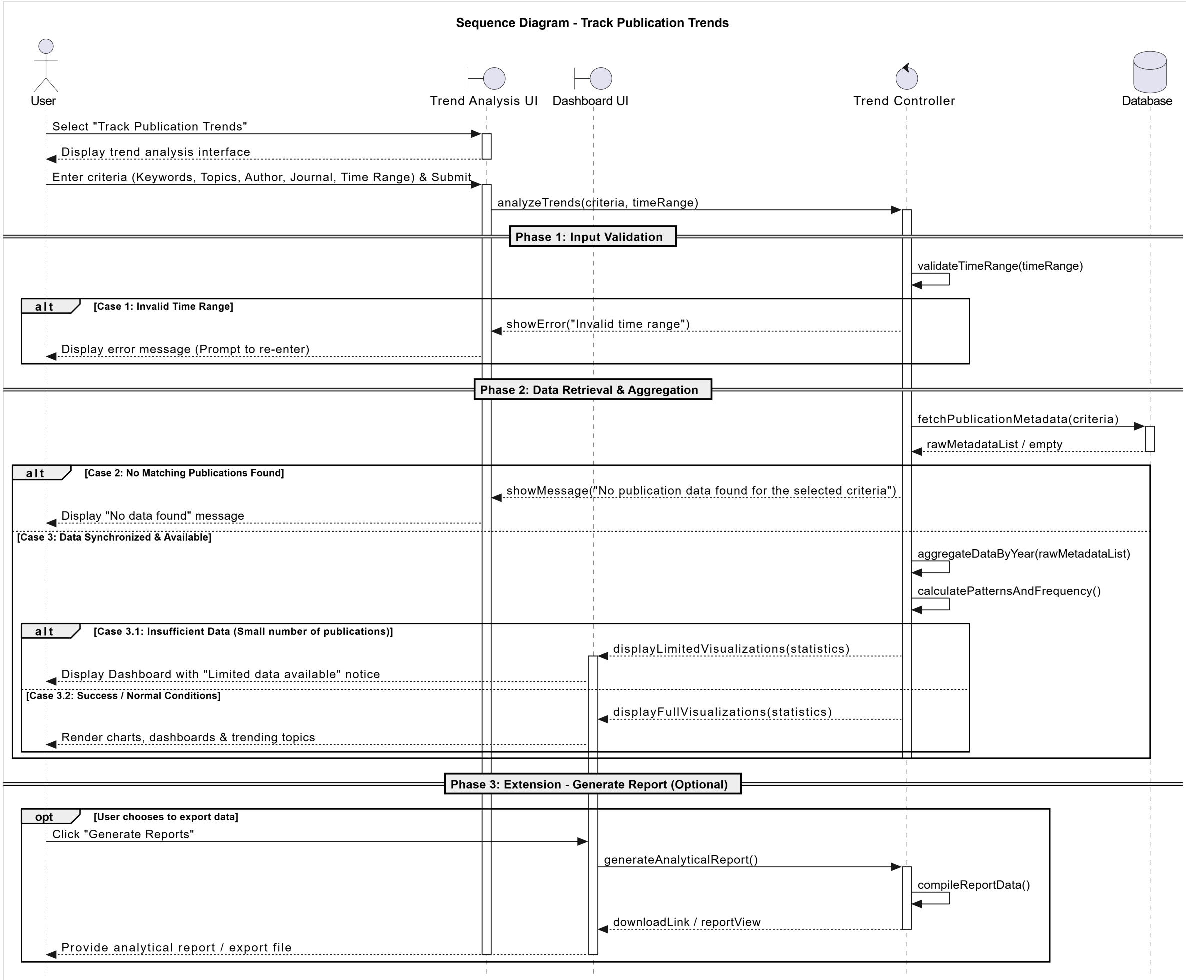
3.6 Track Publication Trends

3.6.1 Activity Diagram



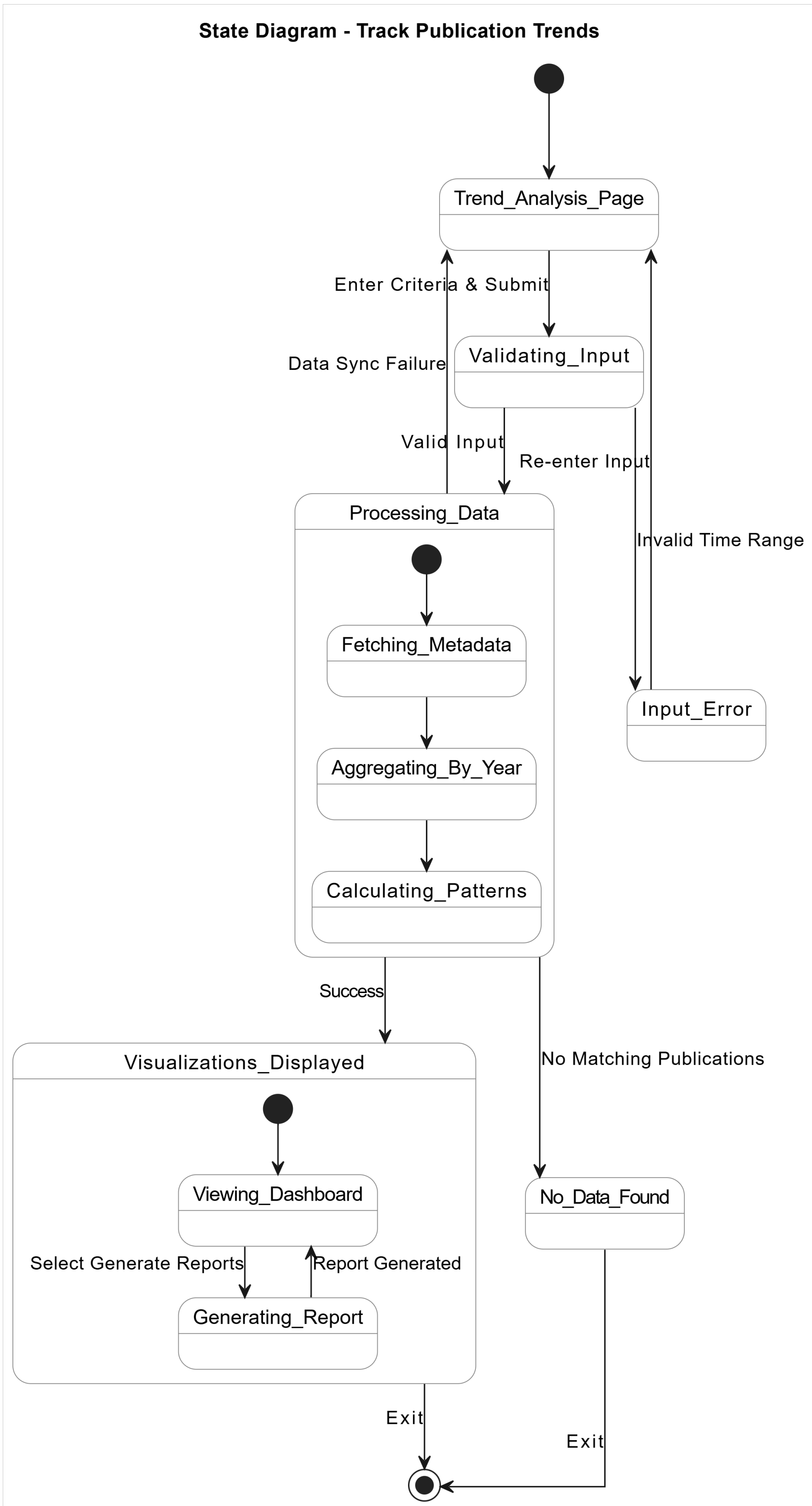
Hình 39: Activity Diagram: Track Publication Trends

3.6.2 Sequence Diagram



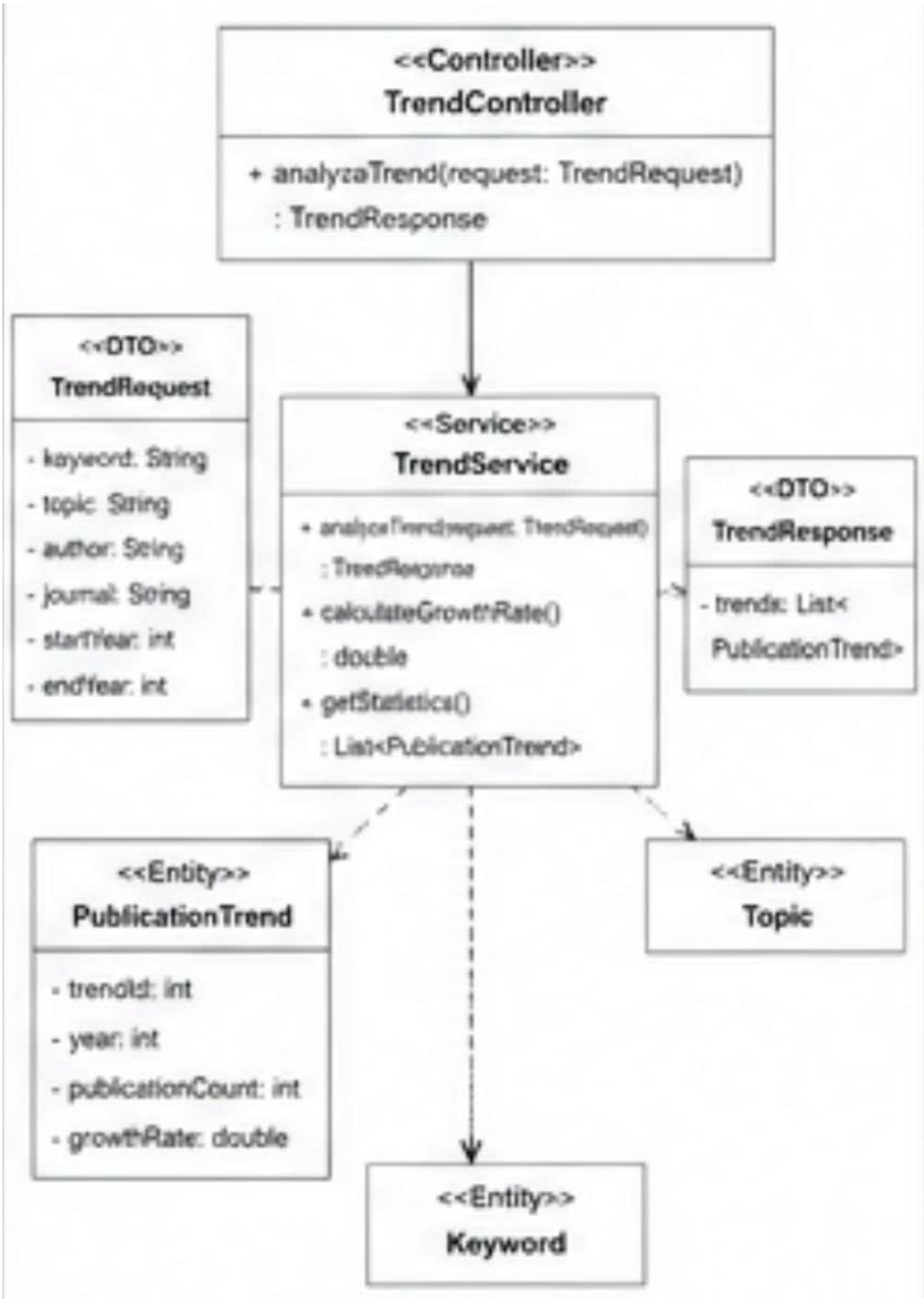
Hình 40: Sequence Diagram: Track Publication Trends

3.6.3 State Diagram



Hình 41: State Diagram: Track Publication Trends

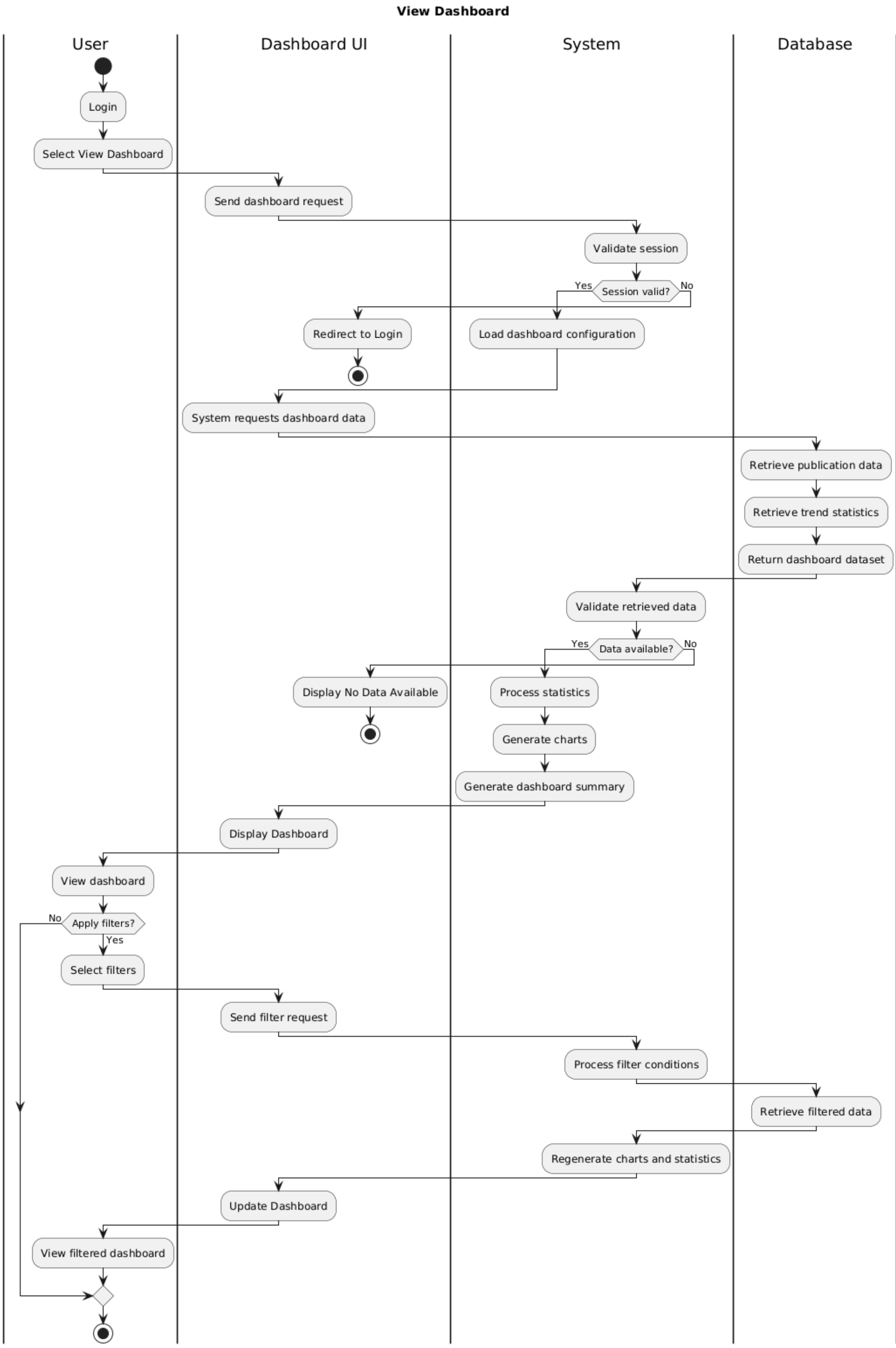
3.6.4 Class Diagram



Hình 42: Class Diagram: Track Publication Trends

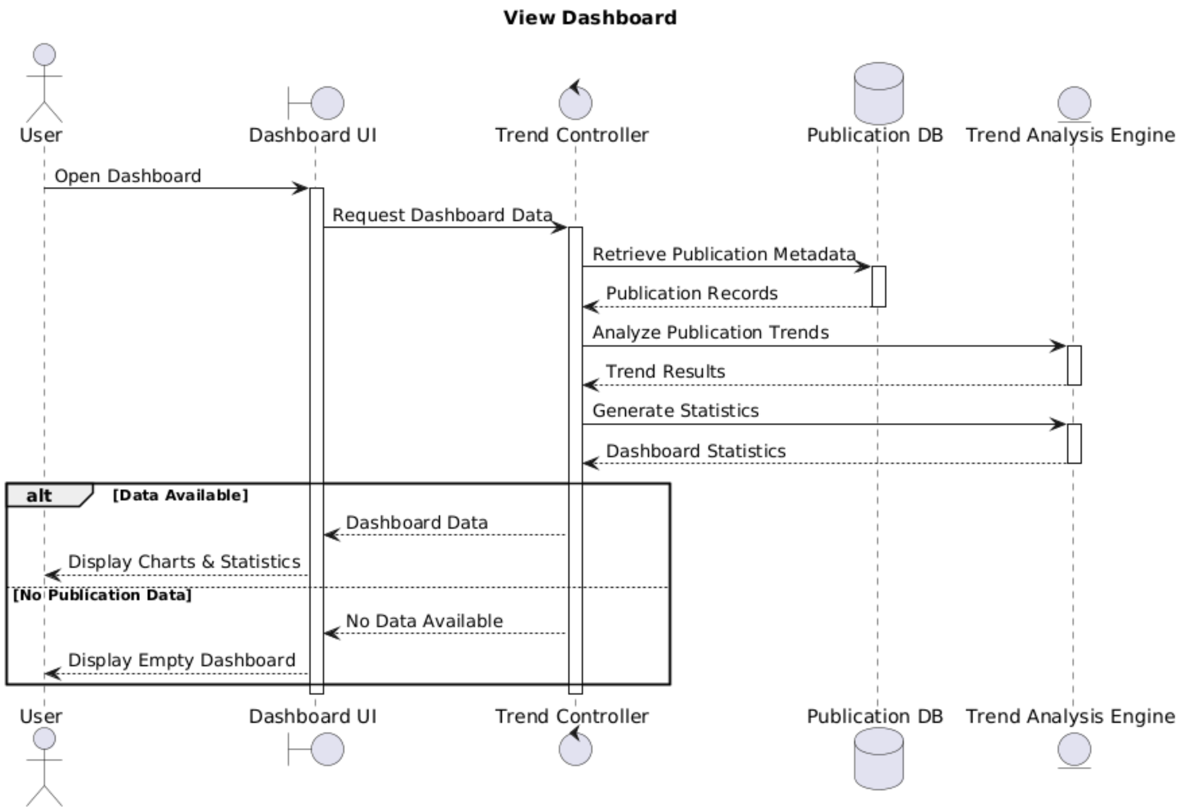
3.7 View Dashboard

3.7.1 Activity Diagram



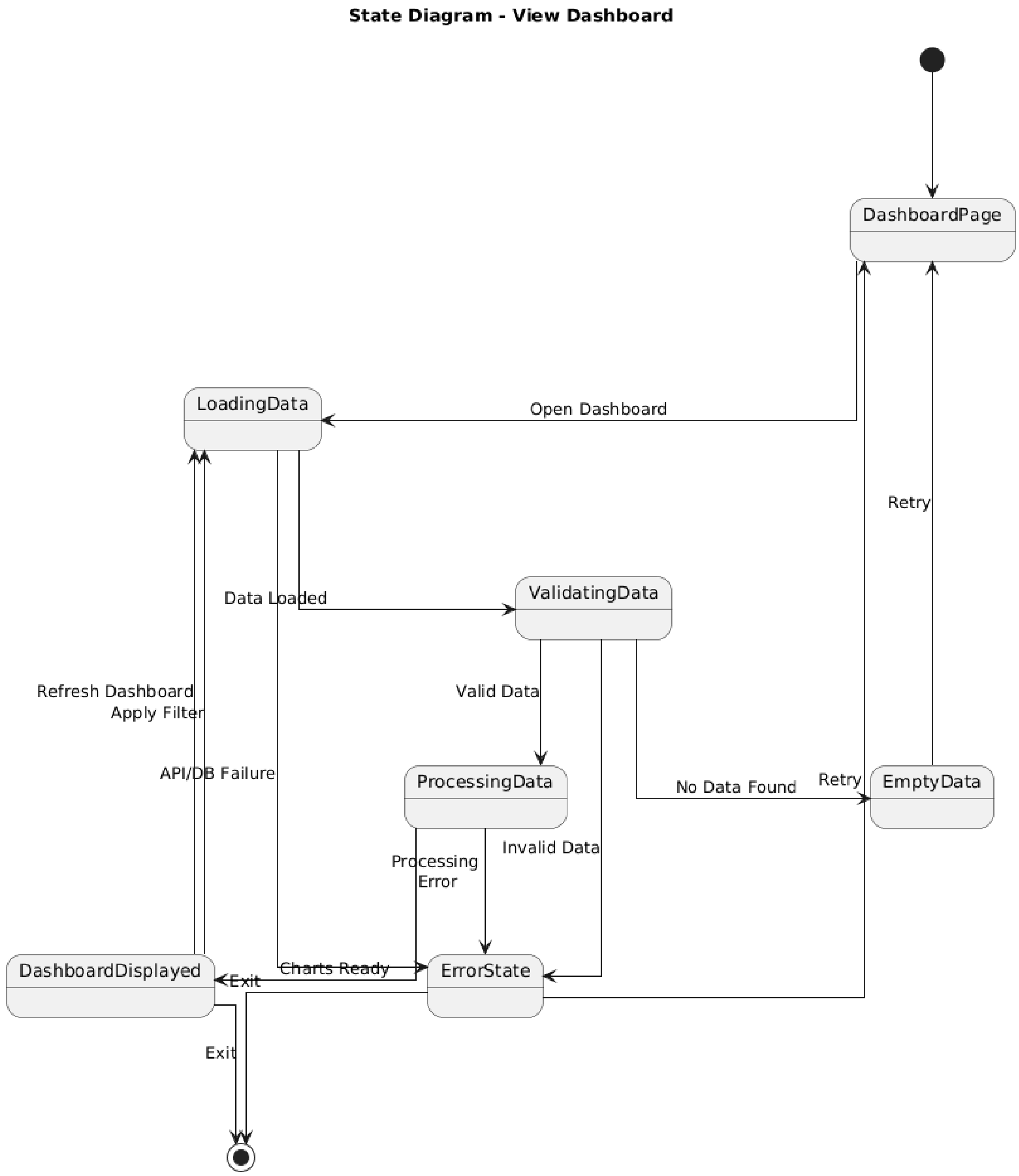
Hình 43: Activity Diagram: View Dashboard

3.7.2 Sequence Diagram



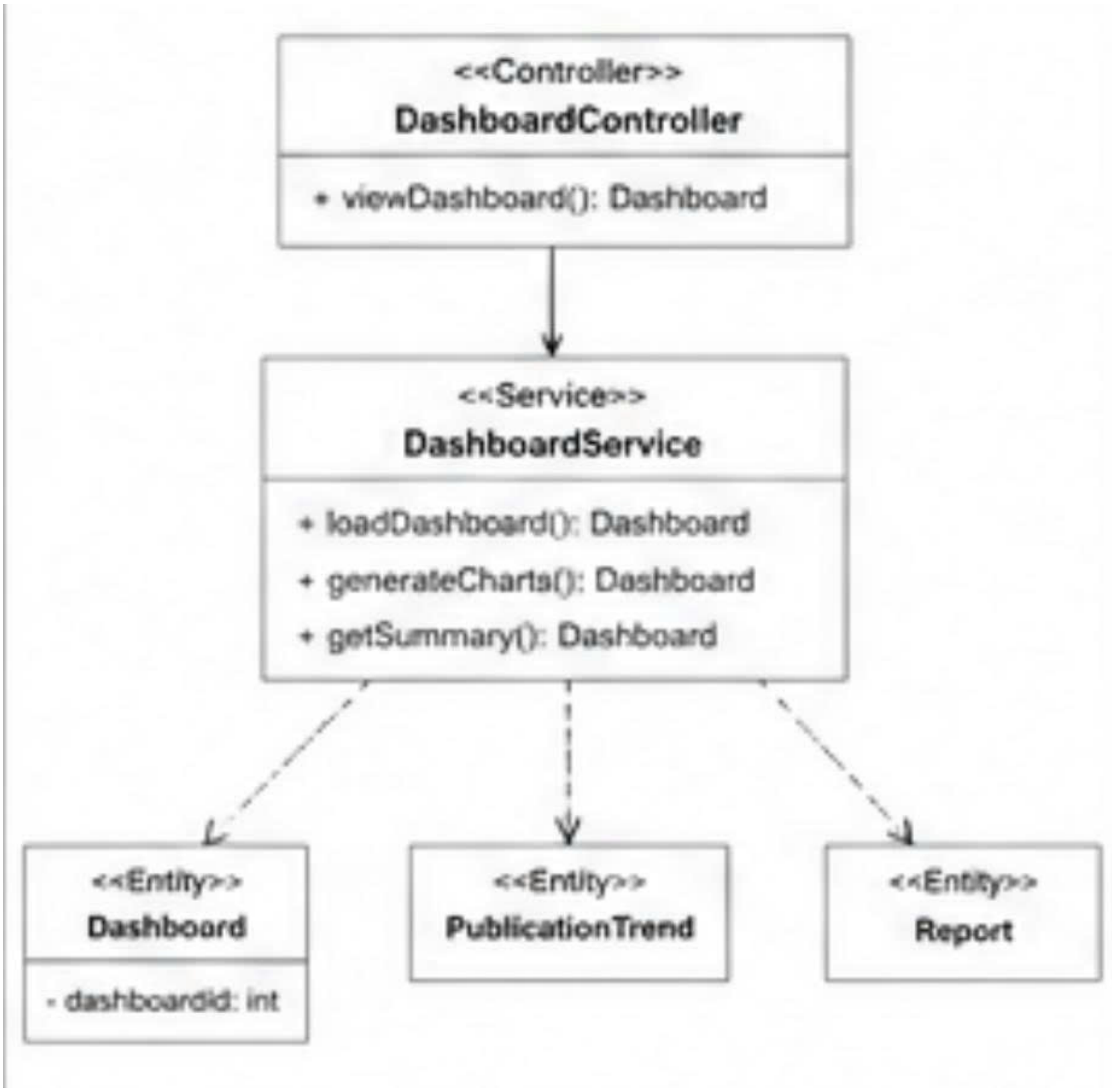
Hình 44: Sequence Diagram: View Dashboard

3.7.3 State Diagram



Hình 45: State Diagram: View Dashboard

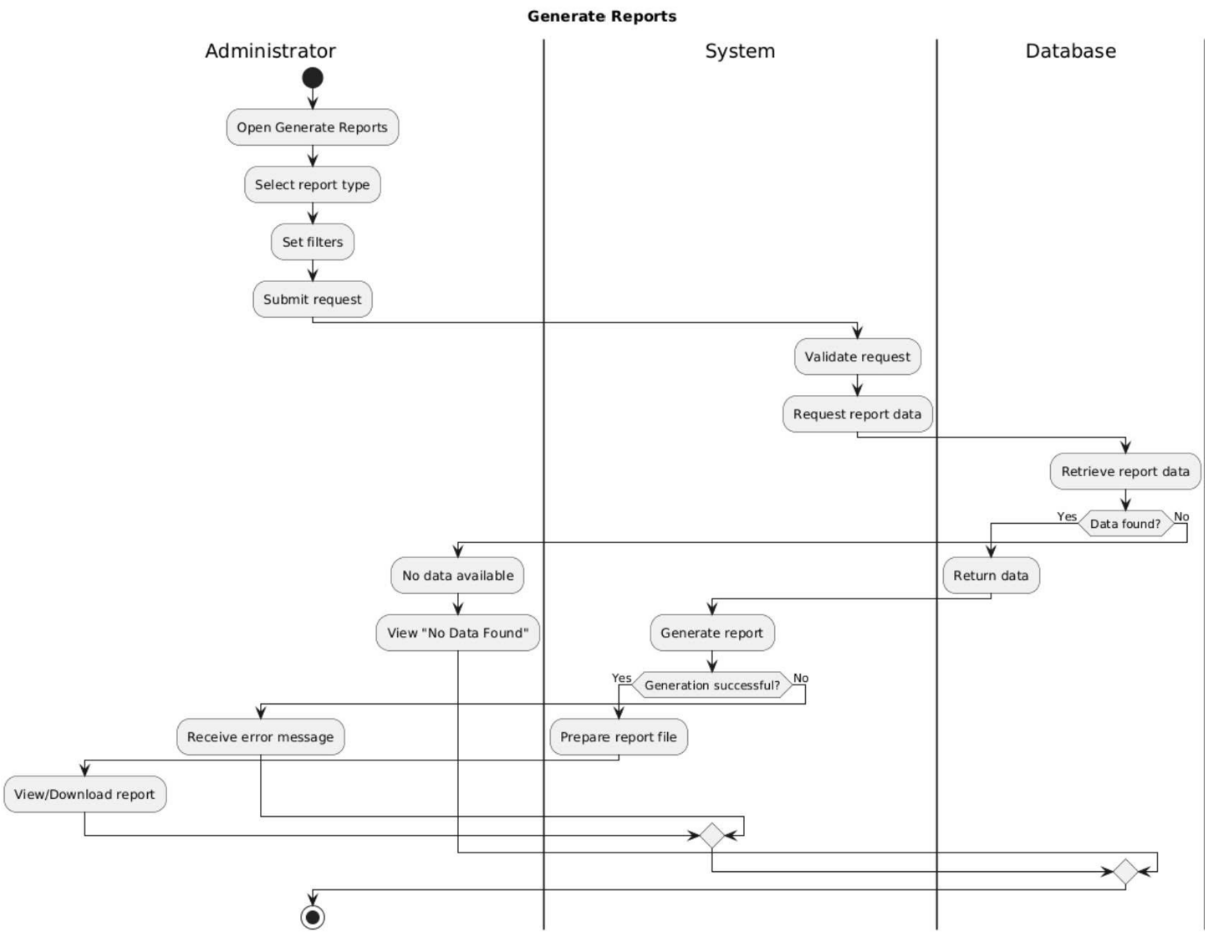
3.7.4 Class Diagram



Hình 46: Class Diagram: View Dashboard

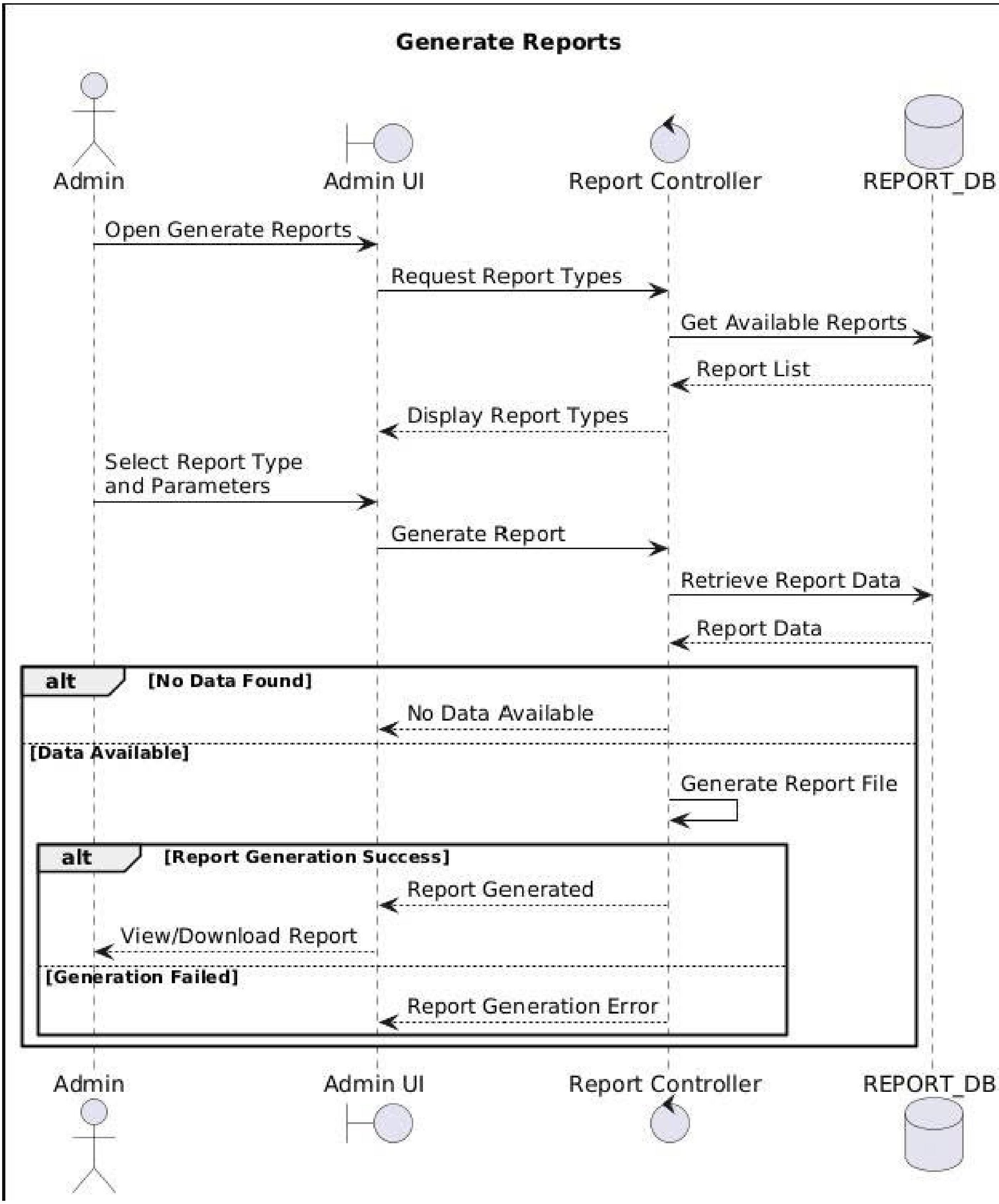
3.8 Generate Reports

3.8.1 Activity Diagram



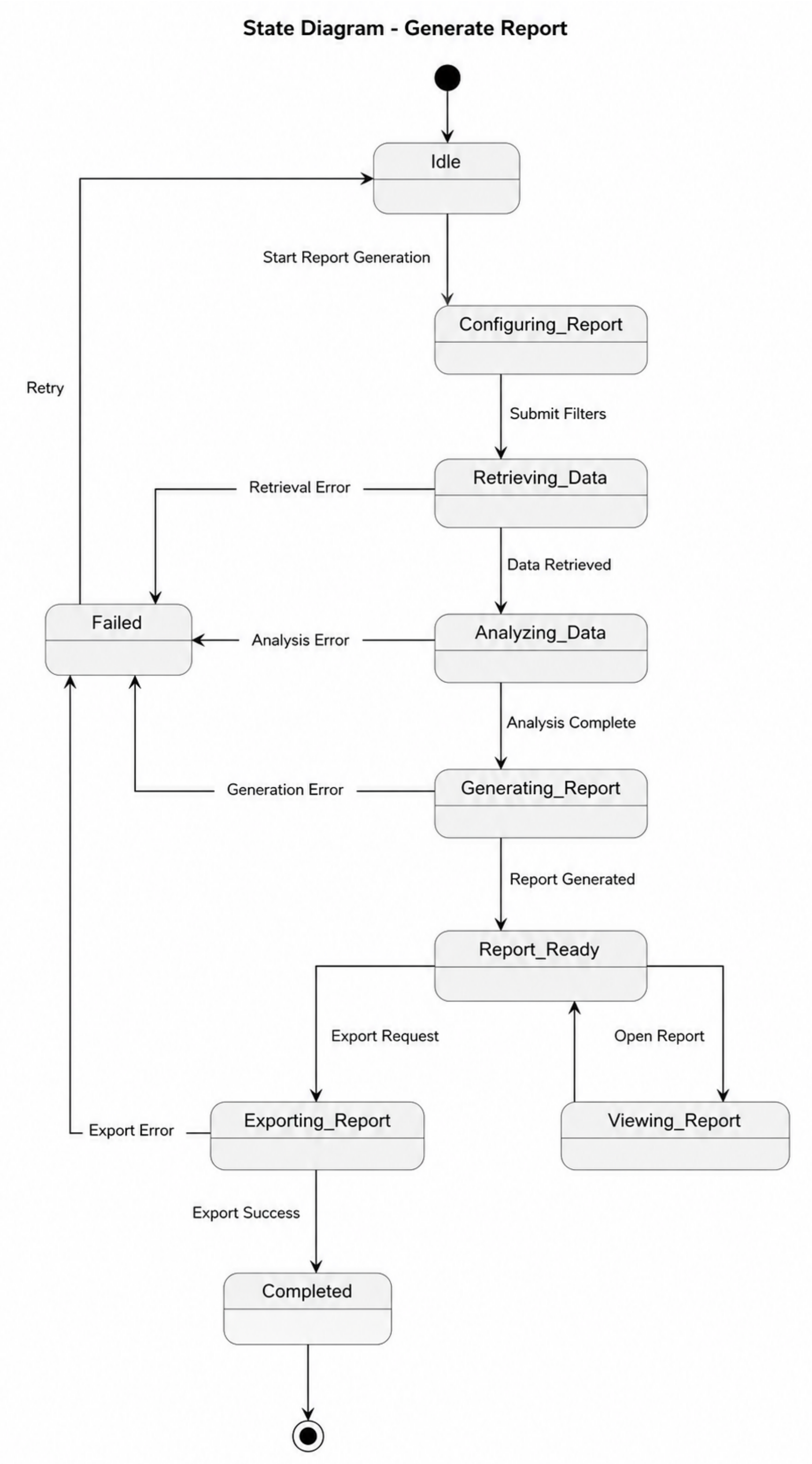
Hình 47: Activity Diagram - Generate reports

3.8.2 Sequence Diagram



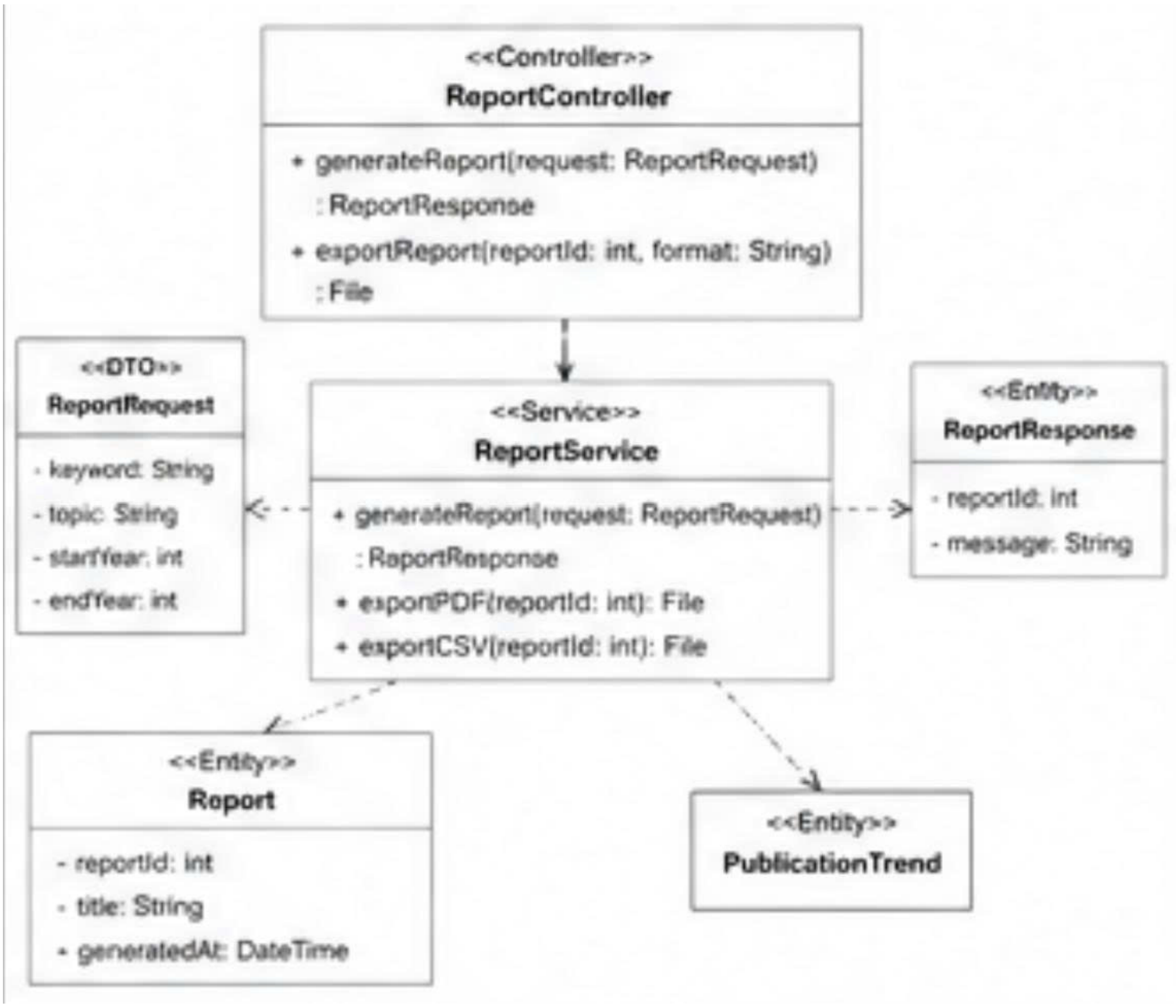
Hình 48: Sequence diagram - Generate reports

3.8.3 State Diagram



Hình 49: State diagram - Generate reports

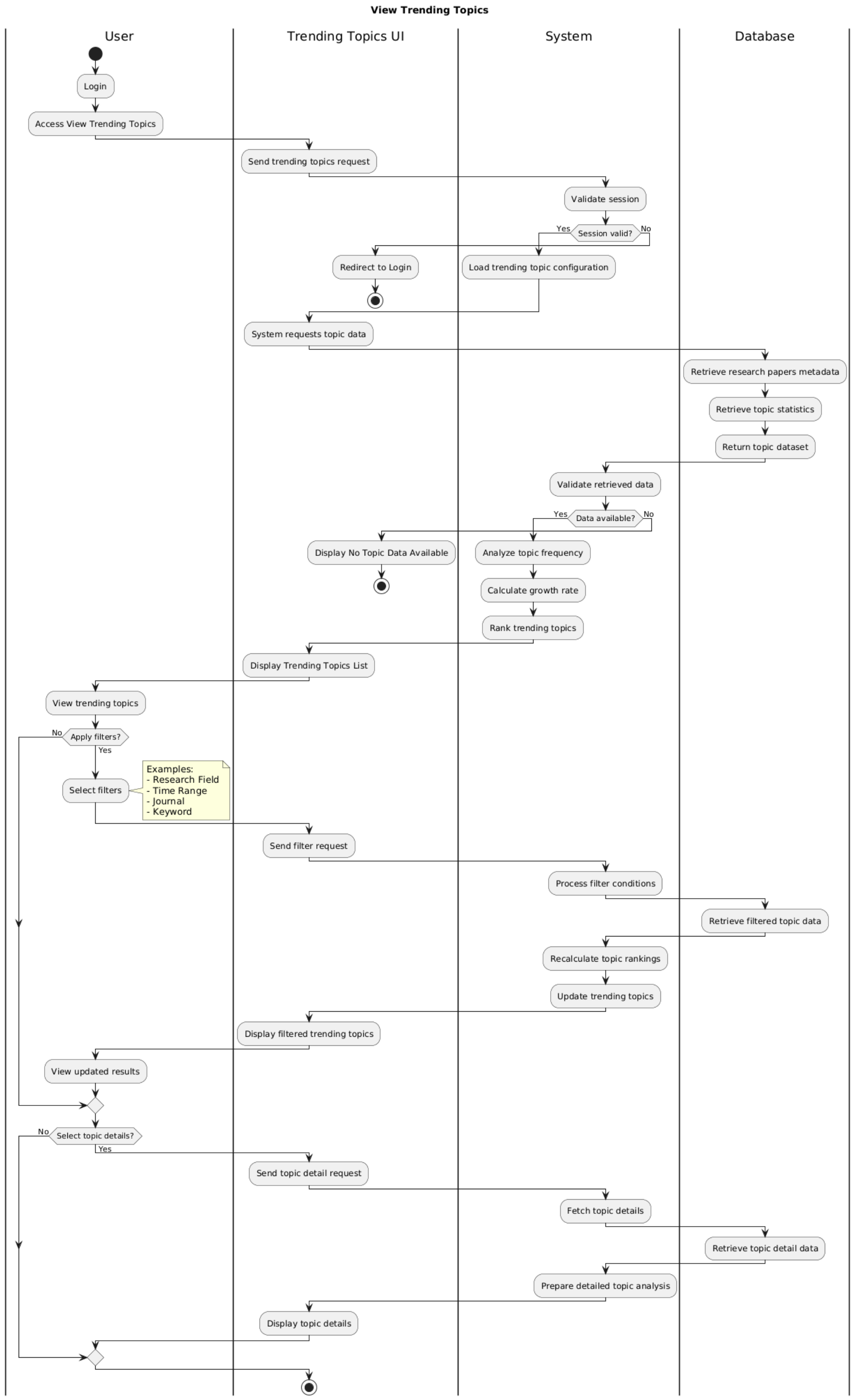
3.8.4 Class Diagram



Hình 50: Class Diagram - Generate Reports

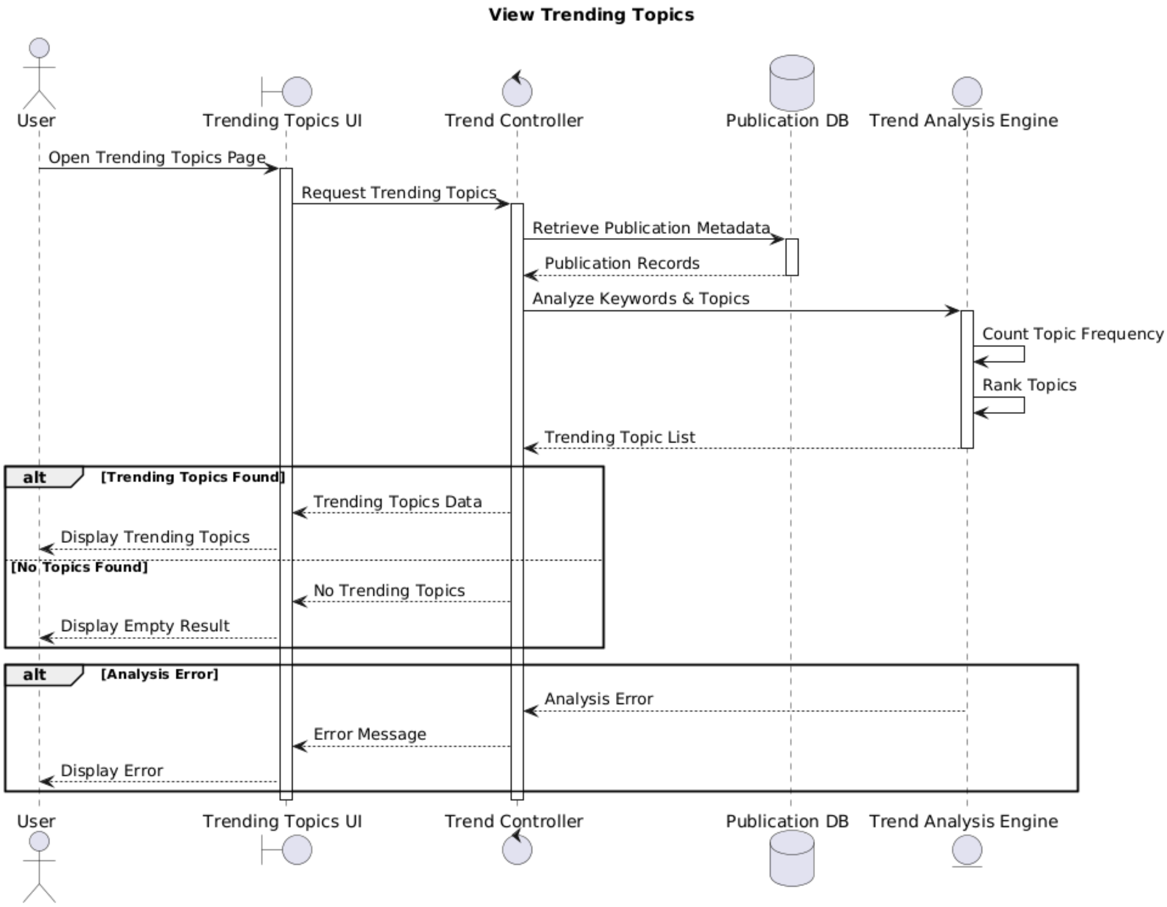
3.9 View Trending Topics

3.9.1 Activity Diagram



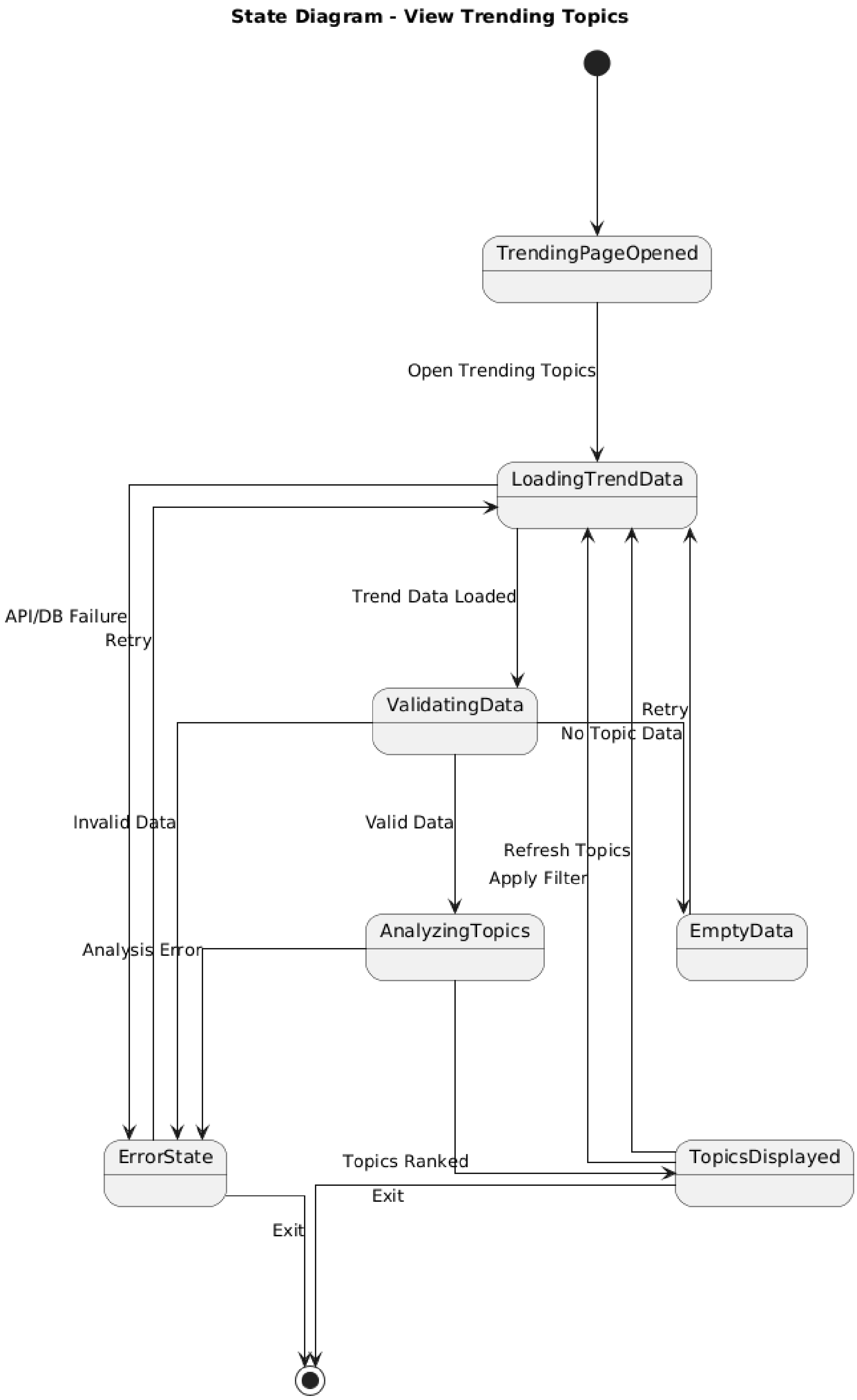
Hình 51: Activity Diagram - View Trending Topics

3.9.2 Sequence Diagram



Hình 52: Sequence diagram

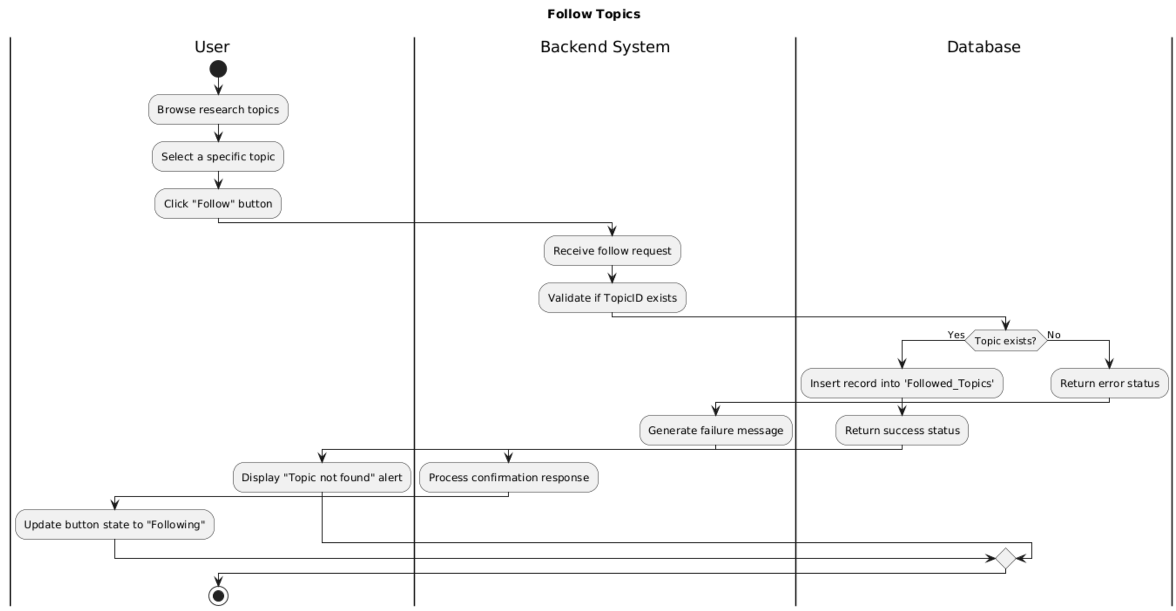
3.9.3 State Diagram



Hình 53: State diagram

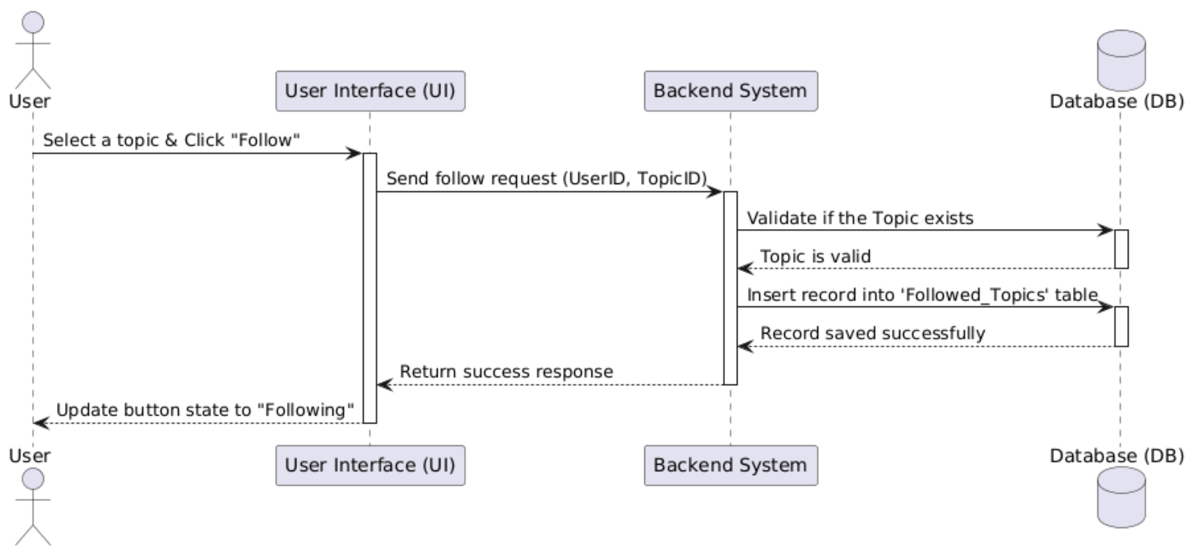
3.10 Follow Topics

3.10.1 Activity Diagram



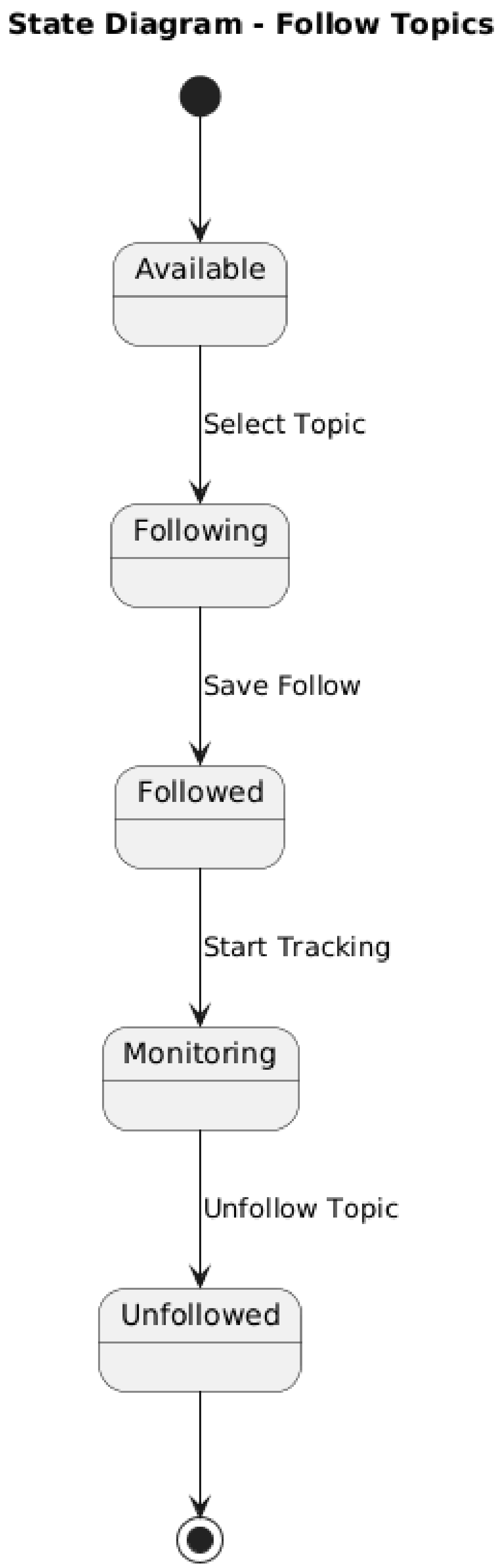
Hình 54: Activity diagram

3.10.2 Sequence Diagram



Hình 55: Sequence diagram

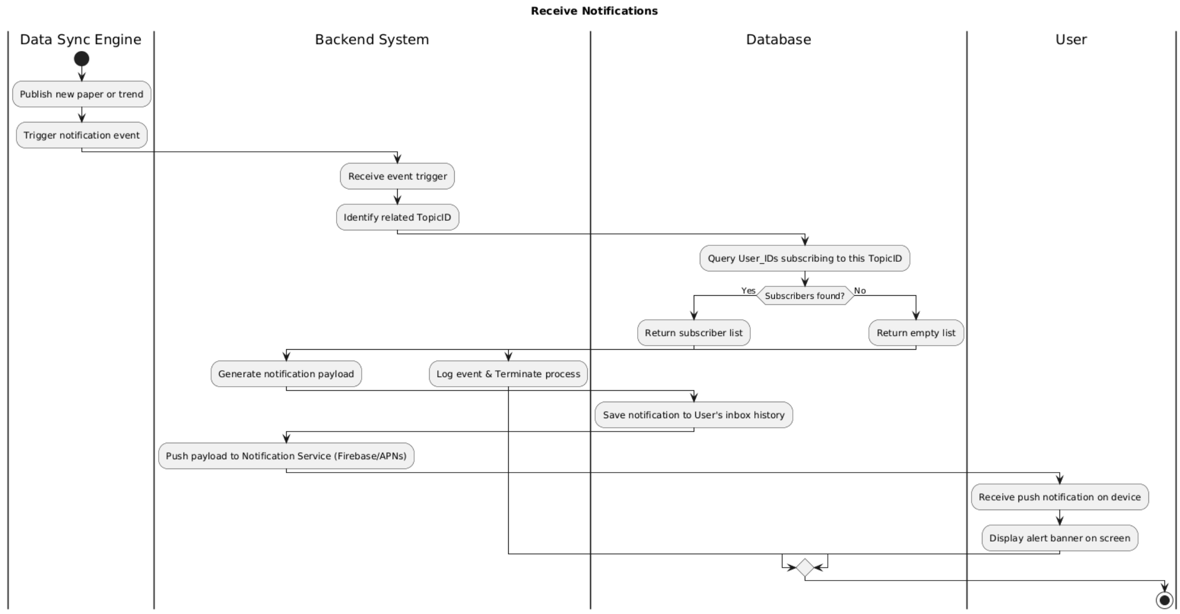
3.10.3 State Diagram



Hình 56: State diagram

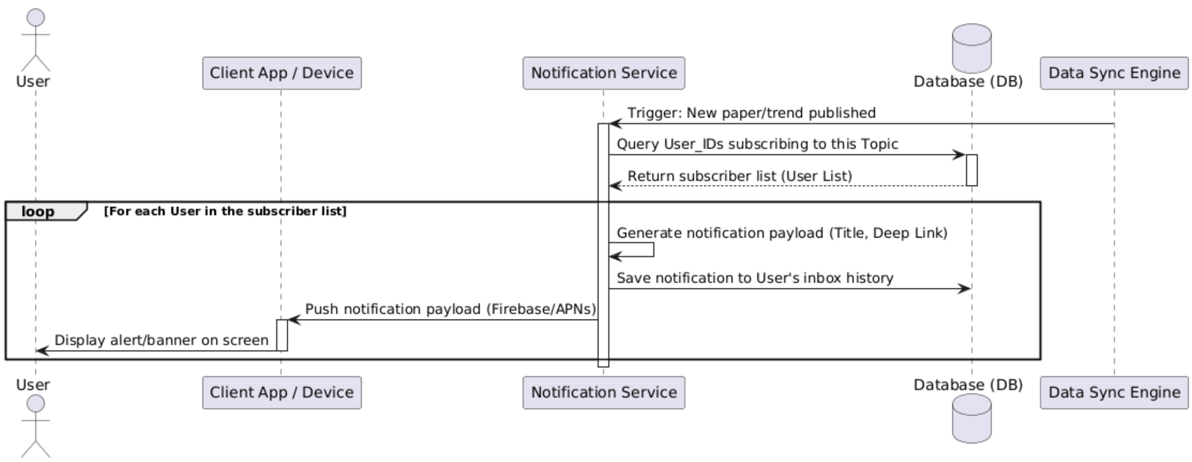
3.11 Receive Notifications

3.11.1 Activity Diagram



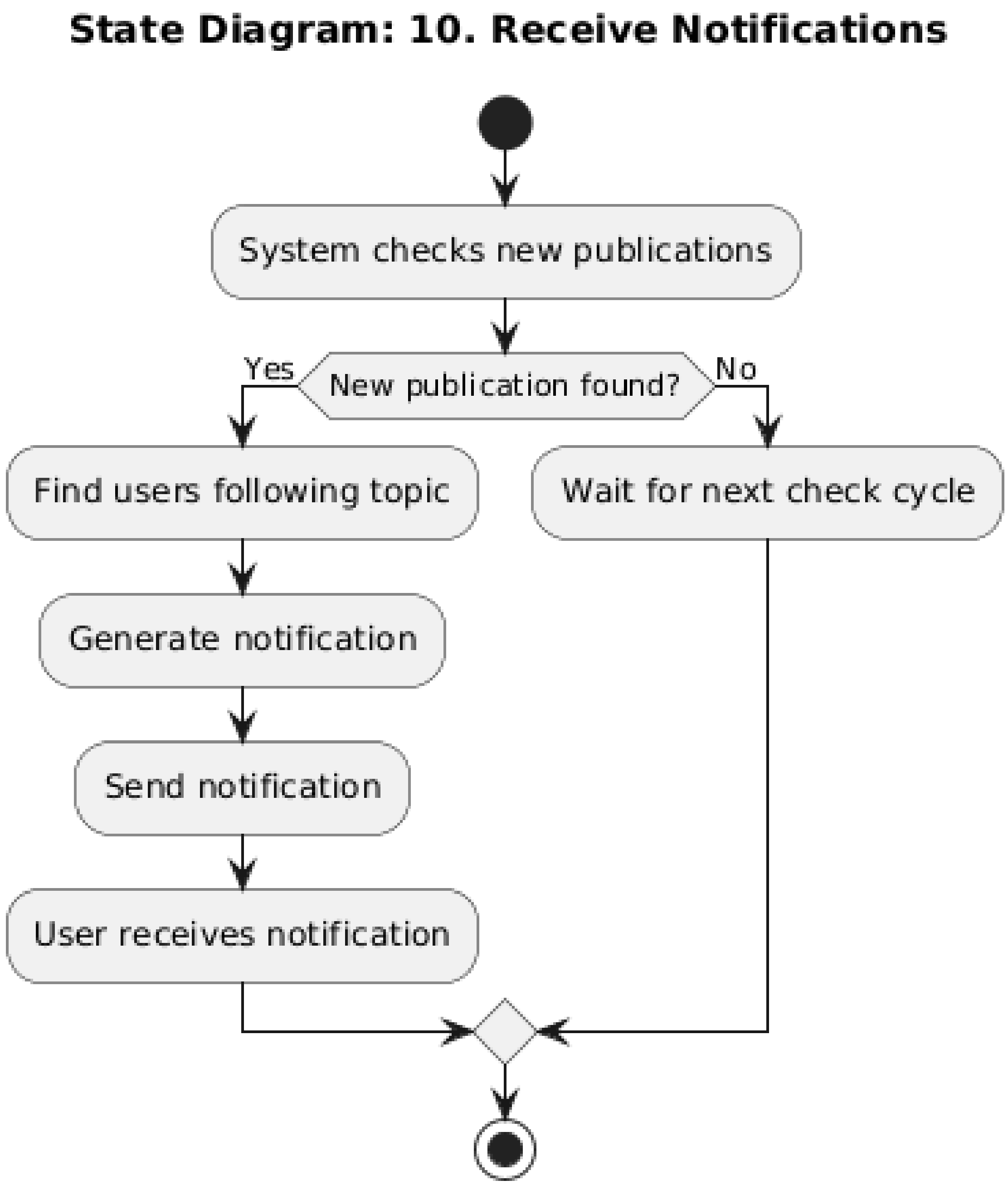
Hình 57: Activity diagram

3.11.2 Sequence Diagram



Hình 58: Sequence diagram

3.11.3 State Diagram



Hình 59: State diagram

3.12 Manage System

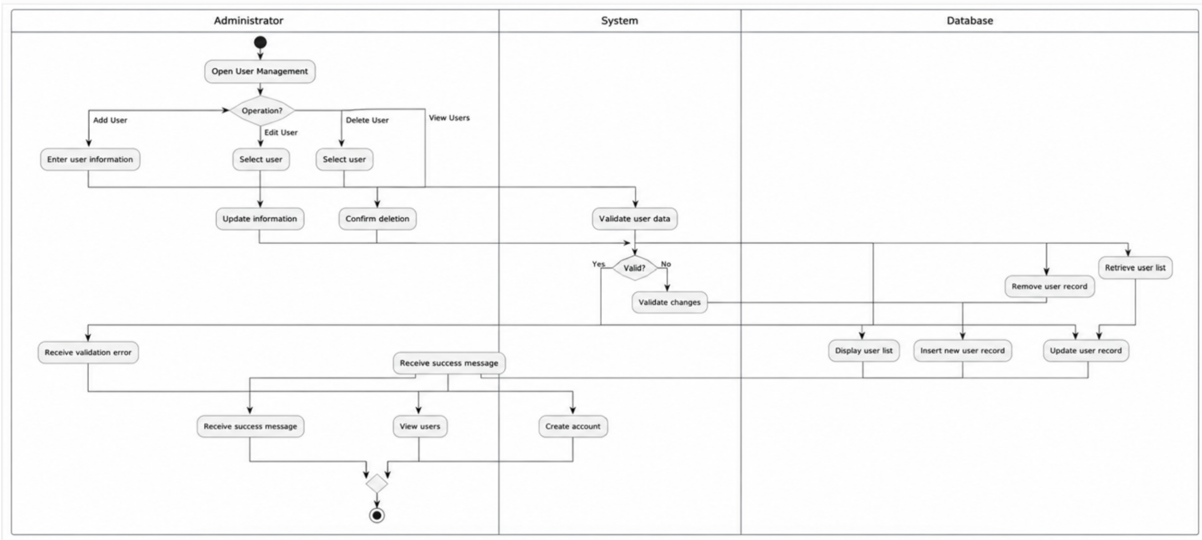
3.12.1 Activity Diagram

3.12.2 Sequence Diagram

3.12.3 State Diagram

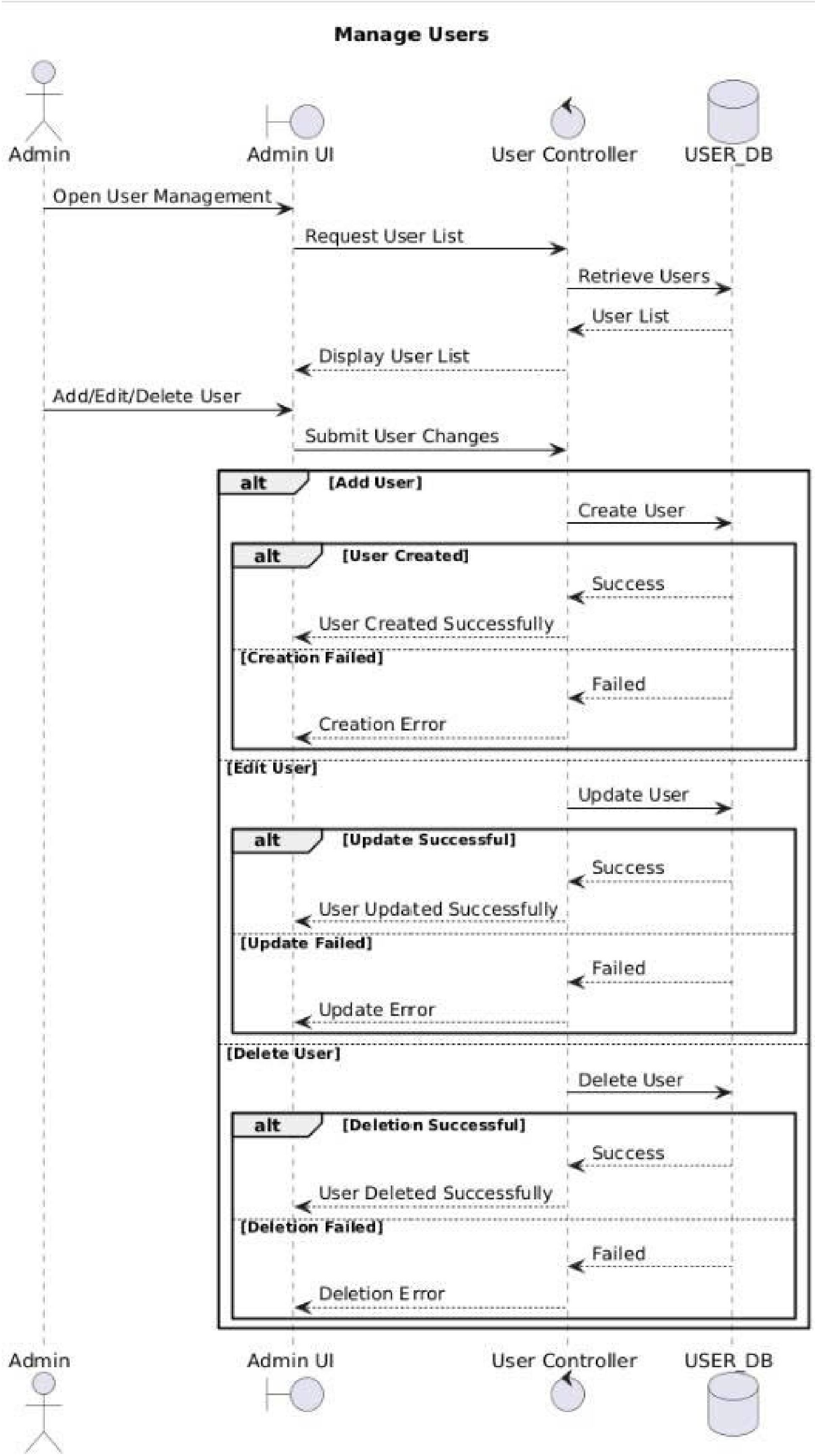
3.13 Manage User

3.13.1 Activity Diagram



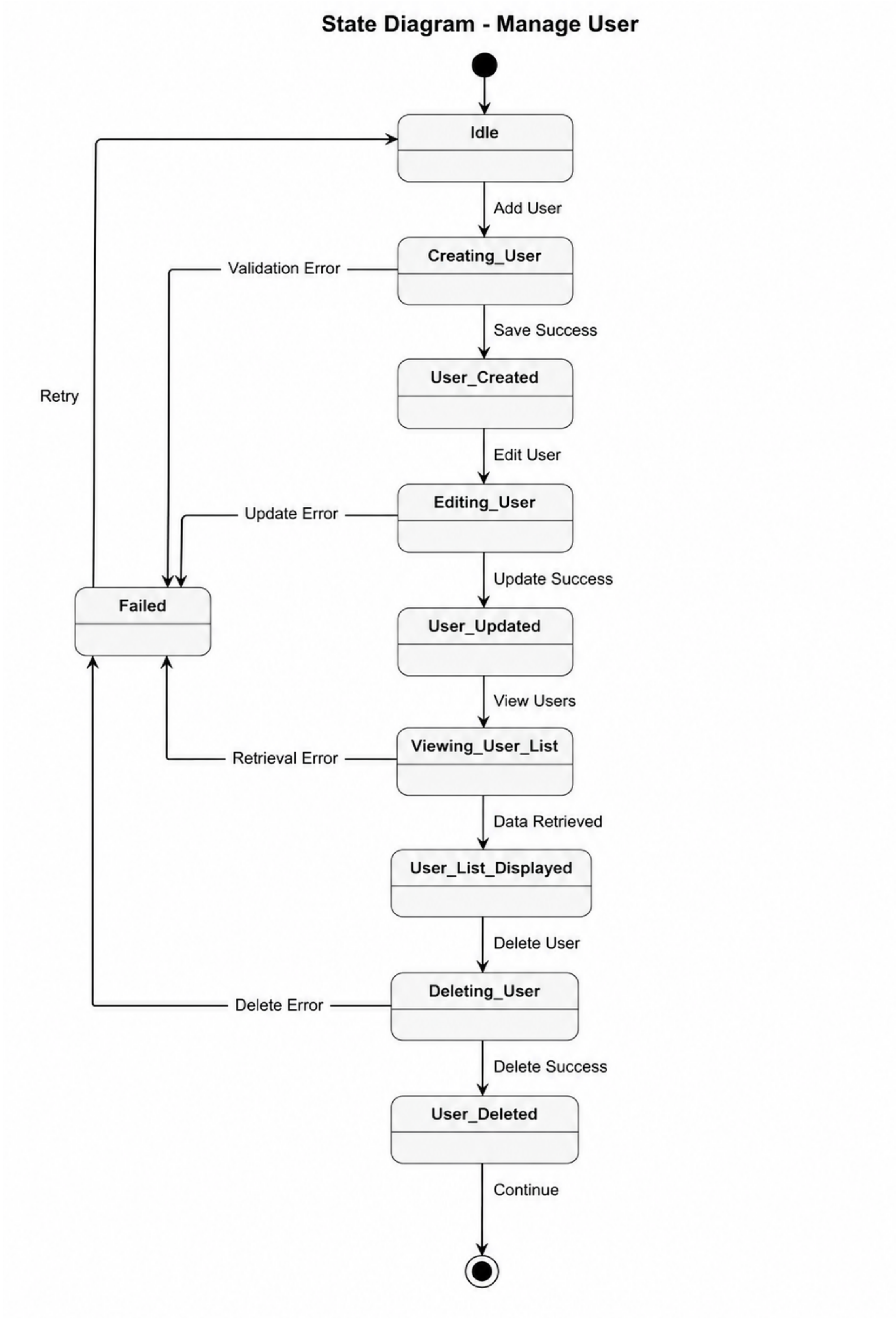
Hình 60: Activity Diagram - Manage Users

3.13.2 Sequence Diagram



Hình 61: Sequence diagram - Manage Users

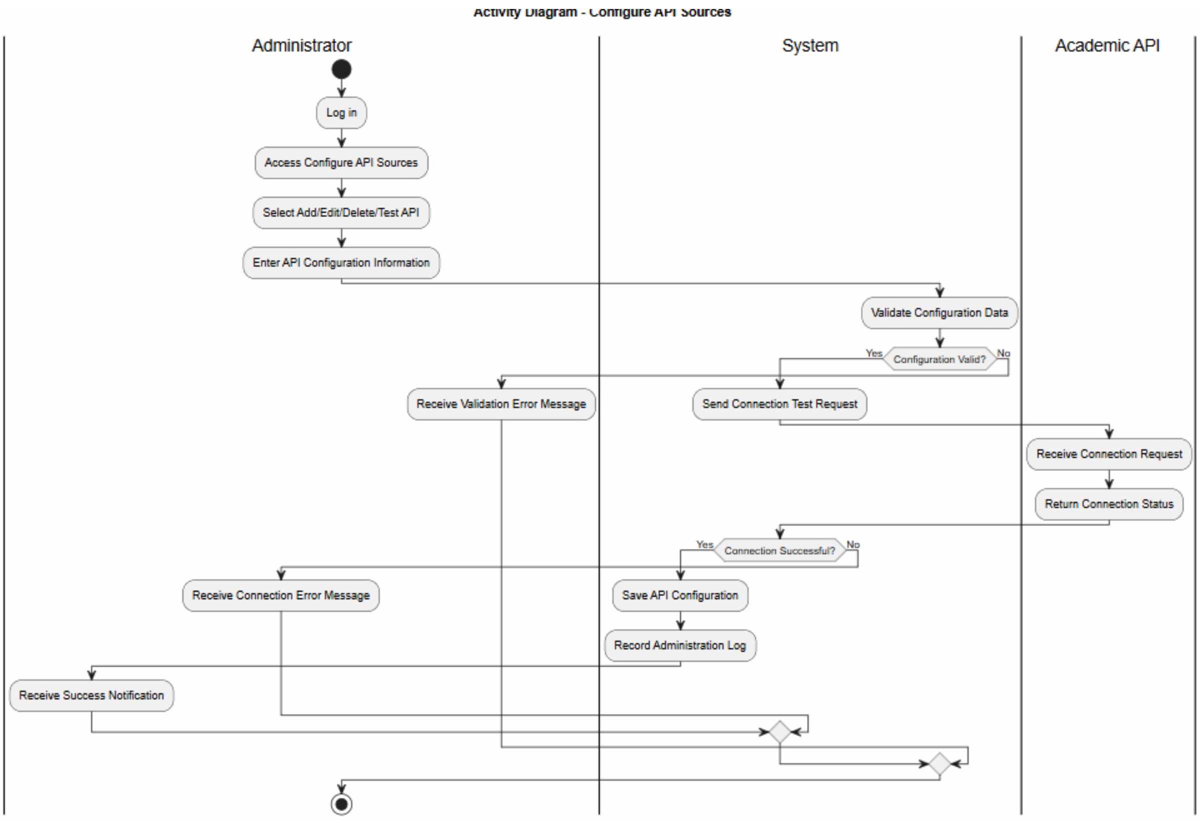
3.13.3 State Diagram



Hình 62: State diagram - Manage Users

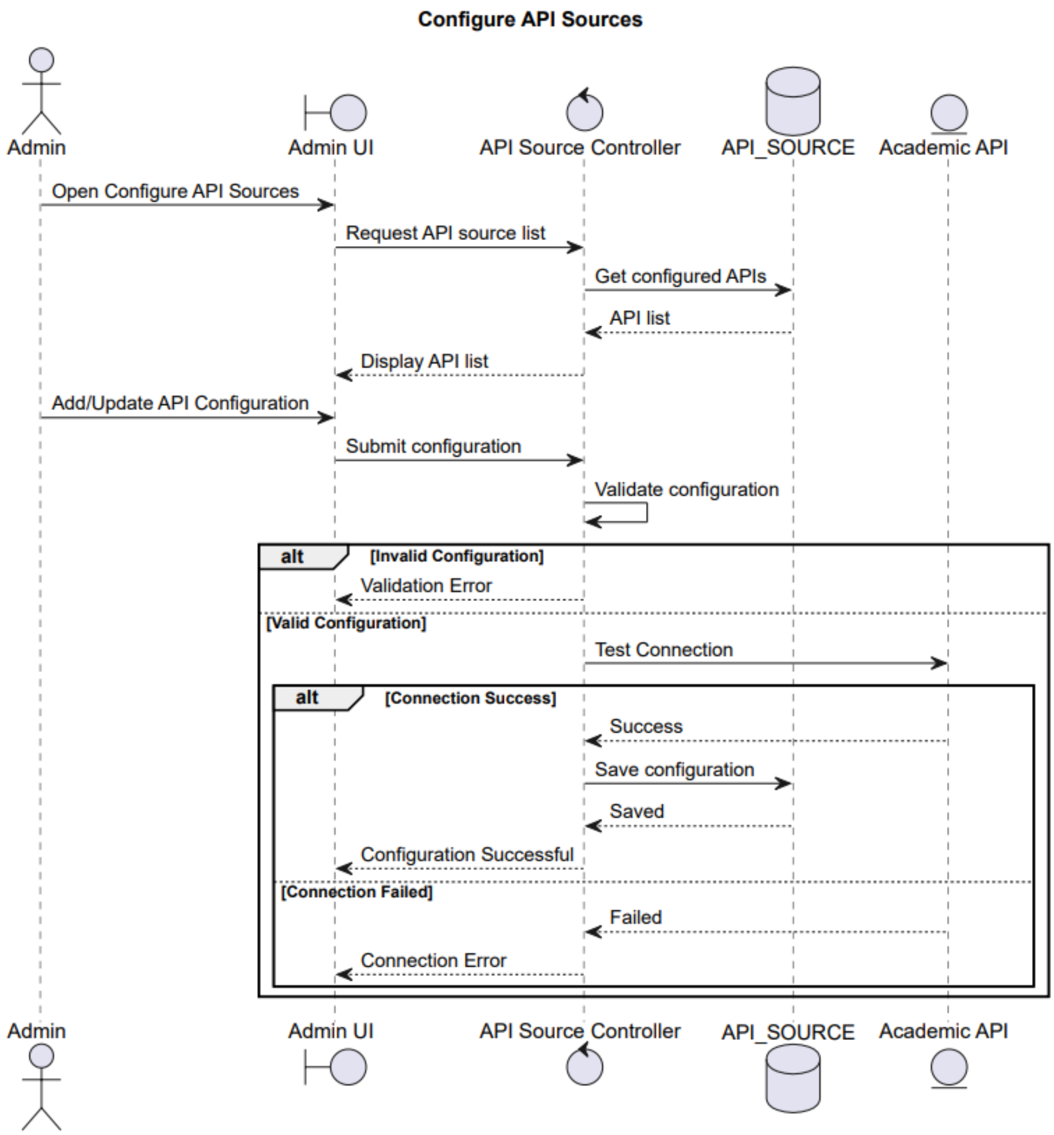
3.14 Configure API Sources

3.14.1 Activity Diagram



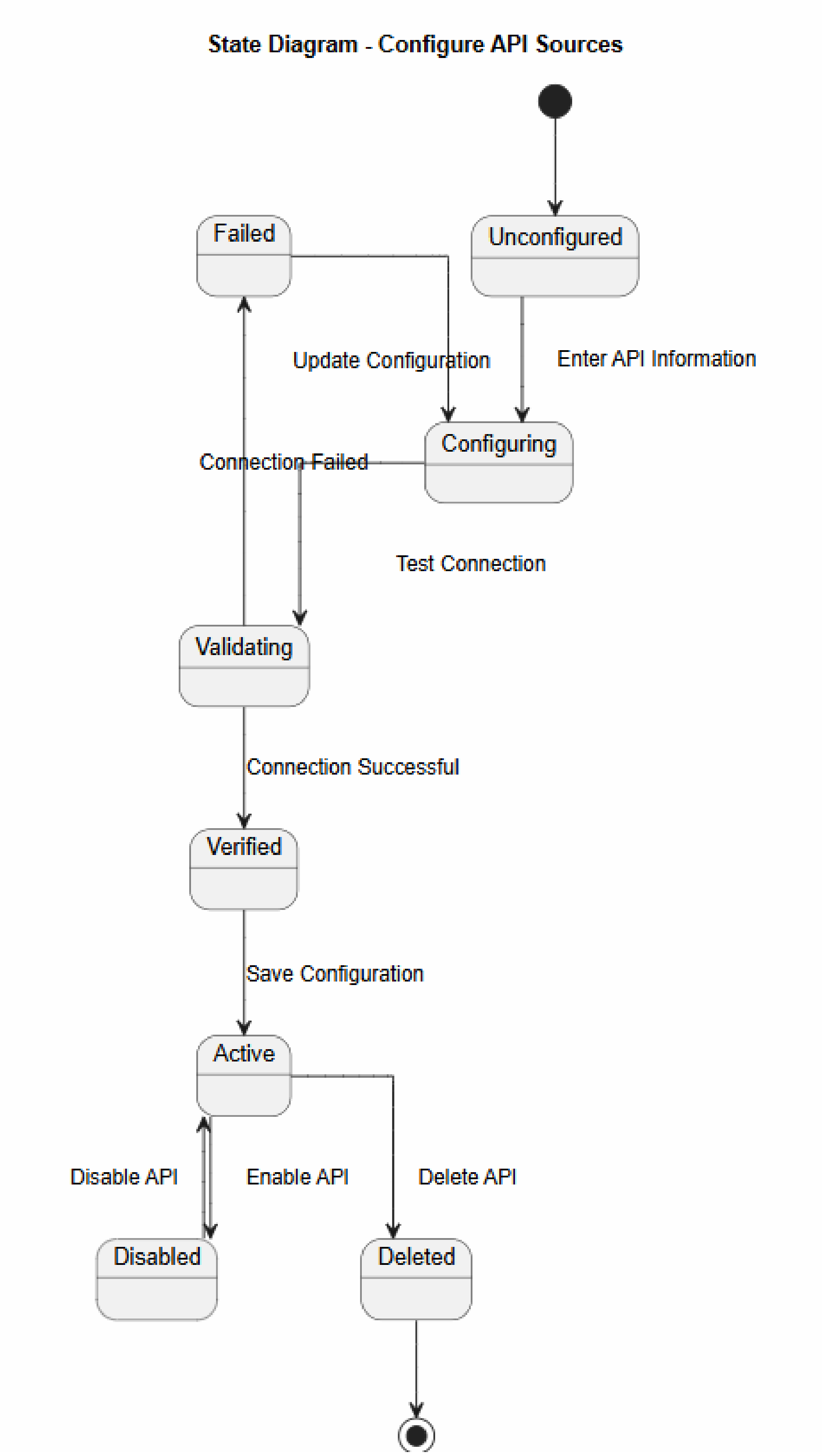
Hình 63: Configure API Sources

3.14.2 Sequence Diagram



Hình 64: Configure API Sources

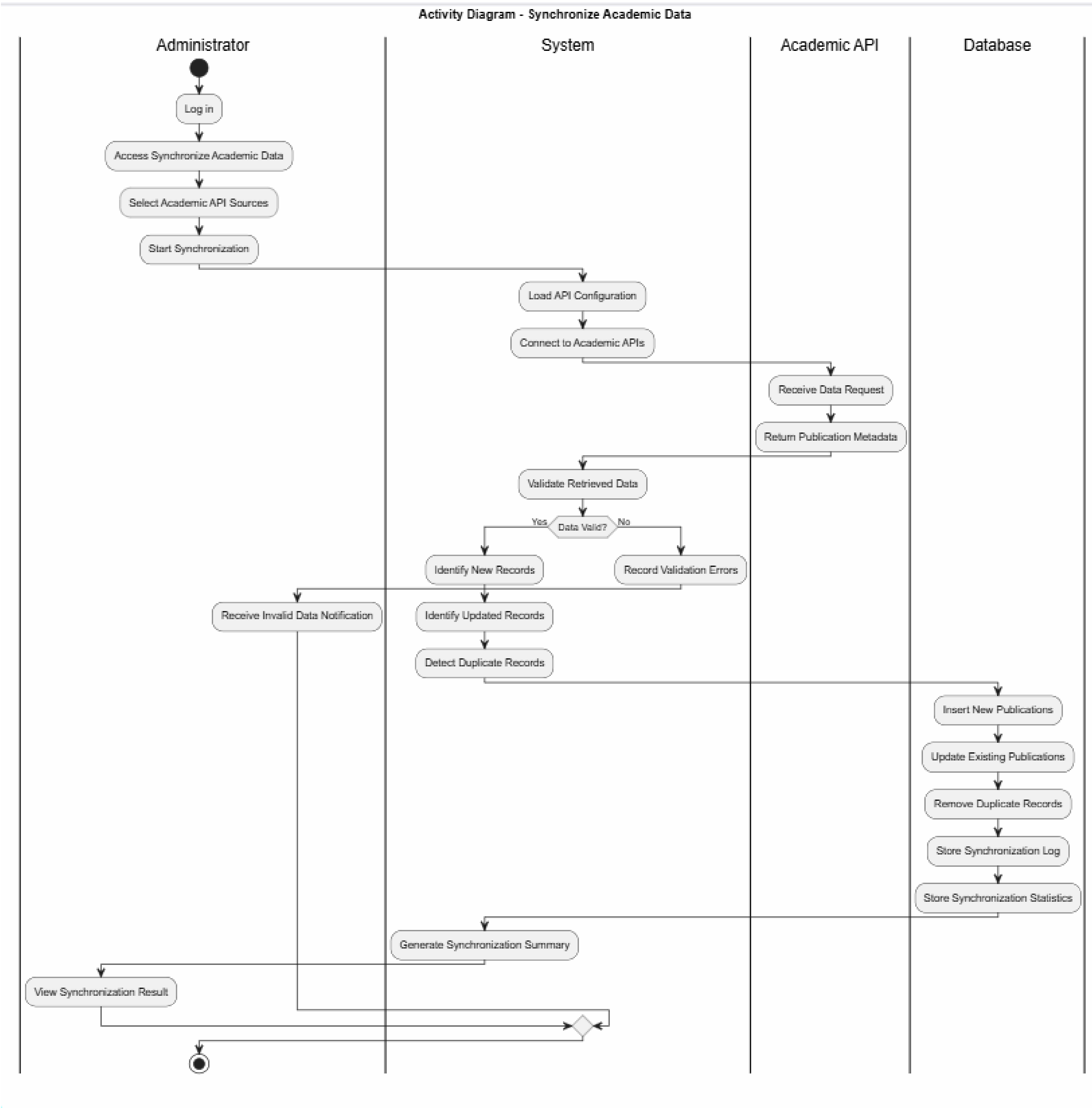
3.14.3 State Diagram



Hình 65: State diagram

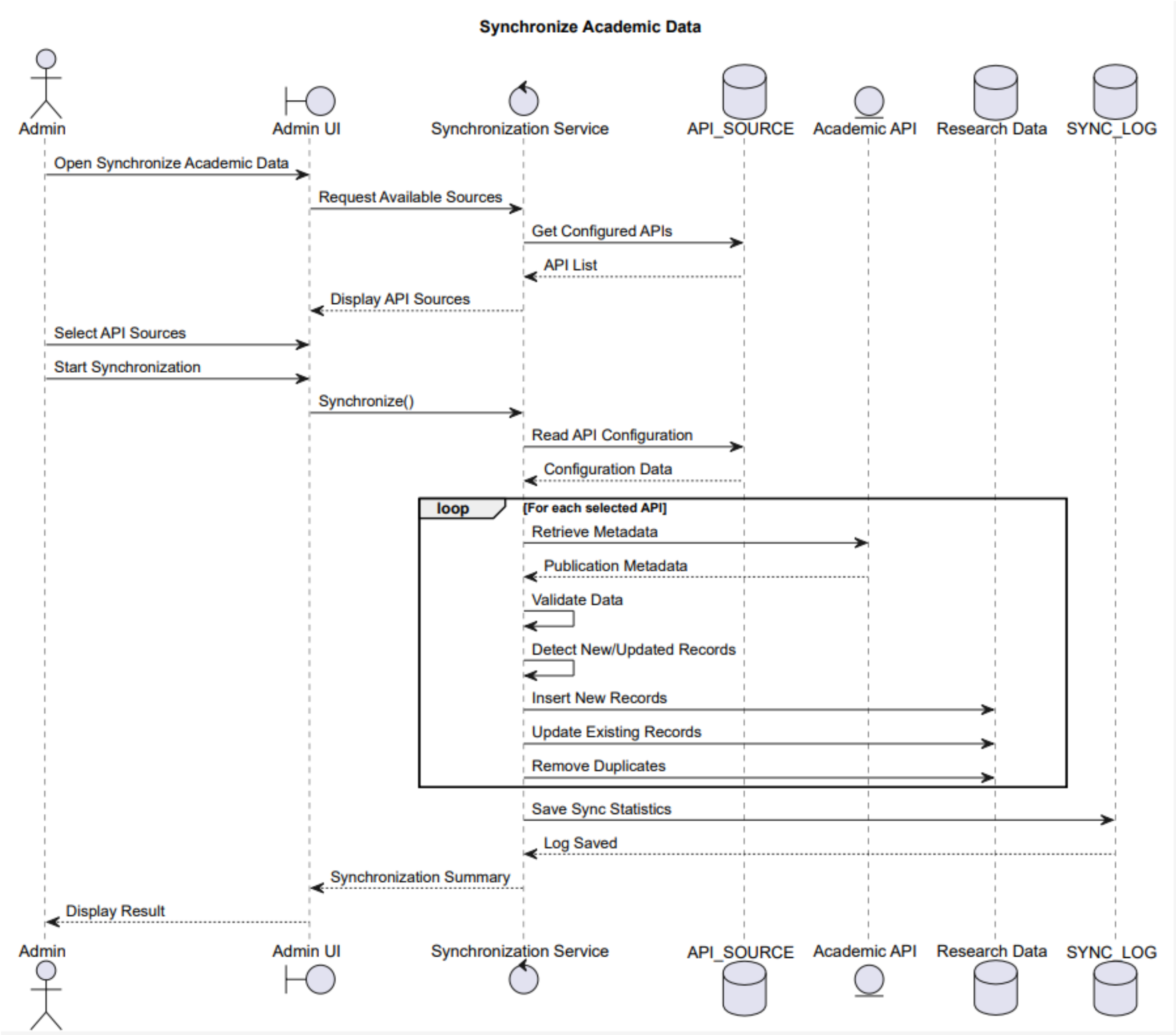
3.15 Sync Academic Data

3.15.1 Activity Diagram



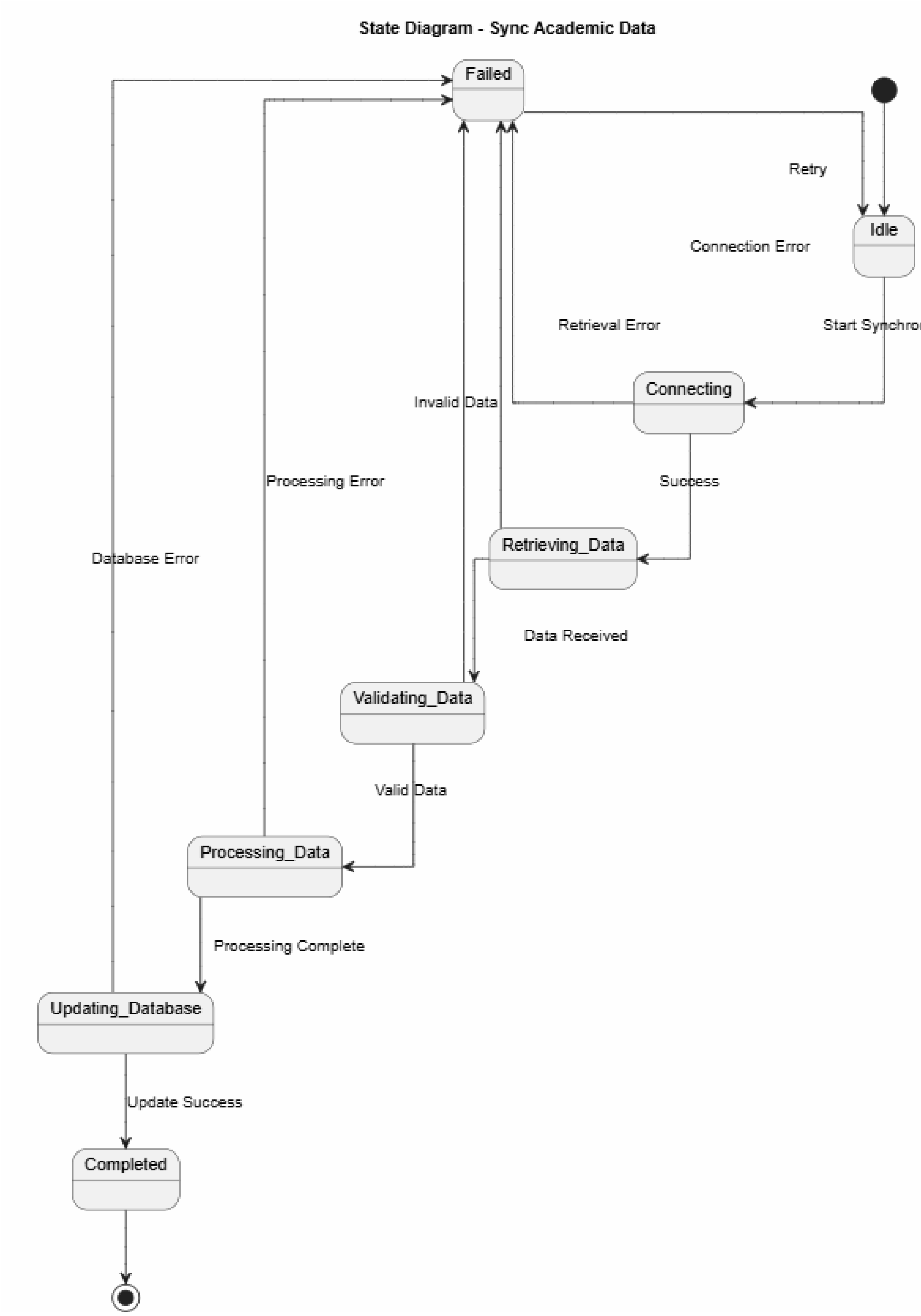
Hình 66: Sync Academic Data

3.15.2 Sequence Diagram



Hình 67: Sync Academic Data

3.15.3 State Diagram



Hình 68: Sync Academic Data